



# Technical Training

## Heat Pump Diagnostics

TRAINERS: Wilson Almeida



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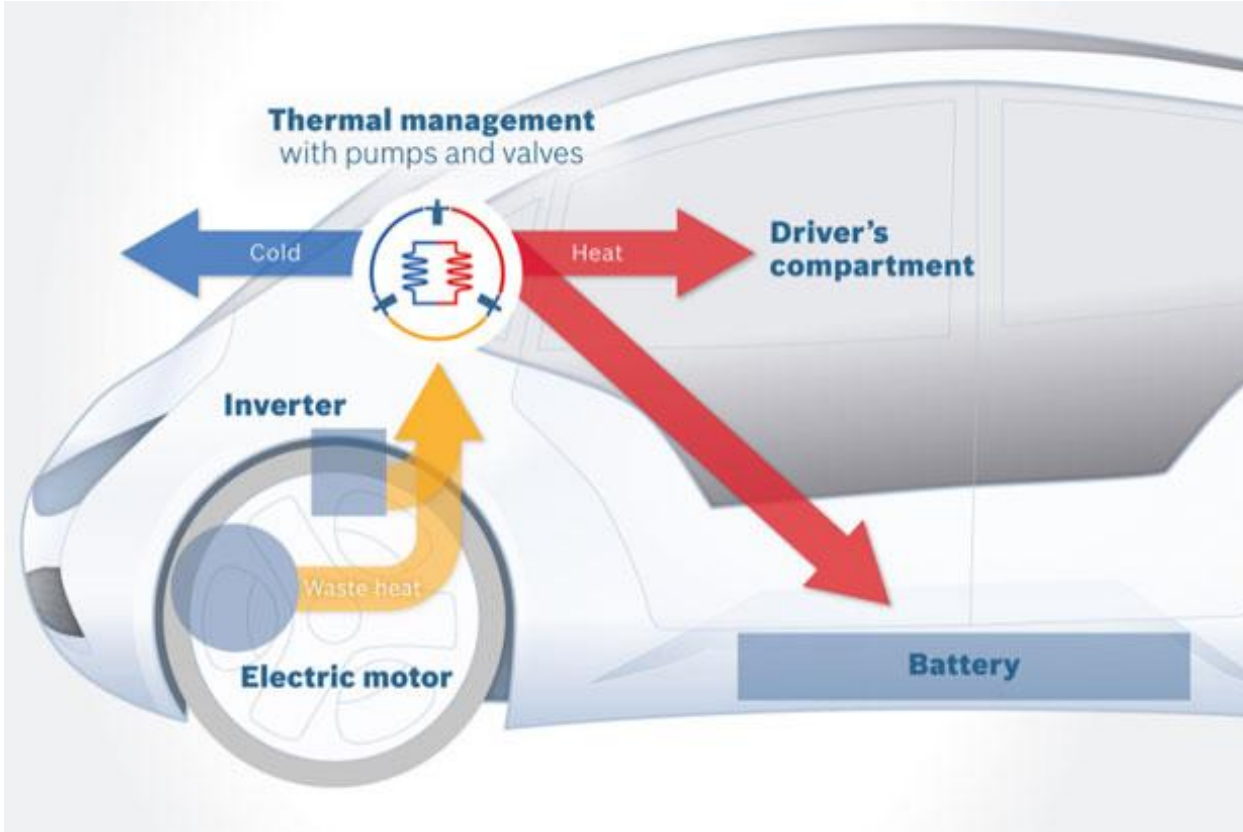
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# Training Objectives:



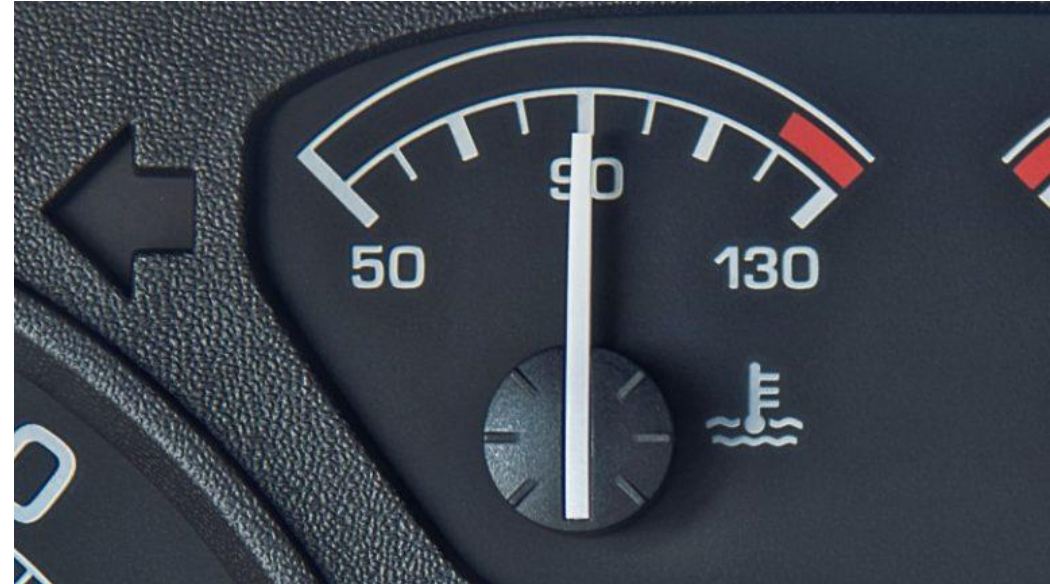
- Explain the different components of a heat pump system
- Explain the operating principle of this type of system
- Explain the work methodology for a heat pump system
- Explain the diagnostic approach for this type of system

# What are the thermal systems in an internal combustion vehicle?

- Coolant, radiator, water pump, heater core...
- Refrigerant, air conditioning
- Oil cooler
- Etc..

# What are the thermal **NEEDS** of an internal combustion vehicle?

- Keep engine coolant at its ideal temperature, so between 90 and 105 °C

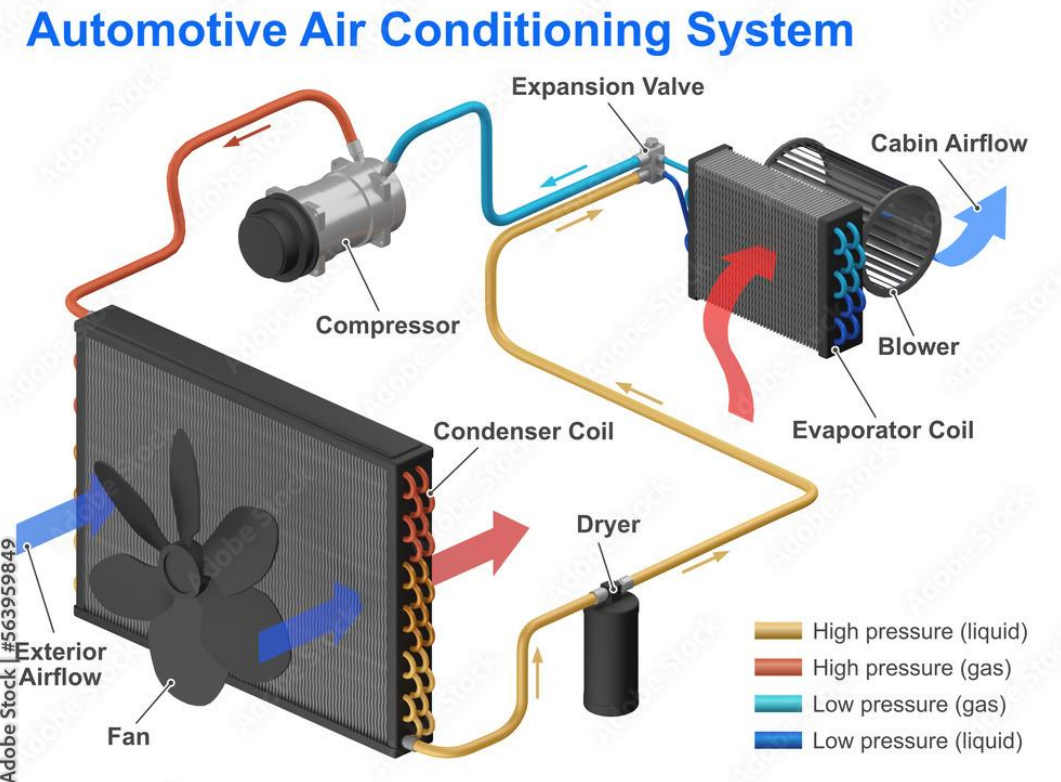
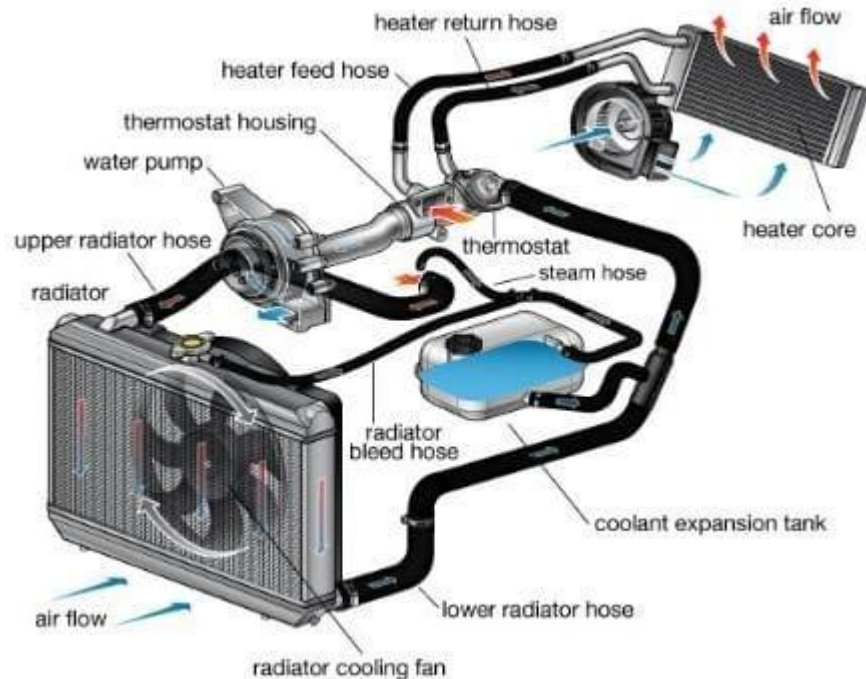


- Keep engine oil at its ideal temperature, so between 70 and 95 °C



# What are the thermal **COMFORT** functions of an internal combustion vehicle?

- Heating the passengers
    - Uses the waste heat produced by the combustion engine (60-70%)
    - Transferred into the cabin through the coolant system and a blower
  - Cooling the passengers
    - Refrigerant, air conditioning compressor
- Automotive Air Conditioning System
- 
- Expansion Valve



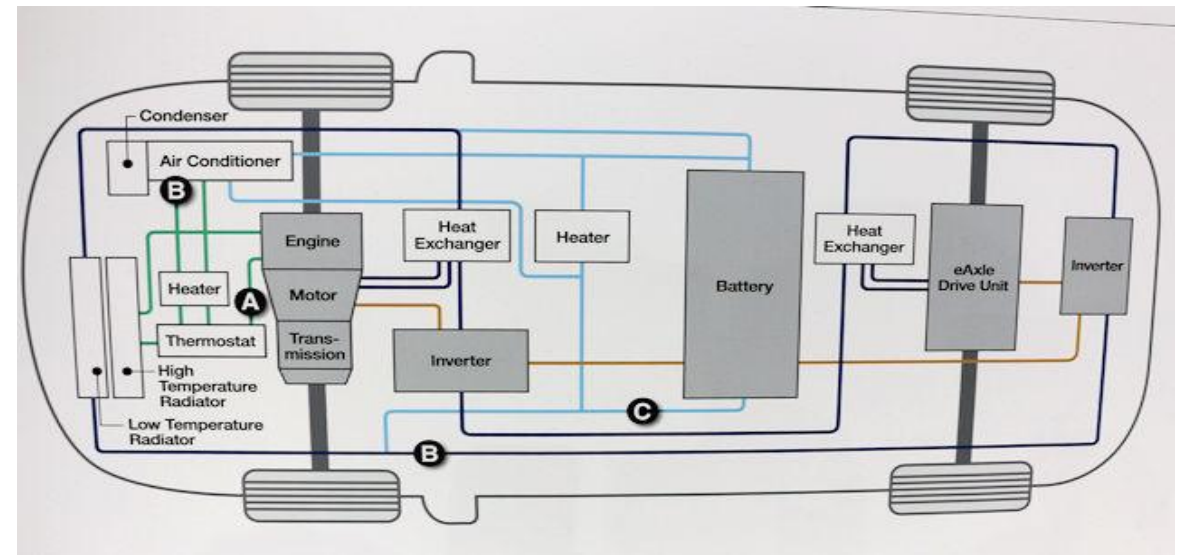
# What are the thermal **NEEDS** of a PHEV?

## Combustion need

- Keep engine coolant at its ideal temperature, so between 90 and 105 °C
- Keep engine oil at its ideal temperature, so between 70 and 95 °C

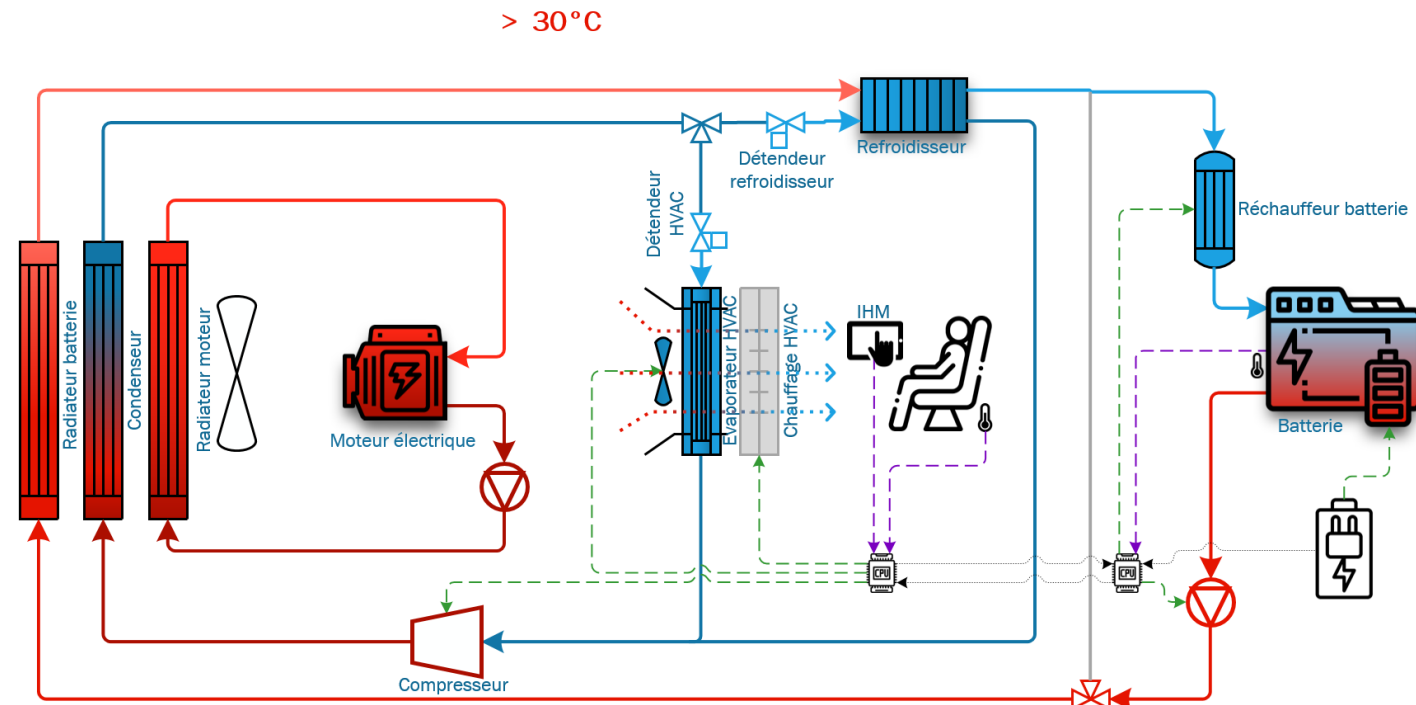
## EV need

- Maintain the ideal temperature for all HV components,
- HV battery 15-35 °C (ideal ~25 °C), electric motor (below its thermal limit), onboard charger, HV-LV converter)

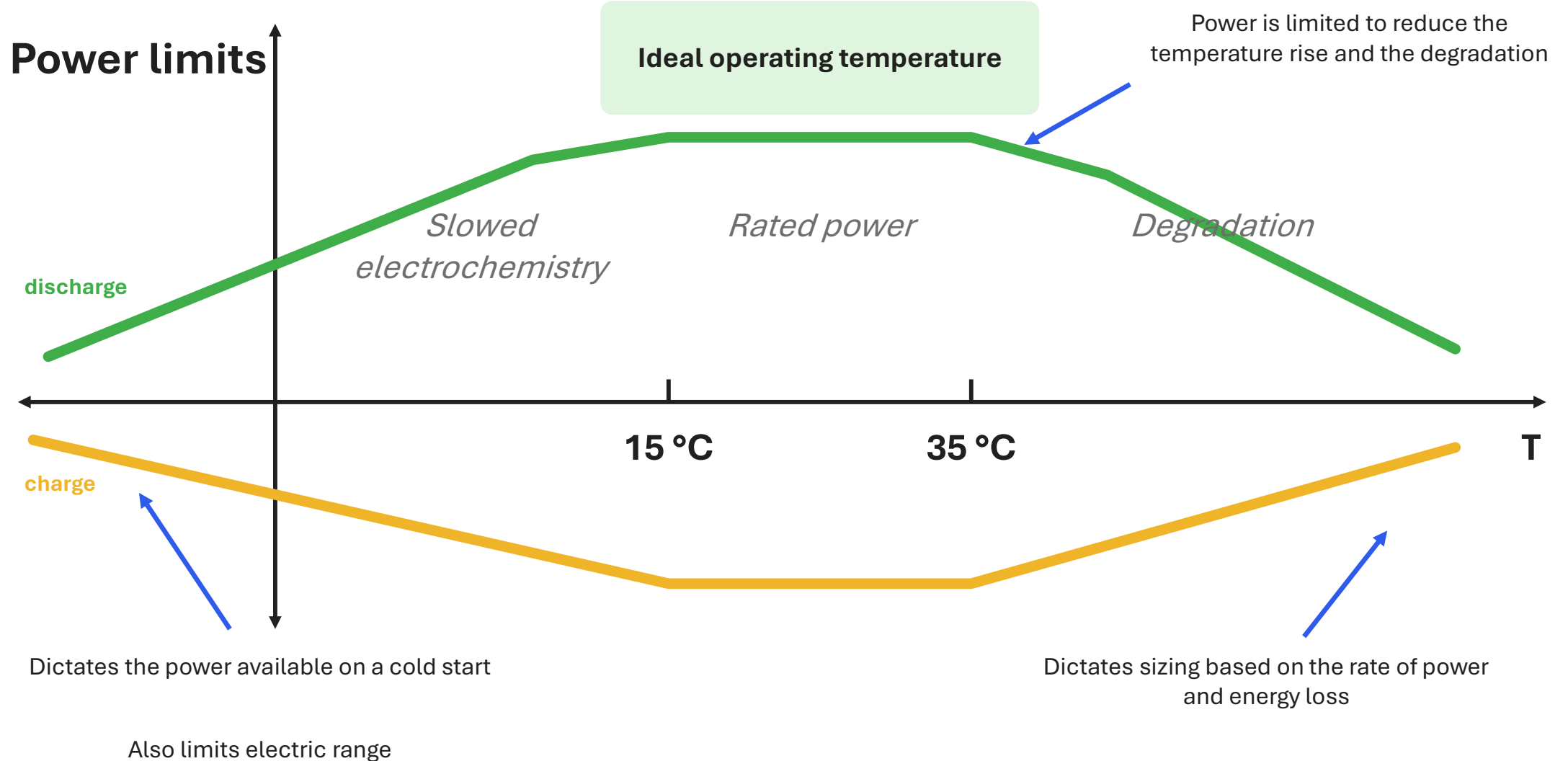


# What are the thermal **NEEDS** of an EV?

- Maintain the ideal temperature for all HV components (HV battery, electric motor, onboard charger, HV-LV converter)



# What are the thermal **NEEDS** of an EV?

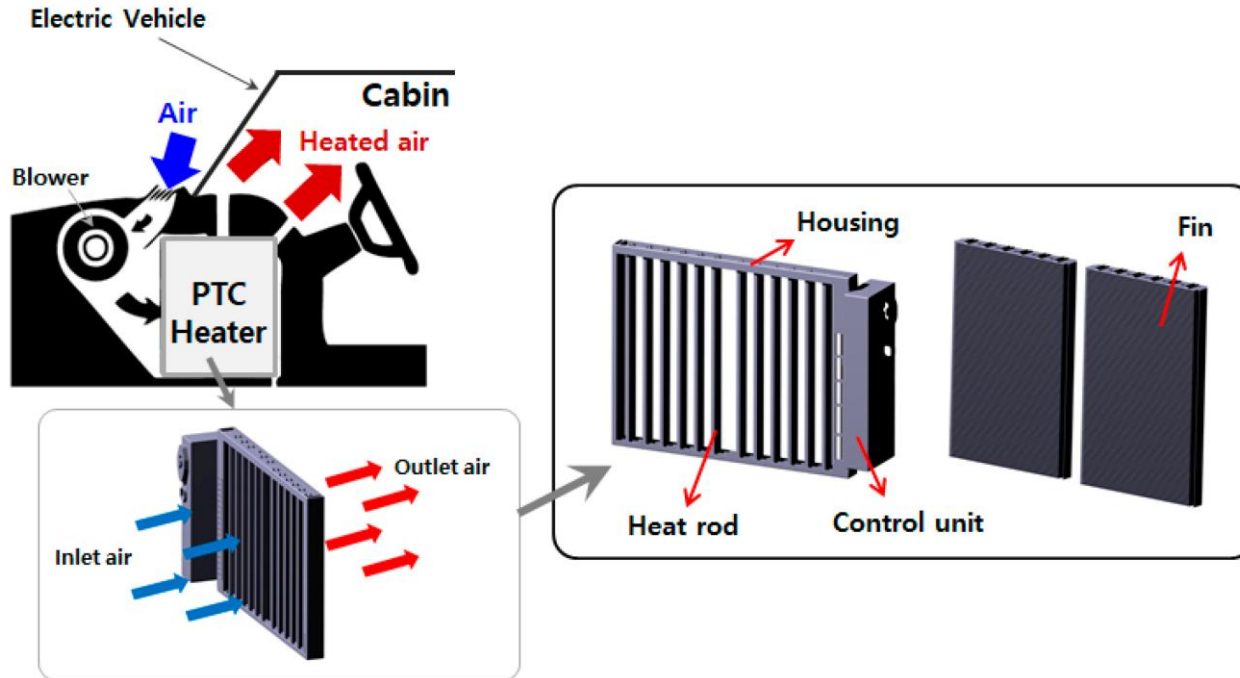


# How do we meet the thermal **NEEDS** of an EV?

- Glycol/coolant
  - Cooled by the radiator
  - Heated by a PTC heating element
- Refrigerant/air conditioning
  - Introduction of the chiller
  - Introduction of the heat pump

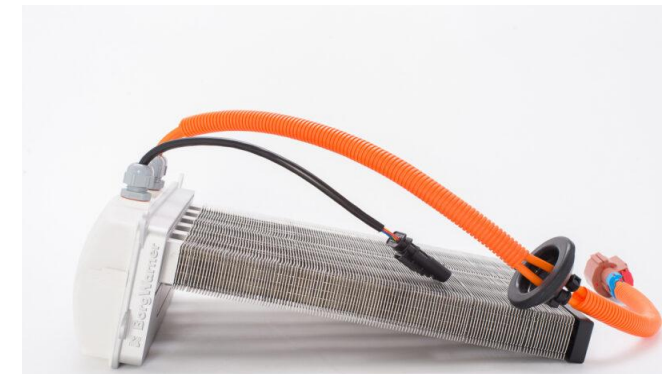
 **KEY POINT** A li-ion battery likes a narrow window: 15-35 °C, ideal around 25 °C. Above 40 °C degradation accelerates; below 0 °C, fast charging risks lithium plating. The whole thermal system exists to keep it inside that zone.

# What is a PTC heating element?



- A PTC heater is an electric heater with a positive temperature coefficient.
- Simple definition: PTC stands for **Positive Temperature Coefficient**. It means that the hotter the element gets, the higher its electrical resistance, which automatically limits the current.

- Cold: low resistance → current flows easily → it heats up fast
  - Hot: resistance rises → current drops → the temperature stabilizes
- Result: **self-regulating heating**, with no risk of overheating.



# Glycol/coolant with a PTC heating element

The PTC heating element: an electric heating element used to warm the coolant in an EV or hybrid. It works like a smart heating element :

- When the coolant temperature is low, it produces more heat, so it draws more power
- When the coolant temperature rises, it automatically reduces its output, with no risk of overheating



In an EV, the PTC heating element quickly warms up the high-voltage battery, the cabin or the electronic components.

# Why a heat pump?

## HEAT PUMP SYSTEMS ARE ESSENTIAL ON ELECTRIC VEHICLES:

- Heat pumps help EVs use energy more efficiently for heating and cooling, which directly affects range
- Thermal management protects the life and the performance of the HV battery, especially in cold weather
- Heat pumps are now common on many EVs. Technicians therefore have to be ready to work on these systems
- Heat pumps bring in several components (HV compressor, EXV, chiller, coolant loop, refrigerant loop and PTC)

# What is a heat pump?

**In an electric vehicle it does not create heat: it moves it.**

## **The high-voltage battery**

Held within its optimal temperature range, summer and winter.

## **The electronic components**

Cooled to prevent overheating during fast charging and sustained driving.

## **The cabin**

Heated or cooled for passenger comfort.

*One thermal system answers all three needs of the vehicle.*

# Heat that already exists

It takes heat where it already is instead of producing it with electricity.

## The available sources

Outside air, even in cold weather.  
Waste heat from the electric motor.  
Heat from the inverter and charger.  
Heat given off by the battery.

## The gain

Far less electricity used for the same comfort in the cabin.  
That is what makes the heat pump energy efficient.

*Nothing is created: heat is captured, moved, then released.*

# Why it matters: driving range

**In winter, heating is the vehicle's largest consumer after propulsion.**

## **Less energy consumed**

The heat pump needs far less electricity than conventional heating for the same heat.

## **More range preserved**

The energy saved stays available for driving, which shows most in cold weather.

*This is the most concrete argument for a Canadian customer.*

# How does a heat pump work?

**It does not produce heat: it moves it. Even when cold, outside air still holds heat.**

## **1 Absorb**

The refrigerant captures heat from the outside air, even in cold weather.

## **2 Compress**

The compressor raises its pressure, and therefore its temperature.

## **3 Release**

The heat is released into the cabin through the condenser.

## **4 Reverse**

The cycle can change its path and cool the cabin instead.

*The source can be the outside air, but also waste heat from the motor, the inverter or the battery.*

# Why it matters: efficiency

A resistance heater turns electricity into heat. A heat pump moves it.

## PTC heater

**1 kWh → 1 kWh**

Efficiency of about 100 %

## Heat pump

**1 kWh → 2 to 4 kWh**

COP above 1, depending on temperature

**For the same electricity drawn from the battery, the heat pump delivers 2 to 4 times more heat.**

# What it changes for the customer

## Battery preservation

In winter, conventional heating drains the battery quickly and cuts the vehicle's range.

The heat pump limits that extra draw and extends the distance between charges.

## Versatility

It heats, but it also cools the cabin and certain electronic components.

And it stays efficient even in cold weather.

*Two arguments to keep in mind when explaining the heat pump to a customer.*

# Limits of the heat pump

**Very efficient and environmentally friendly, but with limits depending on use and conditions.**

## **Extreme cold**

Below  $-20\text{ }^{\circ}\text{C}$  the heat pump captures less heat from the air. A PTC backup heater may become necessary. Consumption then rises sharply.

## **Humidity and source**

Very dry or very humid air reduces its efficiency. Under some conditions the heat flow is not enough to warm things up quickly.

## **Maintenance**

Expansion valve, evaporator, condenser: many components. More parts means more chances of failure and expensive repairs.

*These limits do not call the technology into question, they frame how it is used.*

# Heat Pump COP vs Refrigerant

|       | <b>R134A</b> | <b>R1234YF</b> | <b>R744</b> |
|-------|--------------|----------------|-------------|
| -10°C | 1.6–2.4      | 1.5–2.3        | 1.7–2.6     |
| -20°C | 1.2–1.8      | 1.1–1.7        | 1.3–2.0     |
| -30°C | 1.0–1.4      | 1.0–1.3        | 1.1–1.6     |

|       | <b>R134A + Coolant<br/>PTC</b> | <b>R1234YF +<br/>Coolant PTC</b> | <b>R744 + Coolant<br/>PTC</b> |
|-------|--------------------------------|----------------------------------|-------------------------------|
| -10°C | 1.4–2.1                        | 1.3–2.0                          | 1.6–2.4                       |
| -20°C | 1.1–1.6                        | 1.0–1.5                          | 1.3–1.9                       |
| -30°C | 0.9–1.3                        | 0.9–1.2                          | 1.1–1.6                       |

|       | <b>R134A + Coolant<br/>PTC + Chiller +<br/>Recovery</b> | <b>R1234YF + Coolant<br/>PTC + Chiller +<br/>Recovery</b> | <b>R744 + Coolant<br/>PTC + Chiller +<br/>Recovery</b> |
|-------|---------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------|
| -10°C | 1.6–2.4                                                 | 1.5–2.3                                                   | 1.6–2.5                                                |
| -20°C | 1.5–2.2                                                 | 1.4–2.1                                                   | 1.5–2.3                                                |
| -30°C | 1.4–2.0                                                 | 1.3–1.9                                                   | 1.4–2.1                                                |

# How does all of this work?

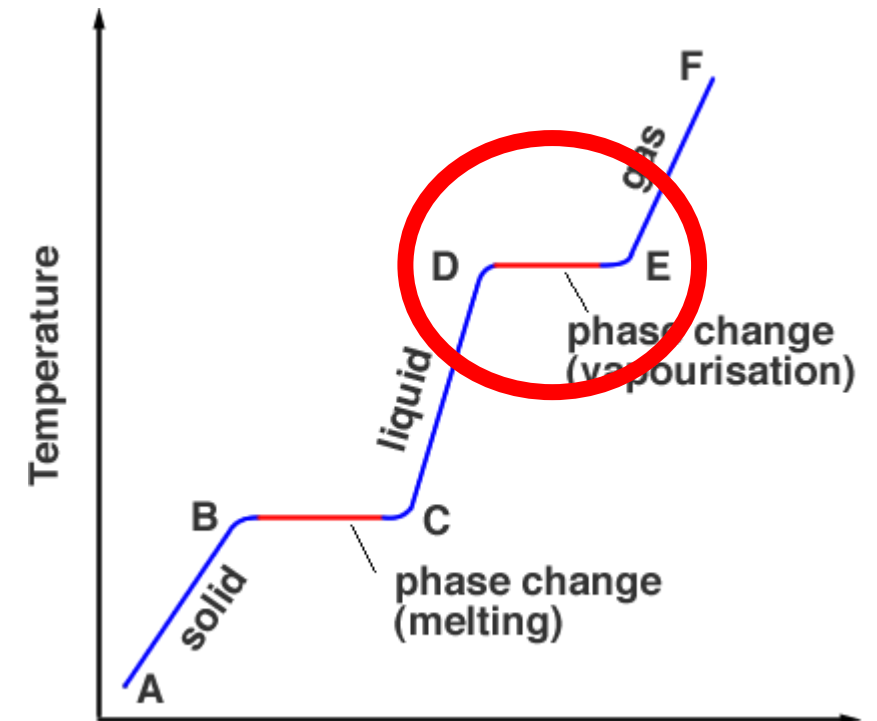
## Latent heat

# What is latent heat?

- **Latent heat** is a thermodynamic concept describing the energy needed to change the state of a substance without changing its temperature. In other words, it is the heat absorbed or released during a phase change such as melting, vaporization or solidification.

In heat pumps, this is what makes refrigerants so powerful:

- Refrigerants absorb a lot of heat (energy) when they evaporate and release it when they condense, all while staying at the same temperature
- That "invisible" energy transfer is what lets electric vehicles heat and cool efficiently

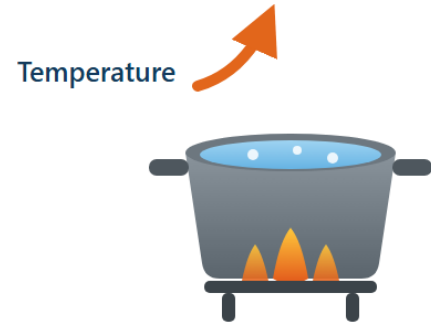


 **TIP** Sweat evaporating cools the skin without changing the temperature of the water: that is latent heat. The heat pump uses exactly that phenomenon, but in a closed, reversible circuit.

# Sensible heat and latent heat

## Sensible Heat

### Temperature Change



Water from 20 °C to 50 °C



The temperature **changes**

## Latent Heat

### Change of State



Water ↔ Steam at 100 °C



Ice ↔ Water at 0 °C

The temperature **stays constant**



# Latent heat at work

**At every change of state the refrigerant absorbs or releases a large amount of energy without changing temperature.**

## **A Evaporation, outside**

The refrigerant arrives as a low-pressure liquid.

It evaporates in contact with the outside air, even when it is cold.

As it evaporates it absorbs a large amount of latent heat.

## **B Condensation, inside**

After compression the refrigerant is a hot gas.

In the condenser it turns back into a liquid.

It then releases the heat captured outside, plus the heat added by compression.

*Even at  $-10^{\circ}\text{C}$  the air holds enough energy to evaporate the refrigerant. That released heat is what warms the cabin.*

# Why latent heat changes everything

The liquid ↔ gas transition carries a huge amount of energy without the refrigerant temperature moving.

## A very high heat of vaporization

Changing state takes, and gives back, far more energy than simply warming a fluid.

That is what makes refrigerants so effective at carrying heat.

## What it makes possible

The compressor only pays for moving the heat, not for creating it.

That is where the COP seen earlier comes from: 1 kWh used, 2 to 4 kWh moved.

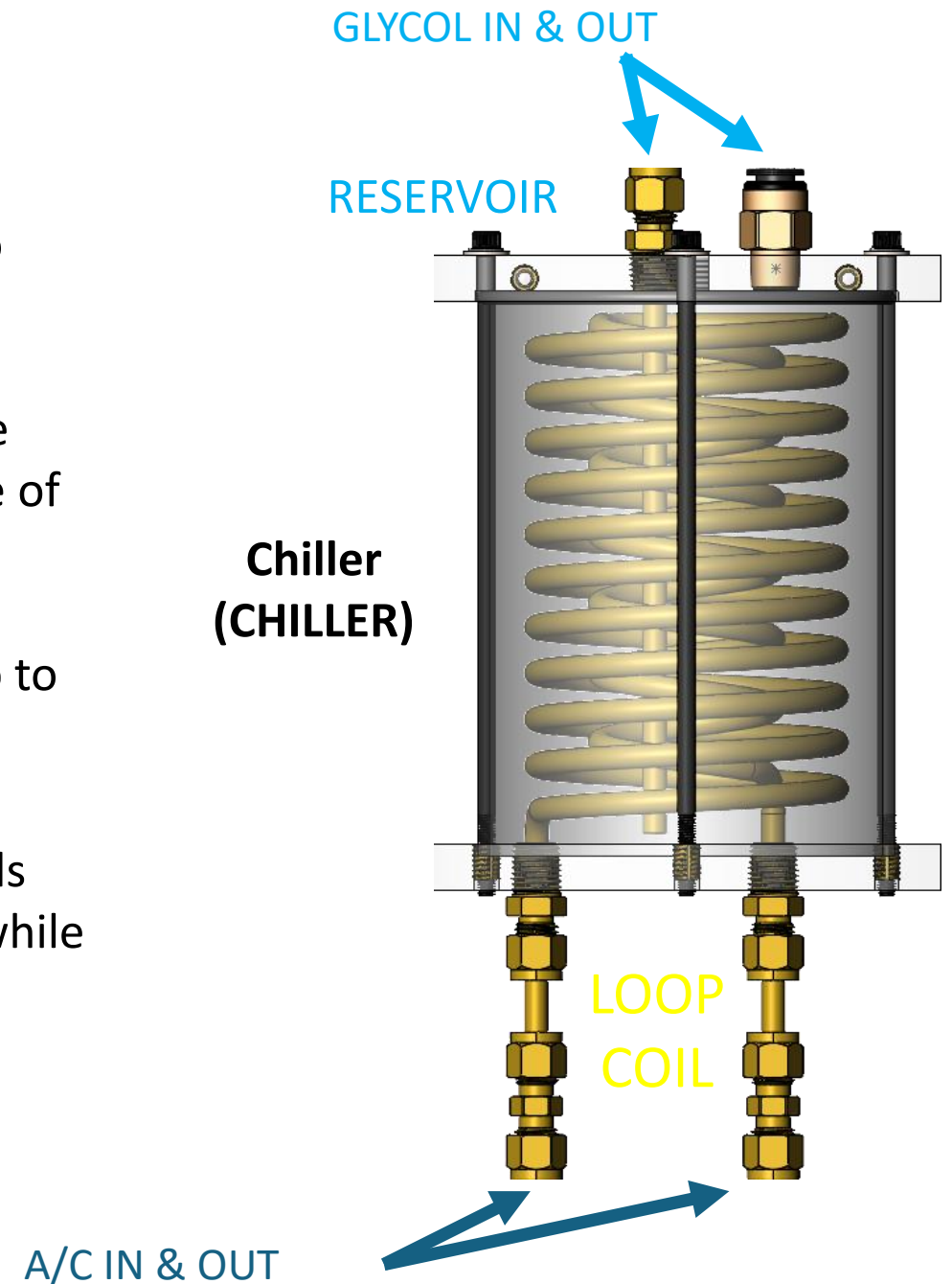
*Without latent heat the heat pump would perform far worse.*

# THE CHILLER EXPLAINED

**THE CHILLER:** a key component that transfers heat between two systems (coolant and refrigerant) inside an EV or hybrid:

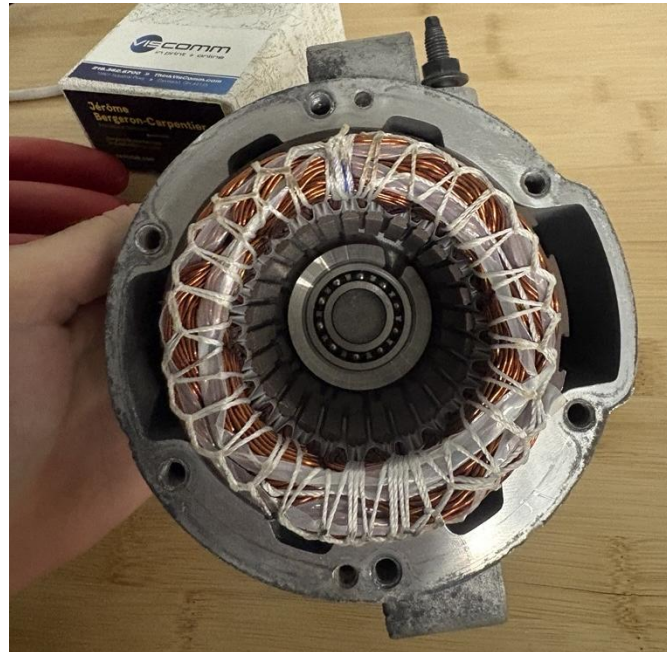
- the refrigerant loop (used to heat/cool the HV battery and the cabin) and the coolant loop (used to manage the temperature of the battery, the cabin and the electronic components)
- Think of the chiller as a bridge that moves heat from one loop to the other

It helps the vehicle stay warm in winter, cool in summer, and holds the HV battery at the right temperature (15-35 °C, ideal ~25 °C) while using energy efficiently



# Mode "lossy" — deliberate loss

Heat recovery by running the electric motors with deliberate losses (current injected without producing useful torque → heat in the windings)



# The vapour-compression refrigeration cycle

## Overview of the 4 stages of the cycle

### ► The cycle uses a refrigerant circulating in a closed loop that changes phase

- Liquid → gas: absorbs heat (evaporation = cooling)
- Gas → liquid: releases heat (condensation = heating)

### ► The 4 stages of the cycle:

- Stage 1: COMPRESSION → the gas is compressed, its temperature rises
- Stage 2: CONDENSATION → the hot gas gives up its heat and becomes liquid
- Stage 3: EXPANSION → the liquid expands, the temperature drops sharply
- Stage 4: EVAPORATION → the cold liquid absorbs ambient heat and turns back into gas

### ► The direction of the cycle determines the operating mode:

- HEATING mode: evaporator outside, condenser in the cabin
- AIR CONDITIONING mode: evaporator in the cabin, condenser outside

### ► Key principle: the compressor is the 'engine' of the cycle

- No compressor = no pressure difference = no heat transfer
- Compressor speed determines the thermal capacity of the system

# Stage 1 – Compression

Heart of the cycle: the electric compressor

## ► Inlet state: refrigerant gas, low pressure, low temperature

- R-1234yf example: suction pressure ~30 to 60 PSI, temperature ~30 °F to 60 °F

## ► The compressor draws in the gas and raises its pressure

- Scroll compressor: the most common type today
- Two nested scrolls compress the gas continuously and quietly
- Variable speed: 800 to 9,000 rpm depending on thermal demand

## ► Outlet state: refrigerant gas, high pressure, high temperature

- R-1234yf example: discharge pressure ~200 to 370 PSI, temperature ~130 °F to 180 °F

## ► The mechanical work supplied to the compressor turns into thermal energy

- $Q_{\text{compression}} = W_{\text{compressor}}$  (electrical energy consumed)
- It is the only external energy input in the system

## ► Technical precautions:

- Always keep enough oil in the compressor
- Avoid drawing in liquid refrigerant (liquid slugging → mechanical damage)
- The compressor sits on the HV circuit (400–800 V) → electrical isolation is essential

# Stage 2 – Condensation

The hot refrigerant releases its heat

## ► The hot high-pressure refrigerant gas enters the condenser

## ► In HEATING mode (heat pump):

- The condenser is the heat exchanger on the CABIN side
- The refrigerant gives up its heat to the air blown into the cabin
- The cabin air is warmed → thermal comfort for the occupants

## ► In AIR CONDITIONING mode:

- The condenser is the outside exchanger (front of the vehicle)
- The refrigerant gives up its heat to the outside ambient air
- An electric fan forces air through the exchanger

## ► Phase change: hot vapour → hot liquid (at high pressure)

- It all happens at a nearly constant temperature (saturation temperature)
- Energy released = latent heat of condensation of the refrigerant
- R-1234yf example at 290 PSI: condensation at about 160 °F (71 °C)

## ► Types of exchangers used in automotive applications:

- Air-to-refrigerant condenser: aluminum tubes and fins (outside, cabin)
- Refrigerant-to-coolant condenser (chiller): brazed plates (battery)

# Stage 3 – Expansion

Pressure and temperature drop

- ▶ **The high-pressure liquid refrigerant enters the electronic expansion valve (EXV)**
- ▶ **The EXV creates a controlled restriction (variable orifice)**
  - Pressure drops sharply from the high side to the low side
  - Example: from 290 PSI 160 °F (71 °C) → 43 PSI 45 °F (7 °C) for R-1234yf
- ▶ **Corresponding temperature drop (the refrigerant's P-T law)**
  - The temperature can fall by 20 °C, 30 °C or more
  - That temperature drop is what makes it possible to absorb heat outside
- ▶ **Start of vaporization: the liquid begins to turn into vapour**
- ▶ **The EXV modulates the flow to hold the optimal superheat at the evaporator outlet**
  - Superheat too low: risk of liquid returning to the compressor
  - Superheat too high: loss of cooling capacity
- ▶ **Types of expansion devices in automotive applications:**
  - EXV (electronically controlled expansion valve): the most common on recent EVs
  - TXV (thermostatic): used on some older vehicles
  - Orifice tube: simple systems, fixed flow (uncommon on modern vehicles)

# Stage 4 – Evaporation

Absorbing ambient or waste heat

- ▶ **The cold low-pressure refrigerant enters the evaporator**
- ▶ **In HEATING mode (heat pump):**
  - The evaporator is the outside exchanger (front of the vehicle)
  - The refrigerant absorbs heat from the ambient air (even at -15 °C or -20 °C)
  - The 'chilled' outside air leaves the exchanger (frost may form)
- ▶ **In AIR CONDITIONING mode:**
  - The evaporator is in the cabin
  - The refrigerant absorbs BOTH heat and moisture from the cabin air
  - The cabin air is cooled and dehumidified
- ▶ **Phase change: cold liquid → vapour (at low pressure)**
  - Heat absorbed = latent heat of evaporation of the refrigerant
  - Evaporation temperature is tied to the low pressure (P-T chart)
  - R-1234yf example at 32 PSI: evaporation at about 32 °F (0 °C)
- ▶ **The gas produced returns to the compressor → the cycle starts again**
- ▶ **Alternative heat sources in heat pump mode:**
  - Waste heat from the motor / inverter (through the coolant circuit)
  - Waste heat from the battery

# Automotive refrigerants

## Evolution and regulation in Canada

### ► R-12 (Freon / dichlorodifluoromethane) — high GWP, high ODP

- Used until 1994 on North American vehicles
- Banned under the Montreal Protocol for destroying ozone
- Still found on older vehicles; requires conversion or replacement

### ► R-134a (tetrafluoroethane / HFC-134a) — GWP = 1,430

- North American standard from 1994 to 2024
- Banned on new Canadian/U.S. models since 2024.
- Still very common in North America (millions of vehicles in service)

### ► R-1234yf (HFO-1234yf) — GWP = 4

- Mandatory on new NA vehicles since 2024 (SAE J2842 standard)
- Slightly flammable (class A2L); requires SAE J2845 certified equipment
- Compatible oil: PAG 46yf, 100yf, etc. (do not interchange with R-134a)

### ► R-744 (CO<sub>2</sub>) — GWP = 1

- Used sparingly by BMW, Korean makes, Audi; very high pressure (up to 2000 PSI)
- Very rare on today's North American market; dedicated equipment required

# Key component: the electronic expansion valve (EXV)

Precise control of refrigerant flow

► **Role: expand the high-pressure liquid refrigerant → low pressure and low temperature**  
and modulate the flow to optimize superheat at the evaporator outlet

► **How it works:**

- Needle driven by a stepper motor or a solenoid
- Position from 0 to 100% open (readable on an OEM scan tool)
- Commanded by the vehicle's thermal module based on measured superheat
- Response time: < 1 second; accuracy:  $\pm 0.5$  °C of superheat

► **Advantages of the EXV over a thermostatic TXV:**

- Independent control of each branch of the system
- Can be calibrated with a scan tool (zero initialization)
- Live data available (% open, calculated superheat)

► **Number of EXVs depending on system complexity:**

- Simple systems (hybrids): 1 to 2 EXVs
- Integrated systems (modern EVs): 3 to 6 EXVs (cabin, battery, outside, recovery)

► **Diagnostics:**

- Check the % open live (it must vary with demand)
- Check that pressure changes as you sweep the % open

# Key component: the sensors

## Monitoring and controlling the system

### ▶ High-side pressure sensor (HP):

- Location: discharge line (between compressor and condenser)
- Signal: 0.5 to 4.5 V (3-wire ratiometric sensor)
- Range: 0 to 500 PSI (R-1234yf / R-134a)

### ▶ Low-side pressure sensor (LP):

- Location: suction line (between evaporator and compressor)
- Signal: same as the HP sensor
- Range: 0 to 150 PSI (R-1234yf / R-134a)

### ▶ Discharge temperature sensor:

- Location: compressor outlet
- Protects the compressor against overheating (typical limit: 250-275 °F) (120-135 °C)

### ▶ Ambient temperature sensor:

- Location: front of the vehicle (ahead of the outside exchanger)
- Drives the mode-switching logic of the heat pump

### ▶ Evaporator / defrost temperature sensor:

- Triggers the defrost cycle if  $T < (23-25\text{ °F})$  ( $-3\text{ °C}$  to  $-5\text{ °C}$ ) (depending on the manufacturer)

### ▶ Battery temperature sensor:

- Several probes inside the high-voltage battery; manages battery heating/cooling mode

### ▶ All sensors are monitored live → available through a scan tool

# Thermal management: ICE vehicle vs electric vehicle

Why the heat pump is essential on an EV

## ► Internal combustion vehicle (ICE):

- The combustion engine rejects ~65–70% of its energy as heat
- That waste heat is used free of charge to warm the cabin
- Engine coolant circuit → heater core → cabin blower
- Energy cost of cabin heating: near zero (the energy was lost anyway)

## ► Electric vehicle (EV):

- The electric motor is very efficient (~95%) → very little waste heat
- At -15 °C, an electric motor may produce only 100–300 W of heat
- To heat the cabin, energy has to be drawn from the high-voltage battery

## ► Impact of PTC heaters (Positive Temperature Coefficient):

- Electric resistance: 3 to 7 kW drawn continuously in cold weather
- On a 75 kWh battery: 30–50% range loss at -15 °C
- Real example: 400 km of range → 200–280 km in winter with PTC alone

## ► The heat pump solves that problem:

- COP 2.5 to 3.5 → delivers 2.5 to 3.5 kWh of heat for 1 kWh consumed
- Range loss cut to 10–25% (depending on temperature and vehicle)

# Coefficient of Performance (COP)

Measuring heat pump efficiency

## ► Definition:

- $\text{COP}_{\text{heating}} = Q_{\text{heating}} / W_{\text{electrical}} = \text{heat delivered} / \text{energy consumed}$
- $\text{COP}_{\text{cooling}} = Q_{\text{cooling}} / W_{\text{electrical}}$

## ► Sample calculation:

- Compressor: 2 kW ( $W_{\text{elec}}$ )
- Heat absorbed outside: 4 kW ( $Q_{\text{ext}}$ )
- Heat delivered to the cabin: 6 kW ( $Q_{\text{cabin}} = 2 + 4$ )
- $\text{COP}_{\text{heating}} = 6 / 2 = 3.0$

## ► Comparison of heating systems:

- PTC heater (resistance):  $\text{COP} = 1.0$  (1 kWh consumed = 1 kWh of heat)
- Heat pump at 0 °C:  $\text{COP} \approx 2.5\text{--}3.5$
- Heat pump at -15 °C:  $\text{COP} \approx 1.5\text{--}2.2$  (degraded by the low temperature)
- Heat pump at -25 °C:  $\text{COP} \approx 1.0\text{--}1.5$  (close to the limit)

## ► Impact on EV range:

- Without a heat pump (PTC alone) at -15 °C: 40–50% range loss
- With a heat pump at -15 °C: 15–25% range loss (a significant saving)
- Example: Tesla Model Y Long Range — 150 km of extra range in winter

## ► Factors affecting COP:

- Outside temperature (the biggest factor)
- Compressor speed · Fouled heat exchangers
- Refrigerant charge level · Oil quality

# Measuring thermal output in the shop

Three simple tools: a thermometer, an anemometer and a ruler

Turn a subjective complaint into an objective number, before taking anything apart.

$$P = Qv \times 0.34 \times \Delta T$$

**$Qv$**

Air flow,  
in m<sup>3</sup>/h.

**0.34**

Volumetric heat capacity  
of air, in Wh/m<sup>3</sup>.°C.

**$\Delta T$**

Temperature at the vents  
minus outside temperature.

*Actual COP = thermal power ÷ electrical power consumed, read on the scan tool or the infotainment screen.*

# The method in six steps

From the tape measure to the vehicle's actual COP

## 1 Vent area

Measure each outlet vent  
(width × height, in metres)  
and add them up.  
→ m<sup>2</sup>

## 2 Settings

Heat at maximum, blower at  
maximum, vent mode, all  
vents open.

## 3 Air speed

With the anemometer, measure  
the speed at the vents and  
average it.  
→ m/s

## 4 Air flow

$Q_v = \text{total area} \times \text{average}$   
 $\text{speed} \times 3,600.$   
→ m<sup>3</sup>/h

## 5 Thermal power

$P = Q_v \times 0.34 \times \Delta T.$   
→ watts

## 6 COP

Thermal power ÷ electrical  
power consumed by the  
compressor.

# Example: 2021 Toyota RAV4 Prime (heating mode)

| Step           | Measurement                                                          | Result                     |
|----------------|----------------------------------------------------------------------|----------------------------|
| Vent area      | 4 identical vents of $9 \times 4.5$ cm ( $0.00405 \text{ m}^2$ each) | $0.0162 \text{ m}^2$       |
| Air speed      | Average of the 4 vents with the anemometer                           | $4.5 \text{ m/s}$          |
| Air flow $Q_v$ | $0.0162 \times 4.5 \times 3,600$                                     | $262 \text{ m}^3/\text{h}$ |
| $\Delta T$     | $57^\circ\text{C}$ at the vents – $17^\circ\text{C}$ outside         | $40^\circ\text{C}$         |
| Thermal power  | $262 \times 0.34 \times 40$                                          | $\approx 3,570 \text{ W}$  |
| COP            | $3,570 \text{ W} \div 1,550 \text{ W electrical}$                    | 2.3                        |

## ► Reading the result:

- COP between 2 and 3: healthy heat pump (depending on outside temperature)
- COP close to 1: the heat pump is not contributing, only the PTC is working → check the EXV, the mode valve, the refrigerant charge
- Low air flow ( $Q_v$ ): plugged cabin filter, weak blower, frosted or fouled evaporator
- Low  $\Delta T$  with good flow: low refrigerant charge, stuck EXV, inefficient inside exchanger

 **TIP** The "summer = winter" rule: the power measured in cooling plus dehumidification in the summer matches the heating capacity available in the winter. Example:  $2,000 \text{ W}$  of cooling +  $1,000 \text{ W}$  of dehumidification  $\approx 3,000 \text{ W}$  of heating. So a heating system can be validated... in the middle of July.

# Thermal challenges specific to EVs

Canadian winters — a challenge of their own

## ► Extreme temperatures

- extreme temperatures of -20 °C to -40 °C in winter
- These temperatures exceed the optimal range of most heat pumps (effective down to -20 °C)

## ► The problem of the whole vehicle cooling down overnight

- Cold battery: internal resistance rises → available power drops
- Oil viscosity rises → more compressor wear at start-up
- Solution: remote preconditioning (plugged in or on battery)

## ► Three thermal challenges at once:

- 1. Heat the cabin for occupant comfort
- 2. Heat the battery to restore performance
- 3. Defrost the outside exchangers (frost = loss of efficiency)

## ► Heat pump systems handle these 3 challenges simultaneously

- Multi-circuit architecture with multiple heat sources
- Dynamic priority based on demand (critical battery > cabin comfort)

## ► Specific case: DC fast charging in cold weather

- A fast charger requires a battery above 10–15 °C to accept high-power charging
- The heat pump preheats the battery to shorten charging time

# The defrost cycle of the outside exchanger

Keeping the heat pump efficient in frosting conditions

## ► Why does frost form on the outside exchanger?

- In heat pump heating mode, the outside exchanger is the evaporator
- Its surface temperature drops below 0 °C (typically -5 °C to -15 °C)
- Humid air in contact with that surface → frost forms
- Critical frosting range: ambient temperature between -5 °C and +5 °C with relative humidity above 70%

## ► Impact of frost:

- Restricts air flow → reduces the ability to absorb heat
- In severe cases: complete blockage → low-side pressure collapses → compressor shuts down

## ► Detecting frost:

- Surface temperature sensor on the outside exchanger (< -3 °C → defrost triggered)
- Or: monitoring low-side pressure (an unexplained rapid drop → indirect detection)

## ► Defrost method 1: cycle reversal

- The cycle is temporarily reversed (A/C mode) → the outside exchanger becomes a hot condenser
- The frost melts in 2–5 minutes; the cabin is kept warm by the cabin chiller

## ► Defrost method 2: integrated electric resistance

- Less common; resistive wire on the exchanger; consumes more energy

**⚠ WARNING** A defrost cycle is not a fault: during one, cabin heating drops temporarily and vapour may escape at the front of the vehicle. That is normal operation: do not diagnose a refrigerant leak on that observation alone.

# The electric compressor — in depth

## Architecture, operation and integration in North American EVs

### ► Construction of a typical e-compressor (e.g. Hanon Systems, Mahle, Denso, Valeo)

- Sealed aluminum housings; high-pressure axial shaft seal
- Integrated PMSM motor (Permanent Magnet Synchronous Motor)
- Scroll mechanism: fixed scroll + orbiting scroll → smooth, continuous compression
- Integrated inverter: converts the HV battery supply (400–800 V DC) into three-phase AC

### ► Control and communication:

- Receives speed commands over the CAN bus from the thermal management module
- Data: actual speed, motor temperature, current draw, fault codes
- Automatic speed limiting if: discharge temp > limit · high-side pressure > threshold · loss of isolation

### ► Characteristics by North American manufacturer:

- Tesla: 400 V compressor (Model 3/Y) or 800 V (Plaid) — supplier: Dana/Sanden
- GM Ultium: 400 V or 800 V compressor (hummer, silverado, etc.) — supplier: Mahle or Hanon
- Ford F-150 Lightning / Mach-E: 400 V compressor — supplier: Denso or Hanon
- Hyundai IONIQ 5/6: 800 V compressor (E-GMP platform) — supplier: Hanon

### ► Compressor diagnostic tests:

- Insulation resistance (500 VDC megaohmmeter): must be > 10 MΩ between phases and ground
- HV supply voltage: check for no voltage drop
- Current draw: measure with an HV clamp meter (non-invasive)

# The electric motor — thermal limits

Why the motor tolerates heat, unlike the battery

## ► The traction motor tolerates heat (unlike the battery)

- It has no narrow optimal window the way the battery does
- Goal: stay below its maximum thermal limits

## ► Thermal limits of the electric motor (PMSM)

- Windings (insulation): withstand up to ~180 °C (class 180-200 °C)
- Permanent magnets (neodymium): irreversible demagnetization from ~150-160 °C (permanent)
- Rule of thumb: every +10 °C above rating cuts insulation life in half

## ► Protection and consequences for the thermal architecture

- Thermal derating: the controller reduces current and torque before the limits are reached
- Motor = a loop that can run hot; battery = a loop that must stay cool (15-35 °C)
- That is why the battery has its own loop with a chiller, separate from the motor

 **KEY POINT** Two opposite needs: the motor can run hot, the battery has to stay cool (15-35 °C). That is exactly why the battery has its own loop with a chiller, separate from the motor's.

# Thermal management module

The brain of the heat pump system

► **Role: coordinate every thermal actuator in the vehicle in real time**

► **Actuators controlled by the thermal management module:**

- E-compressor (speed, on/off)
- EXV (% open)
- 3-way valves in the coolant circuit
- Electric coolant pumps (cabin, battery, motor)
- Fan(s) on the outside exchanger
- Auxiliary PTC heater (backup)
- Any other component the manufacturer chooses to add

► **Inputs (monitored sensors):**

- High-side and low-side pressures · Discharge, suction and ambient temperatures
- Battery temperatures (BMS) · Motor / inverter temperatures
- Requested cabin temperature · Vehicle speed

► **Communication:**

- CAN or CAN FD bus
- Receives requests from: BMS, HVAC module, ECM, BCM, etc.

► **Names by North American manufacturer:**

- Tesla: 'Thermal Control Unit (TCU)' · GM: 'BCM and BECM'
- Ford: 'HVAC, ACCM, HPVCM' · Hyundai/Kia: 'ACCM'
- Toyota: 'Air Conditioning Amplifier Assembly (ACAA)'

# Direct battery cooling

## Thermal protection and fast-charging optimization

### ► Why cool the battery directly?

- DC fast charging (DCFC) generates significant heat (internal losses)
- High charging power: 150–350 kW → cells can reach 45–55 °C
- Above 40 °C: accelerated cell degradation; the BMS cuts power

### ► Refrigerant path (battery cooling mode):

- Compressor → outside exchanger (condenser) → battery EXV → battery chiller (evaporator) → compressor

### ► Advantage of direct cooling (refrigerant) over indirect cooling (coolant only):

- Two to three times the cooling capacity
- Target temperature reached faster
- Allows longer charging sessions without power reduction

### ► North American manufacturers using direct cooling:

- Tesla (every model since 2012) · GM Ultium · Ford F-150 Lightning · Hyundai E-GMP

### ► Typical trigger thresholds (example):

- Battery temp > 30–35 °C: active cooling triggered by the BMS
- Battery temp > 40 °C: charging power automatically reduced

### ► Link to diagnostics:

- If the battery chiller fails → DCFC charging is power-limited
- Possible DTC in the BMS: 'Battery Cooling Fault'

# Preconditioning the battery and the cabin

Best use before departure or before a DC fast charge

▶ **Definition: heating or cooling the battery and the cabin before use, ideally while plugged in**

▶ **Types of preconditioning:**

- On a home charger (EVSE): free energy from the grid; does not affect range
- On battery (not plugged in): uses up range; to be used sparingly

▶ **How it is triggered:**

- Mobile app (Tesla, GM myChevrolet, FordPass, etc.)
- Scheduled timer (automatic preconditioning)
- GPS navigation to a DCFC station (automatic on recent Tesla, Ford, GM Ultium)

▶ **Measured benefits (Canadian winter conditions):**

- Battery preconditioned to 20 °C vs a cold battery at -20 °C:
- Range recovered: +10–20% (less cabin heating needed on the road)
- DCFC charging power: +50–100% (accepts full power sooner)
- Battery life: better (avoids charge cycles at too low a temperature)

💡 **TIP** The Quebec winter argument: schedule preconditioning the night before, with the vehicle plugged in. Battery and cabin at temperature at departure, maximum range and more efficient fast charging, even after a night at -30 °C.

# Integration with DC fast charging (DCFC)

Thermal management during and after a high-power charging session

## ► The DCFC network in North America:

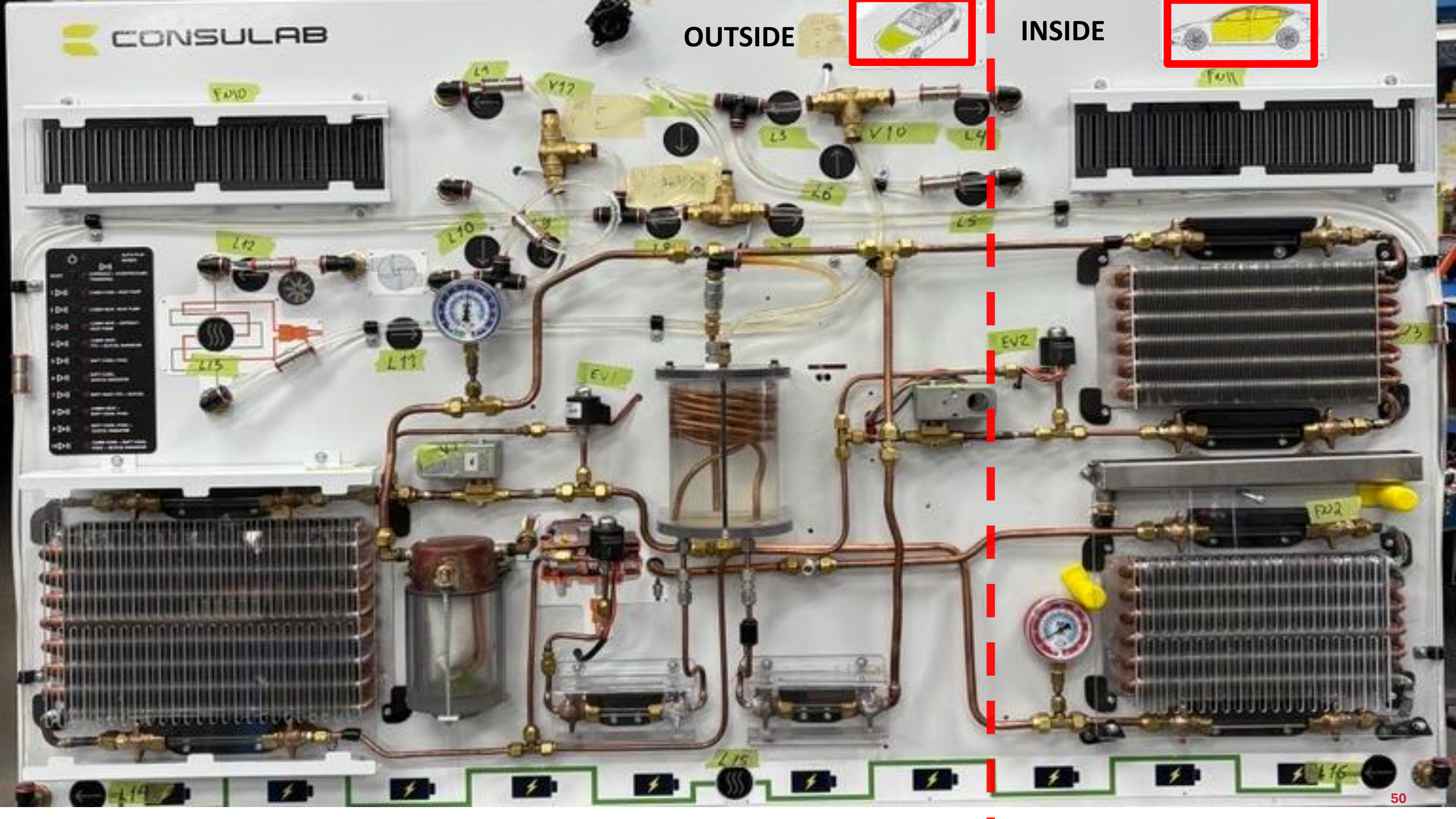
- Tesla Supercharger (V3/V4: 250–350 kW) · Circuit Électrique · ChargePoint · EVgo
- Ford Charge Network · GM Ultium Charge 360 · Toyota / Hyundai / Kia networks

## ► How a DCFC session unfolds thermally:

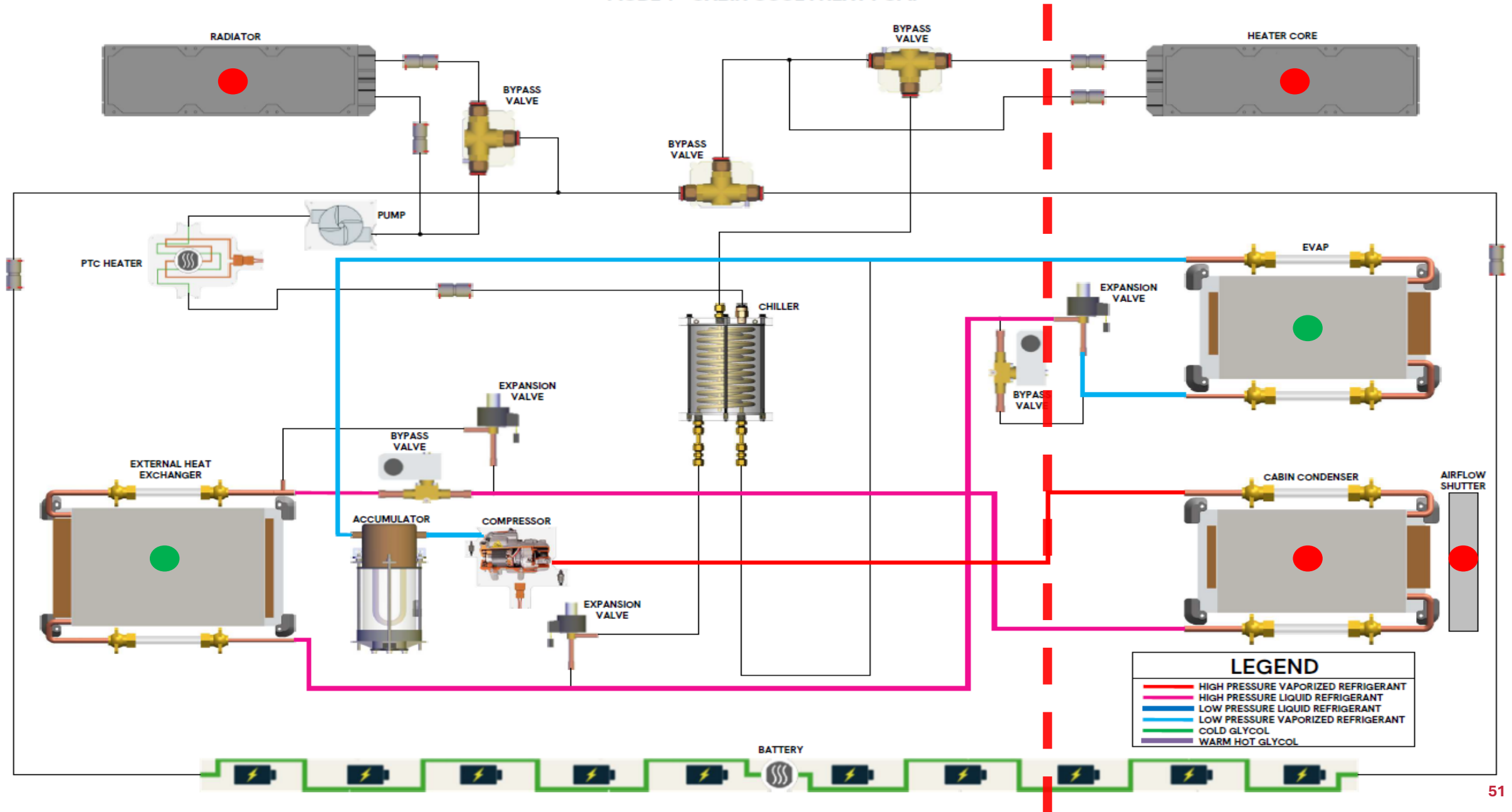
- On arrival: if the battery is cold ( $< 15\text{--}20\text{ }^{\circ}\text{C}$ ), charging is limited by the BMS (e.g. 50 kW instead of 250 kW)
- Active preconditioning: the heat pump warms the battery before arrival
- During charging: heat pump in battery cooling mode to hold the temperature below  $35\text{--}40\text{ }^{\circ}\text{C}$
- End of charge (above 80%): power reduced by the BMS  $\rightarrow$  less heat generated

## ► Symptoms of a battery cooling system failure during DCFC:

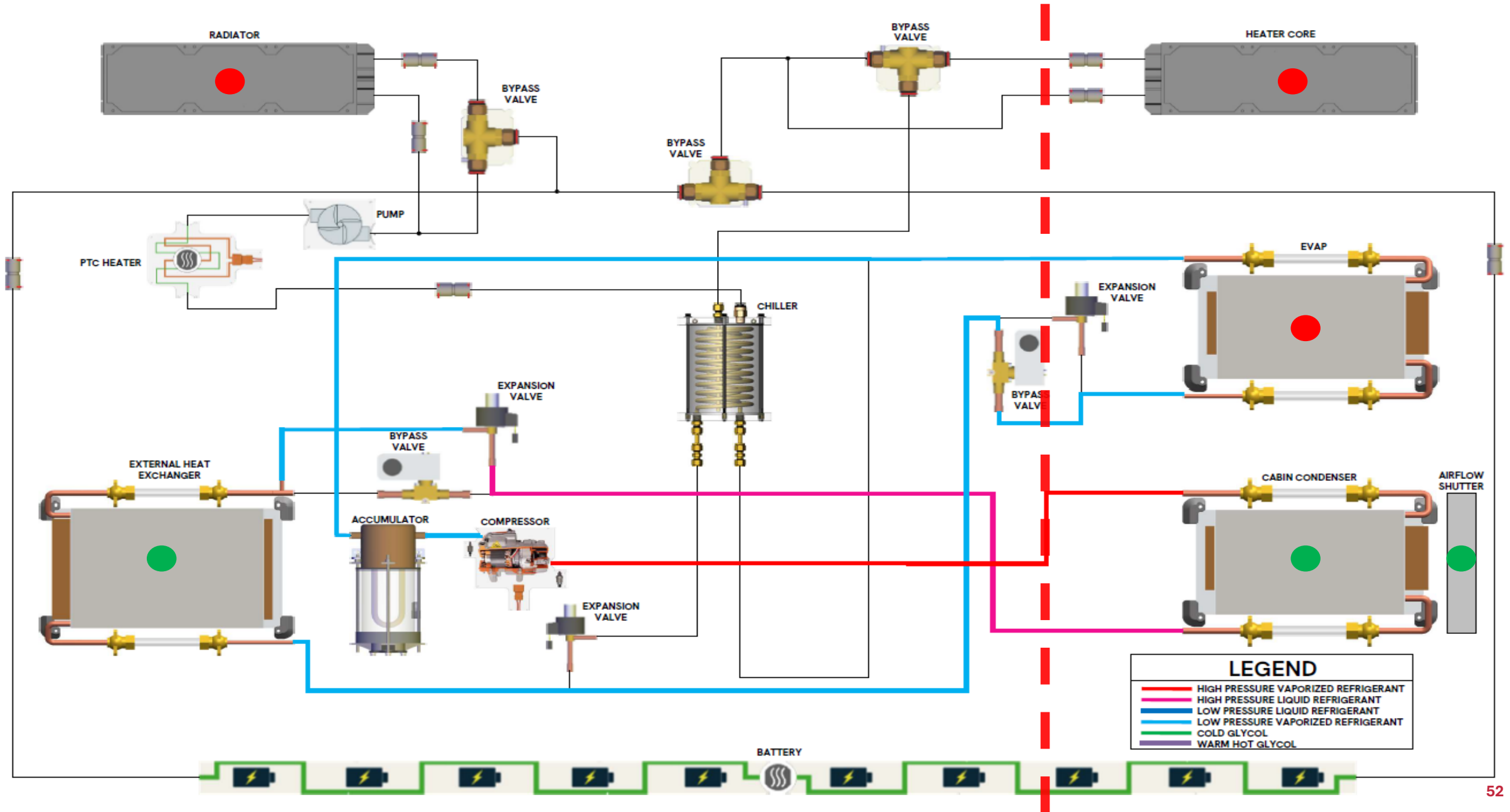
- Charging power limited or slow even at a low state of charge
- Warning message: 'Battery Cooling System Fault' or 'Charging Power Reduced'
- Battery temperature climbing quickly on OBD data with no active cooling



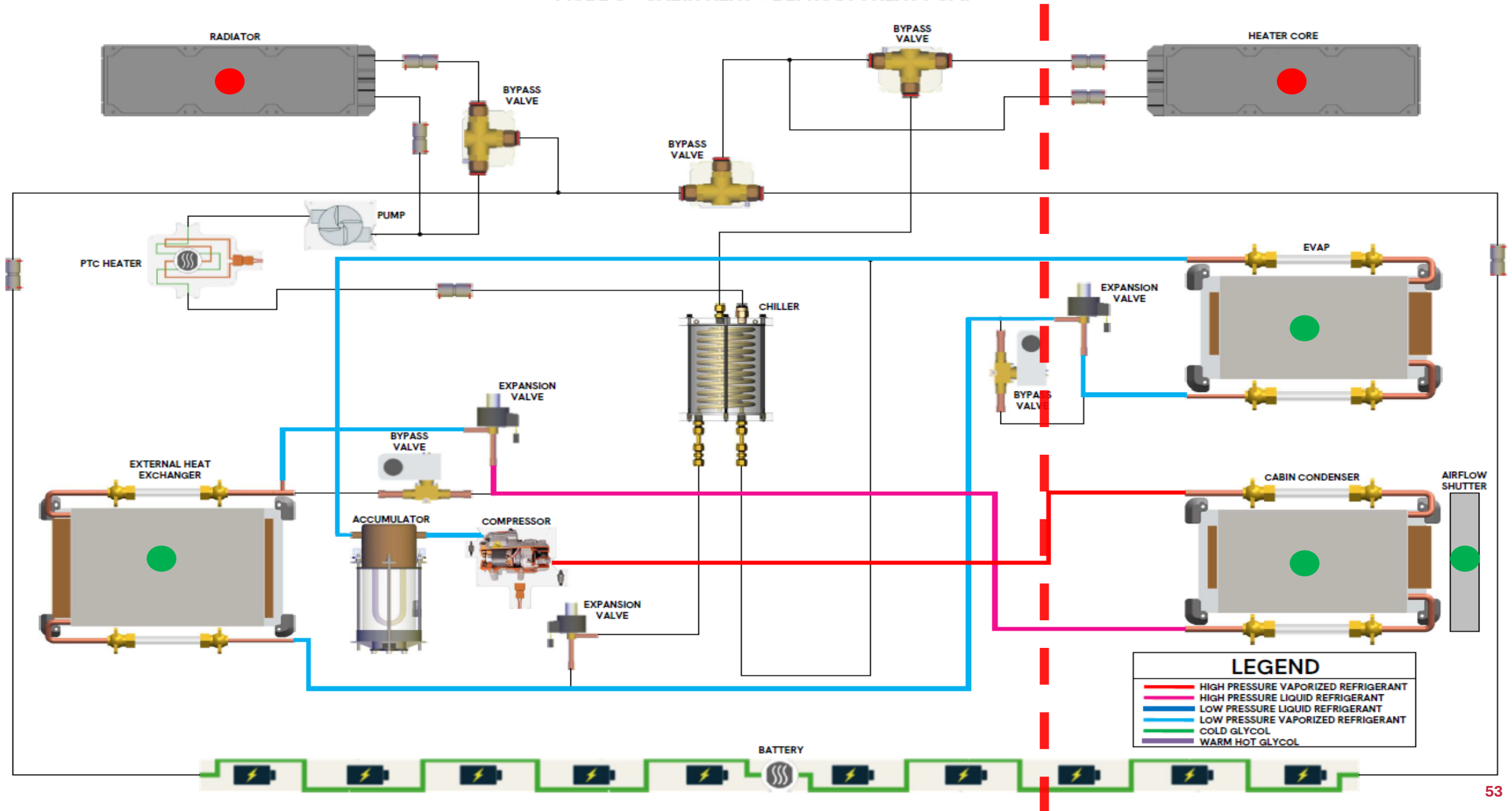
- MODE 1 = CABIN COOL : HEAT PUMP



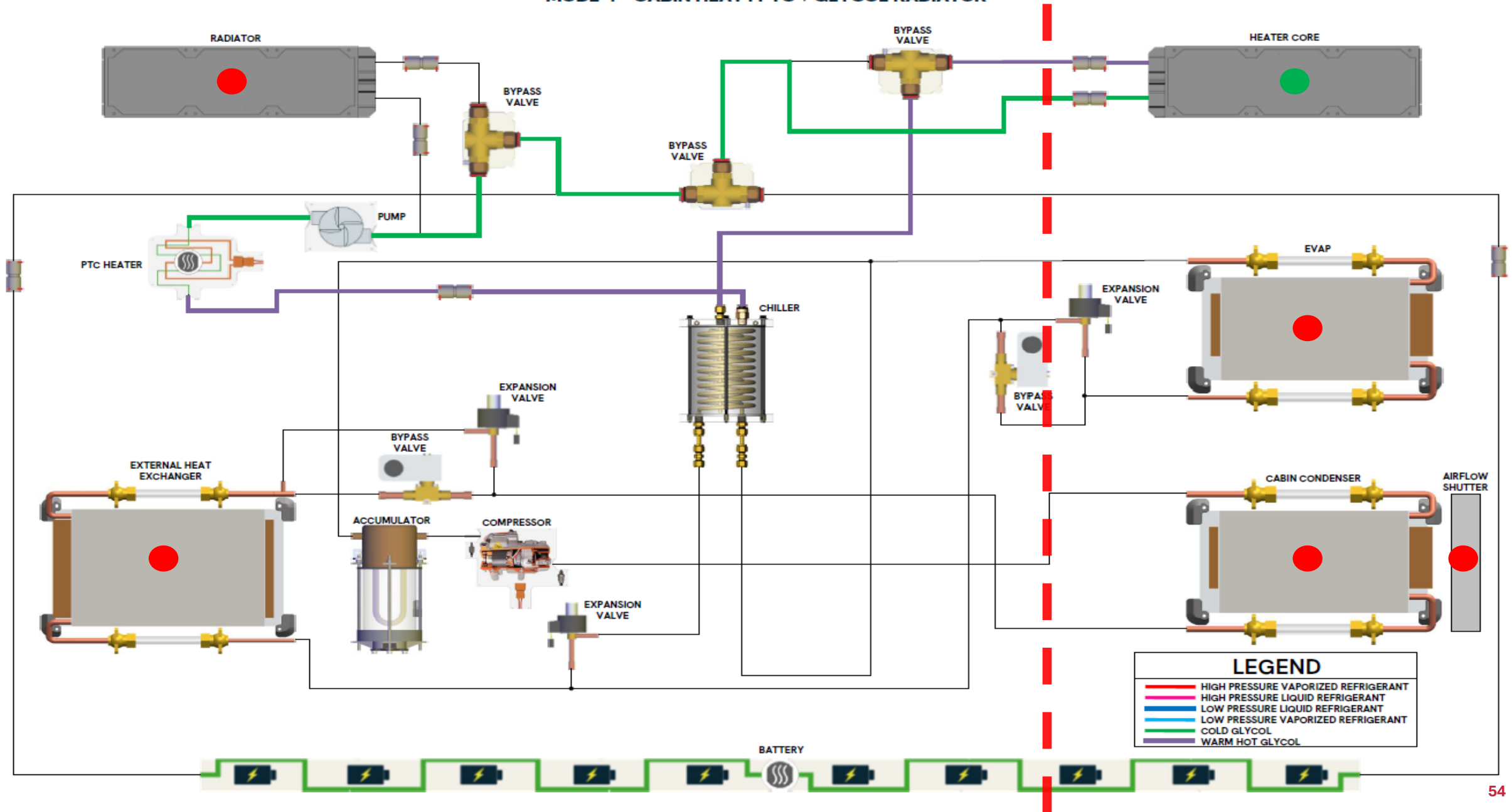
# - MODE 2 = CABIN HEAT : HEAT PUMP



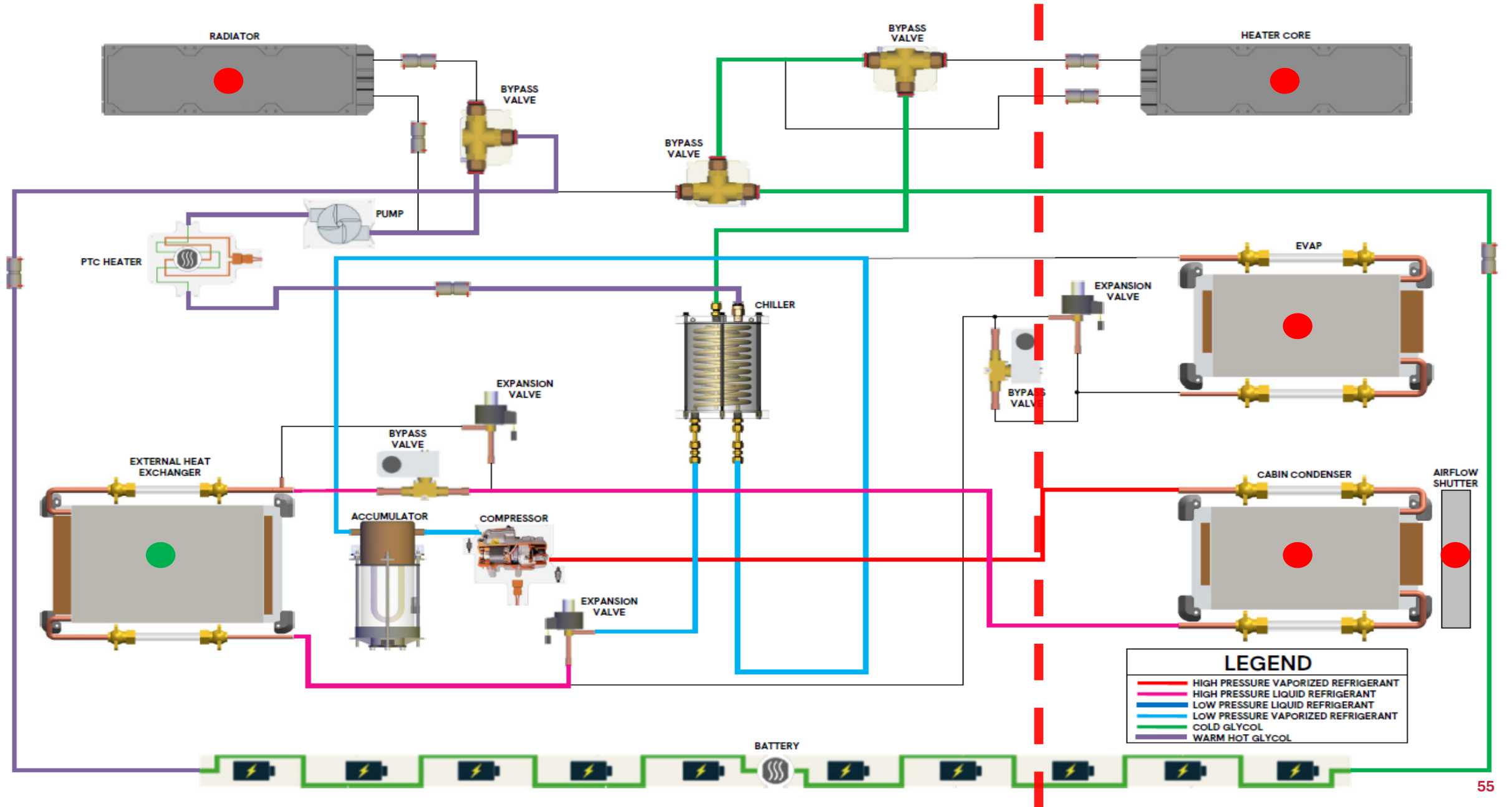
- MODE 3 = CABIN HEAT + DEFROST : HEAT PUMP



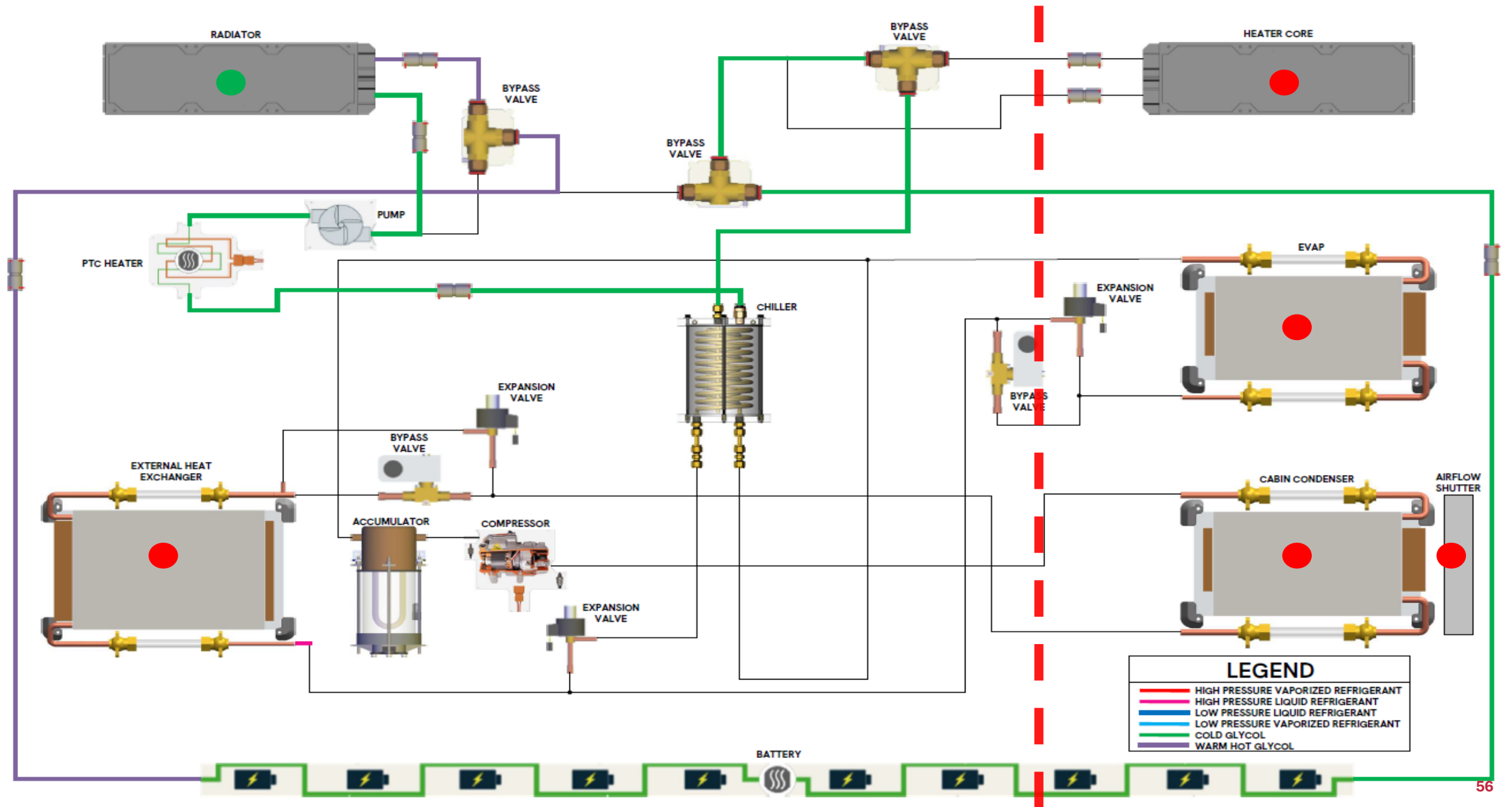
- MODE 4 = CABIN HEAT : PTC + GLYCOL RADIATOR



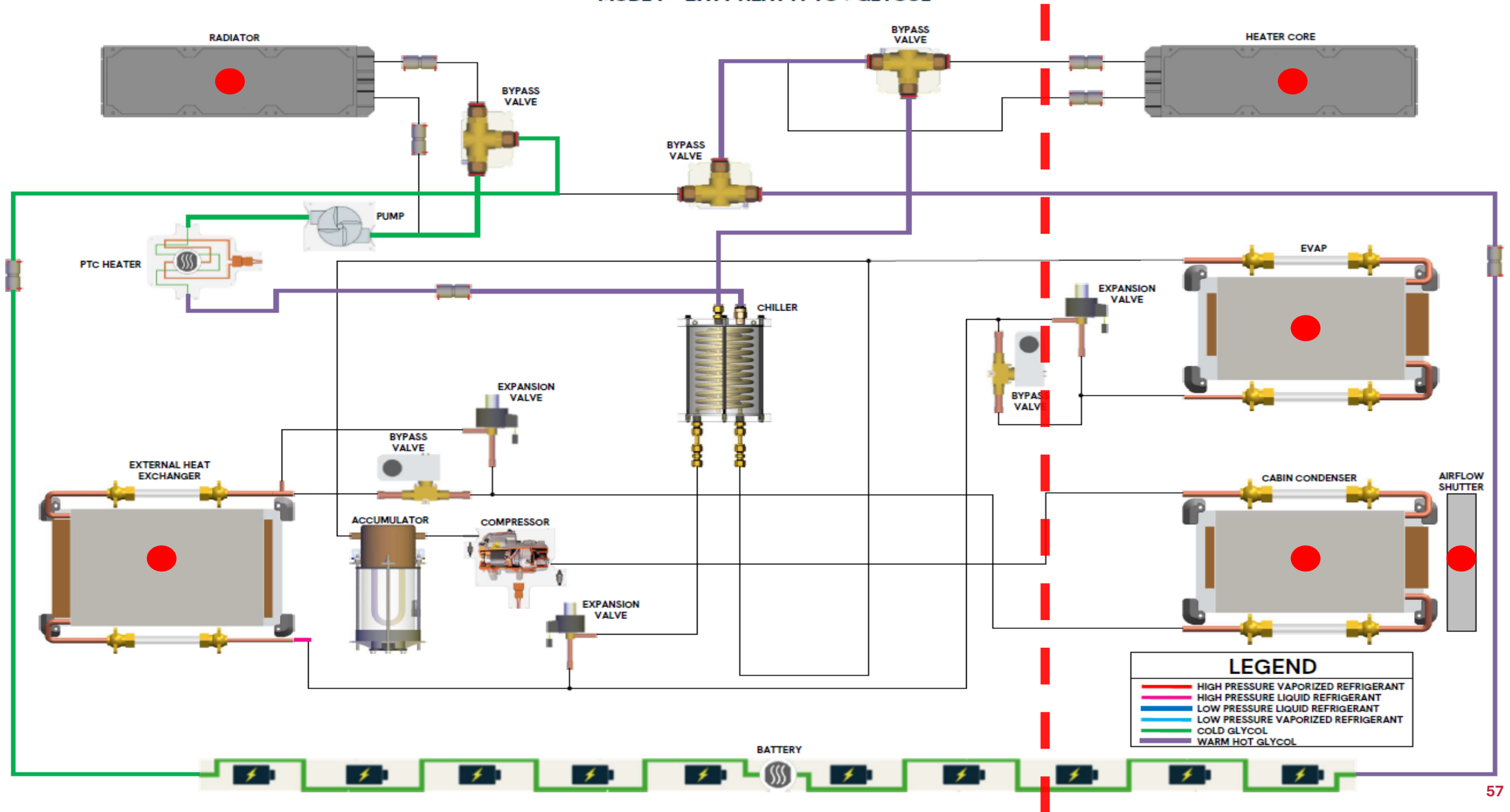
# - MODE 5 : BATT COOL : HVAC



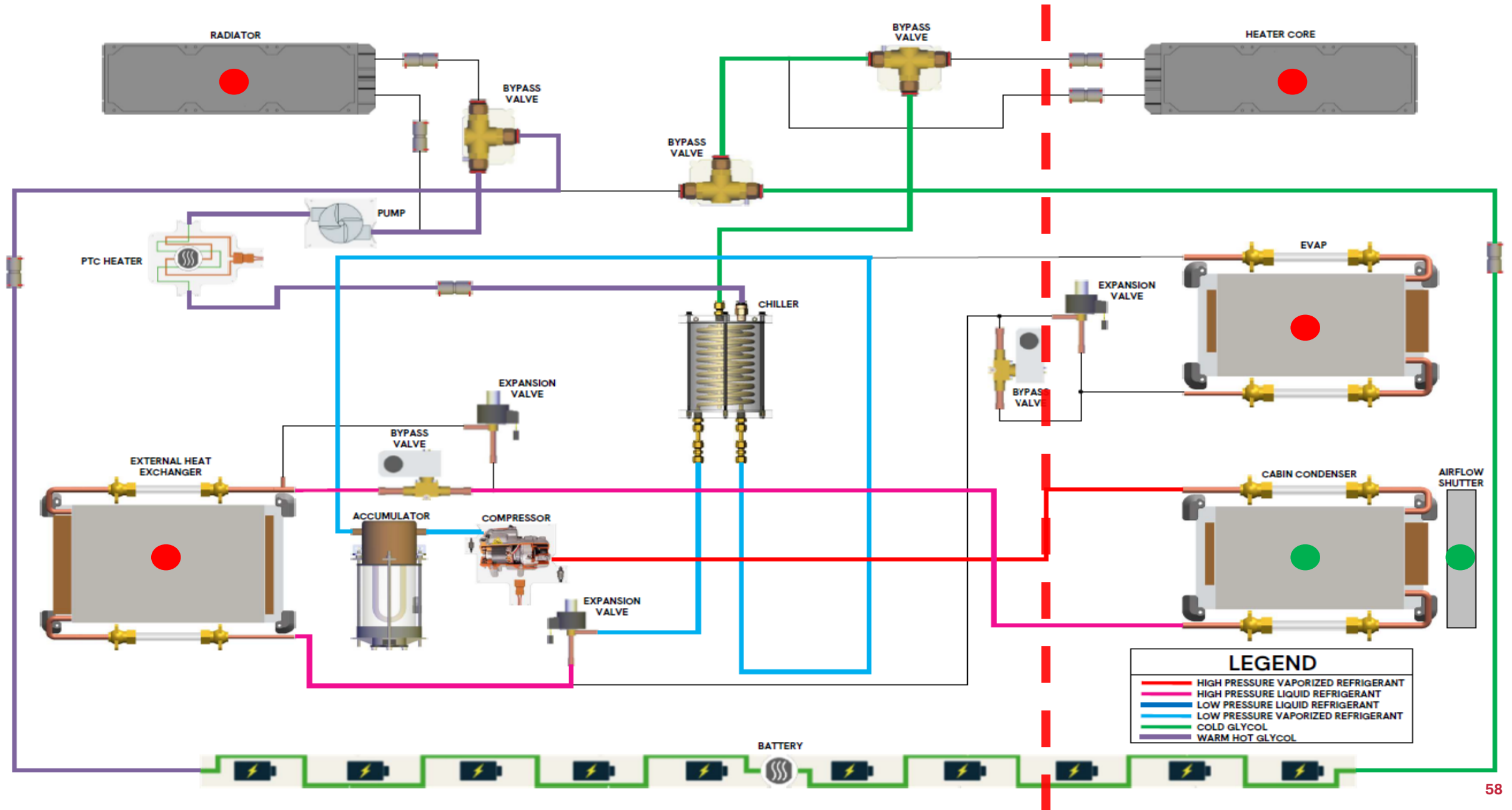
# - MODE 6 = BATT COOL : GLYCOL RADIATOR



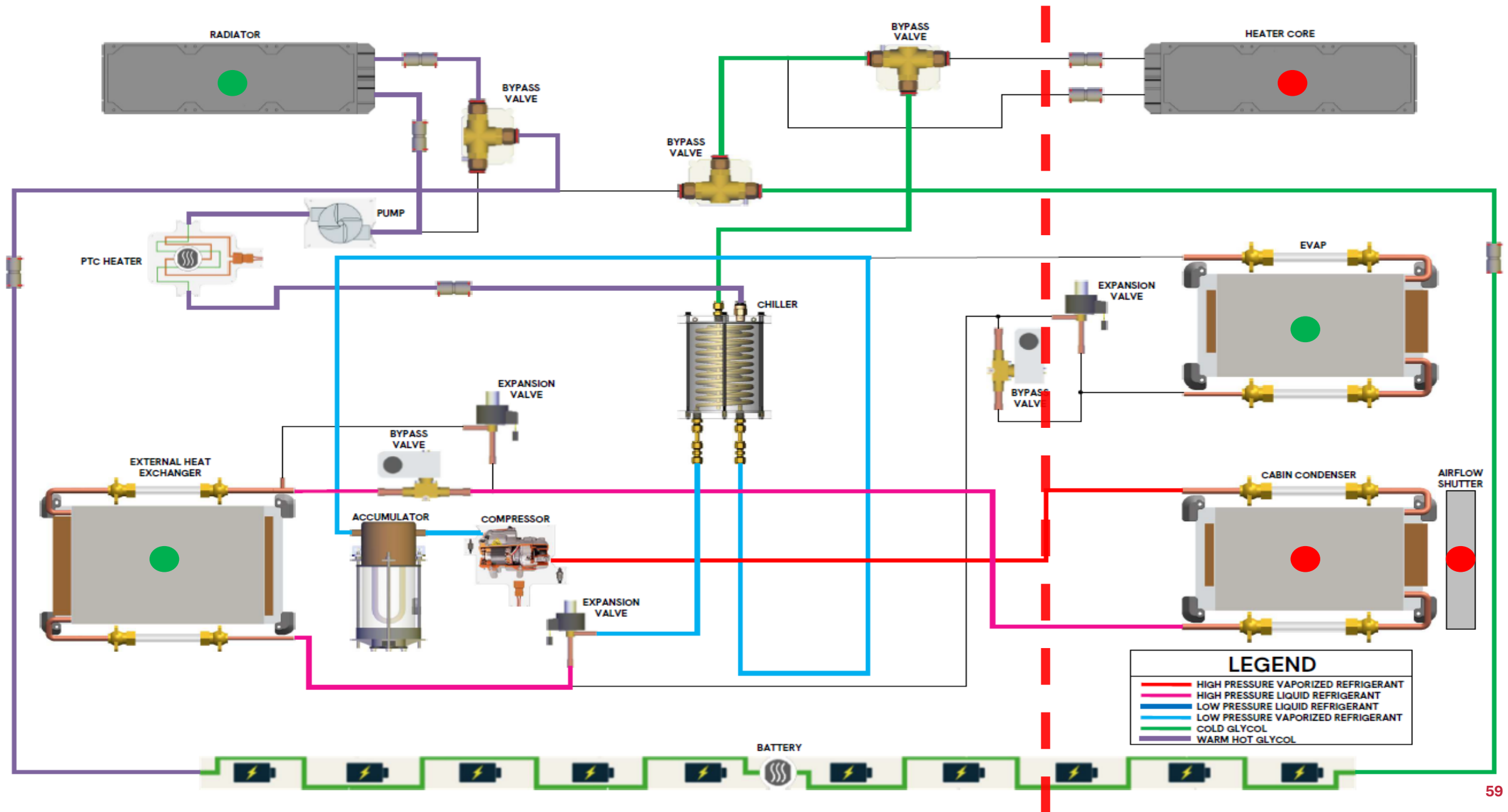
- MODE 7 = BATT HEAT : PTC + GLYCOL



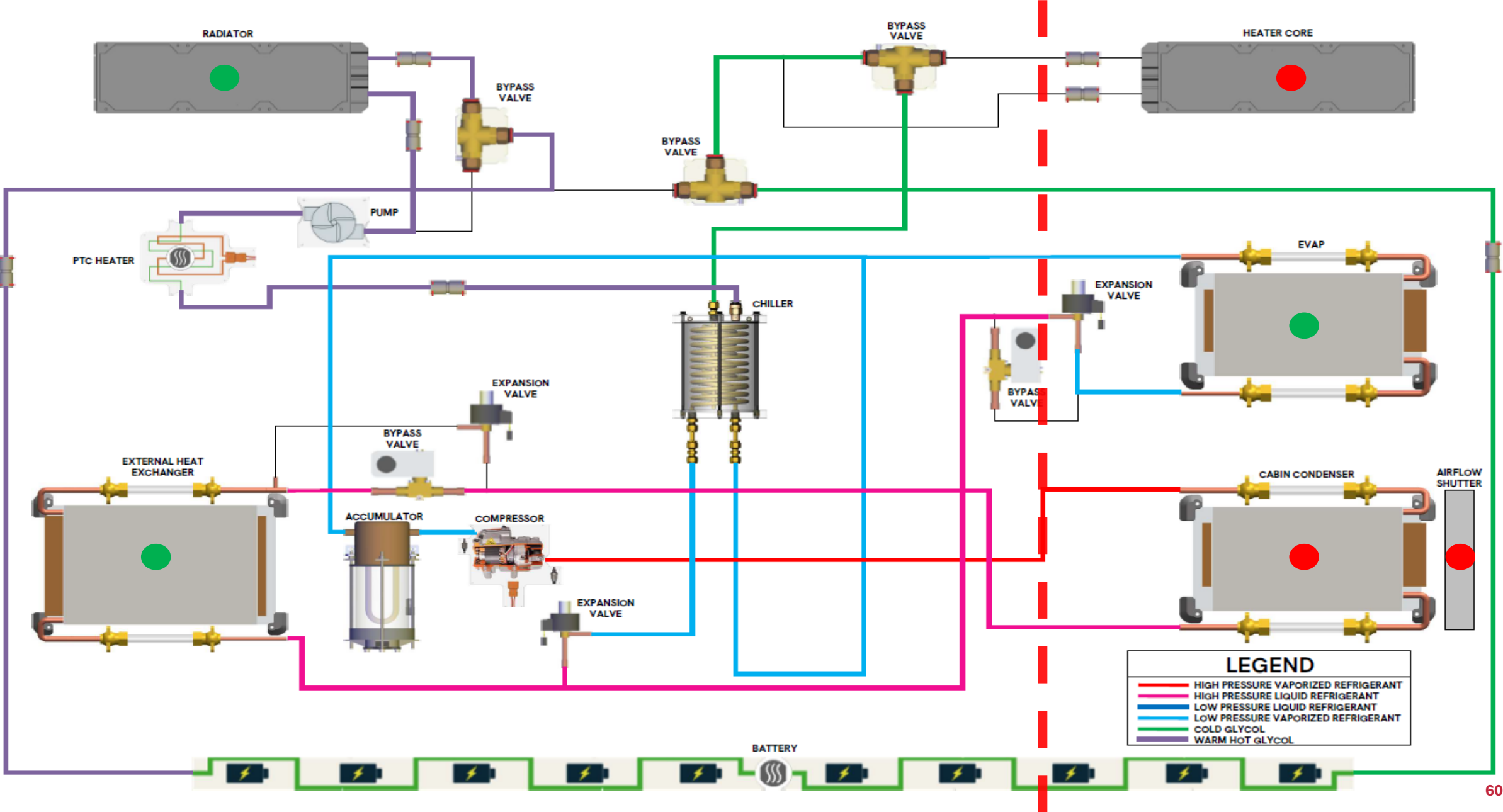
# - MODE 8 = CABIN HEAT AND BATT COOL : HVAC



- MODE 9 = BATT COOL : HVAC + GLYCOL RADIATOR



- MODE 10 = CABIN COOL + BATT COOL : HVAC + GLYCOL RADIATOR





# Tesla – Thermal system overview

The leader in integrated thermal management

## ► Models concerned: Model 3 • Model Y • Model S • Model X • Cybertruck

## ► Evolution of the Tesla thermal system:

- 2012 (Model S): basic heat pump + PTC heater; indirect battery cooling
- 2017 (Model 3): introduction of the 'Super Manifold' — combines pump, valve and reservoir
- 2021 (Model Y / refreshed 3): 'Octovalve' — a revolution in thermal architecture
- 2023 (Cybertruck): 48 V system + advanced thermal management for the 4 motors

## ► Refrigerant:

- R-134a or R-1234yf: always confirm with the VIN

## ► Distinctive Tesla characteristics:

- Fully software-driven control; OTA updates change the thermal strategy
- Limited diagnostic access for non-Tesla shops (Toolbox required)
- Very detailed data history in the Tesla Service portal

# Tesla Model Y – Complete thermal architecture

The best-selling EV in Canada in recent years

## ► Thermal system (2021+):

- Compressor: Hanon/Sanden or Denso
- Octovalve: manages 6 coolant circuits
- 4 EXVs for the 4 branches of the refrigerant circuit
- Heat sources: outside air + front motor + rear motor + onboard charger

## ► Heat pump performance according to Tesla:

- Heating at 0 °C: COP  $\approx$  3.0; saves ~100 km of range vs PTC alone
- Heating at -15 °C: COP  $\approx$  2.0; saves ~60–80 km vs PTC alone
- Lower limit of the air-source heat pump: -25 °C; the PTC takes over below that

## ► Refrigerant specifications:

- Oil:
  - **Denso** compressor: ND-11 A/C oil POE84
  - **Sanden/Hanon** compressor: RB100EV A/C oil POE84

## ► Known issues:

- Refrigerant leaks at the fittings (O-rings)
- Octovalve actuator failure

# Tesla – The Octovalve

## ► What is the Octovalve?

- An 8-port fluid valve combining several 3-way valves in a single component
- Driven by 3 internal rotary actuators (stepper motors)
- Lets 6 separate coolant circuits be connected or disconnected

## ► The 6 circuits managed by the Octovalve:

- Battery circuit (cooling/heating) · Front motor circuit · Rear motor circuit
- Cabin chiller circuit · Main radiator circuit · PTC heater circuit

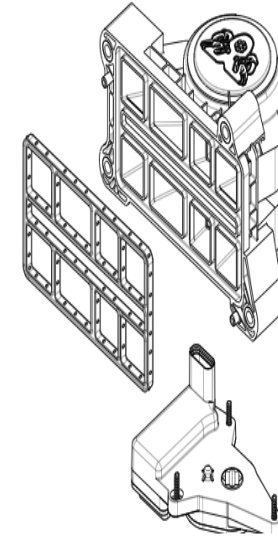
## ► Advantages of the Octovalve:

- Replaces 8–12 separate 3-way valves → fewer connections, fewer potential leaks
- Smooth, instant control by software; complex modes are easy to program
- Allows waste motor heat to be sent to the battery or the cabin with a single command

## ► Diagnosing the Octovalve:

- Angular position sensors → values visible in Tesla Toolbox
- Fault code if measured position ≠ commanded position: Octovalve Position Error
- Actuator test available in Tesla Toolbox: manual command of each rotor

⚠ **The Octovalve is replaced** – Estimated replacement cost:  
85\$ CAD



## 4. OCTOVALVE ASSEMBLY

1506859-00-D

CA\$85.00

Quantity

- 1 +

Required for Repair: 1.

Part must be ordered in multiples of this quantity: 1.

Add to Cart

# Tesla – Diagnostic points and fault codes

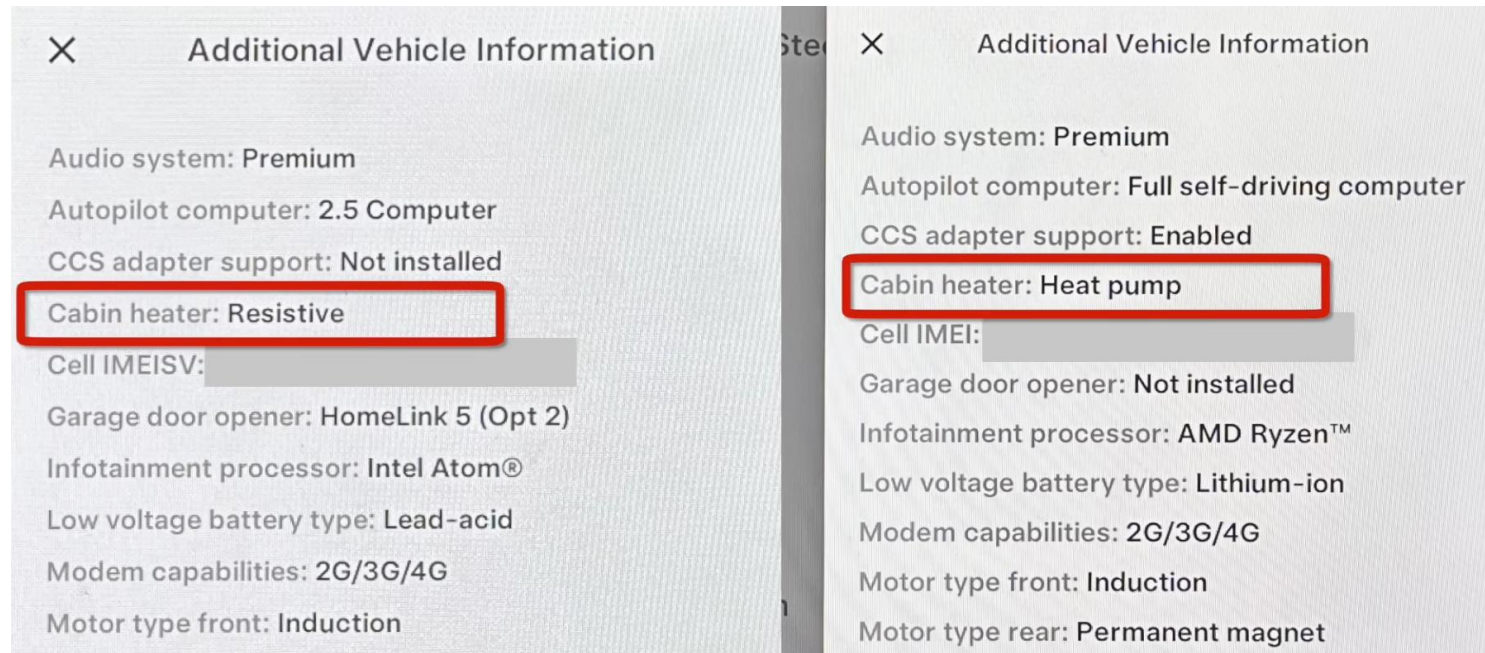
## Tools, common DTCs and procedures

### ► Live data available in Toolbox:

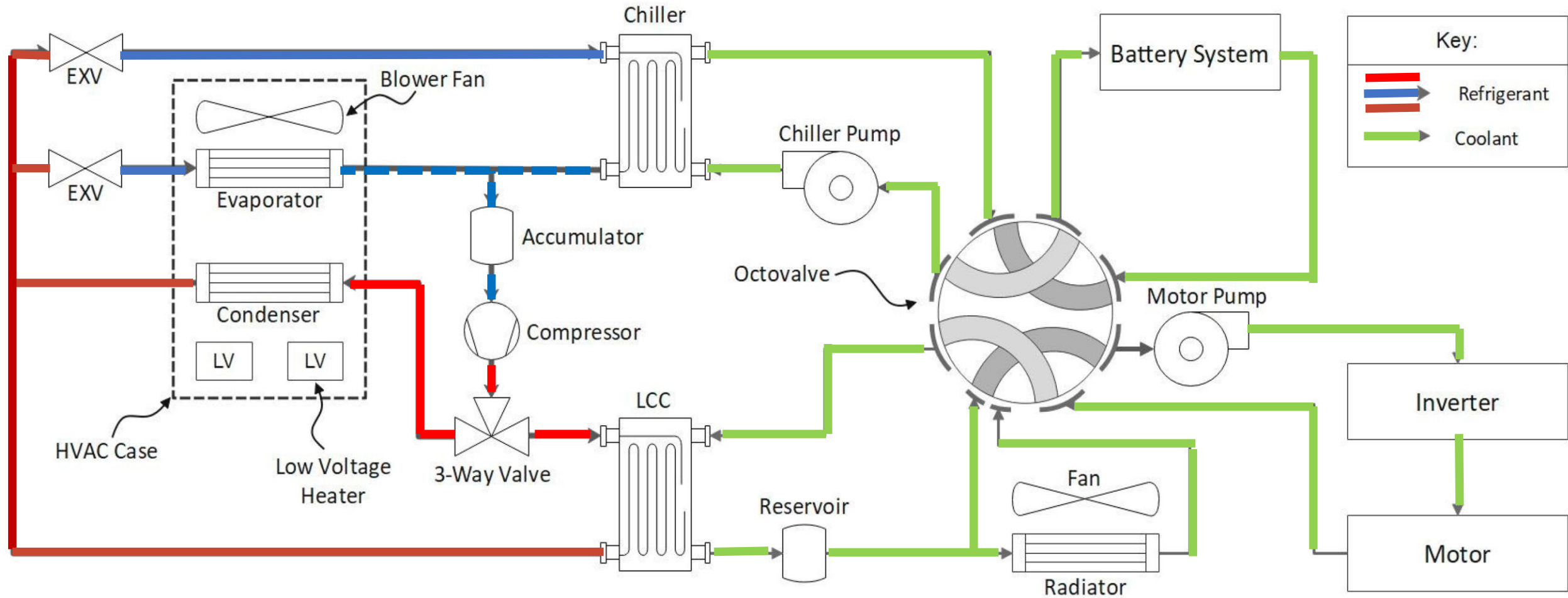
- HP/LP pressures · Compressor discharge temp · Compressor speed (rpm)
- % open of the 4 EXVs · Octovalve position · Battery temp (min/max/average)

### ► Tesla-specific procedures:

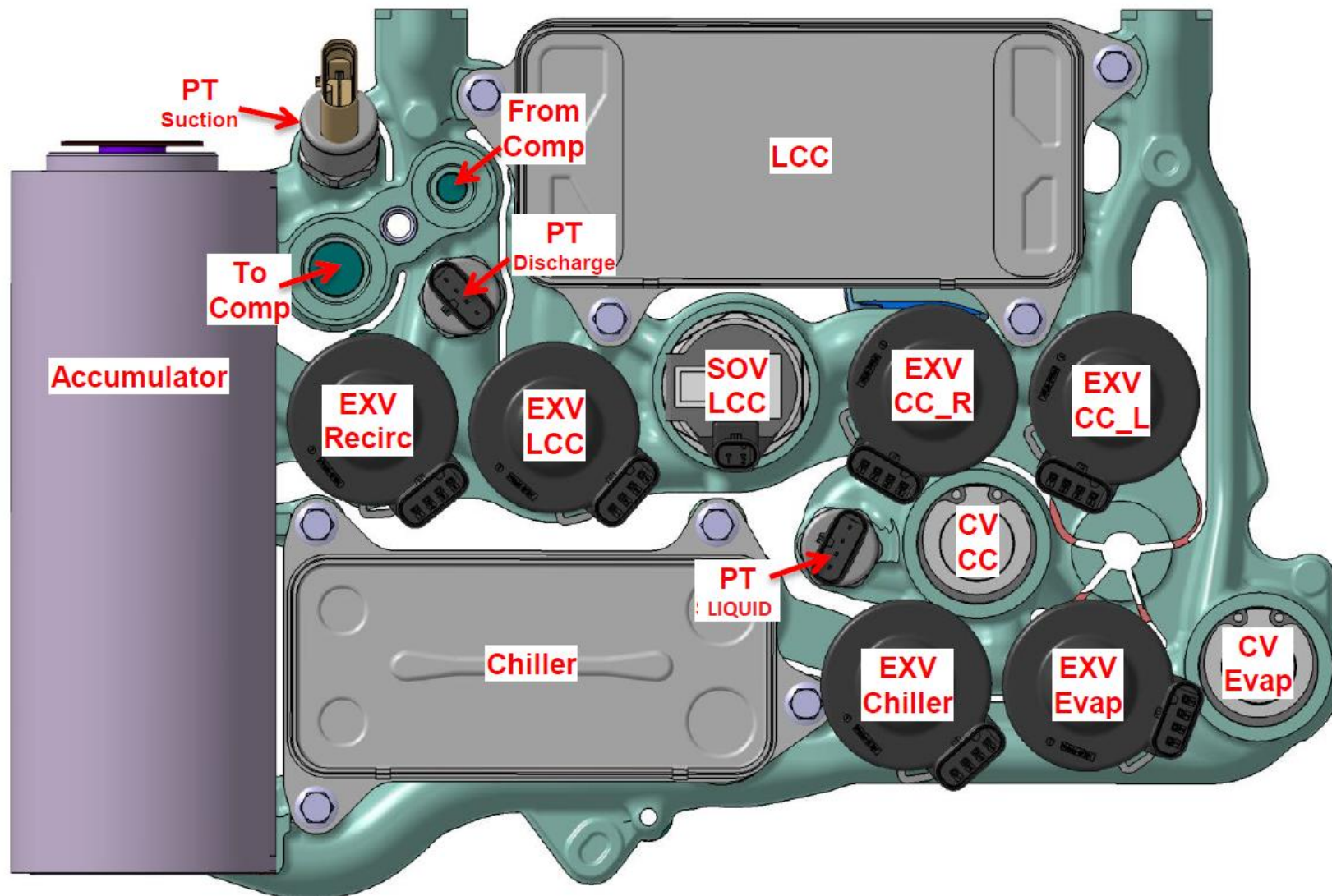
- Compressor replacement: 'Compressor Oil Fill' required through Toolbox
- Refrigerant recharge: 'Refrigerant Charge' procedure with the specified quantity
- After Octovalve replacement: 'Octovalve Calibration' in Toolbox



# TESLA Heat Pump Circuit – all models



# SUPER MANIFOLD



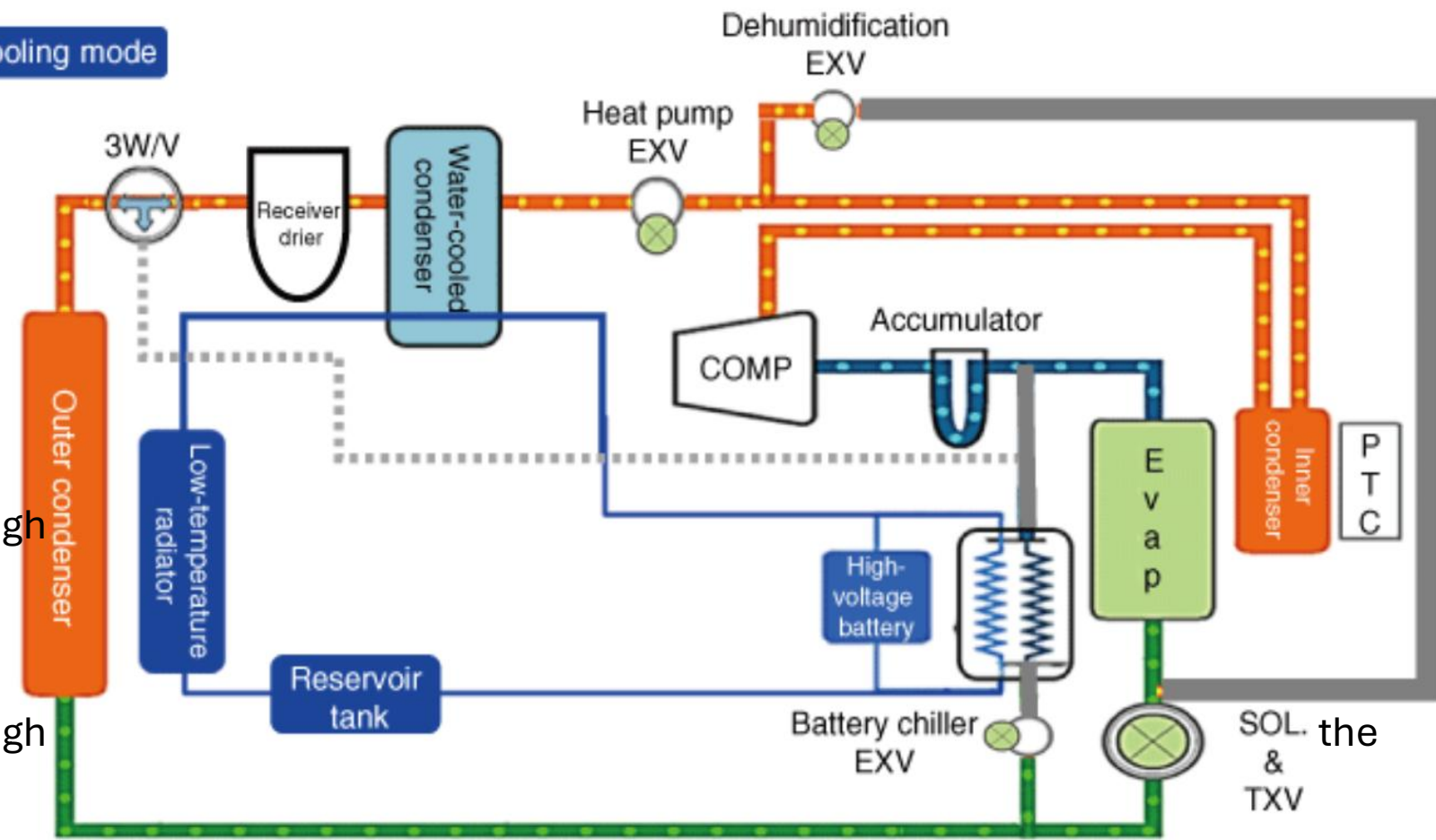


# The six operating modes

The key: the outside condenser is reversible (condenser in A/C, evaporator in heating) · Adapted from CPCPA MAP-265 (2024)

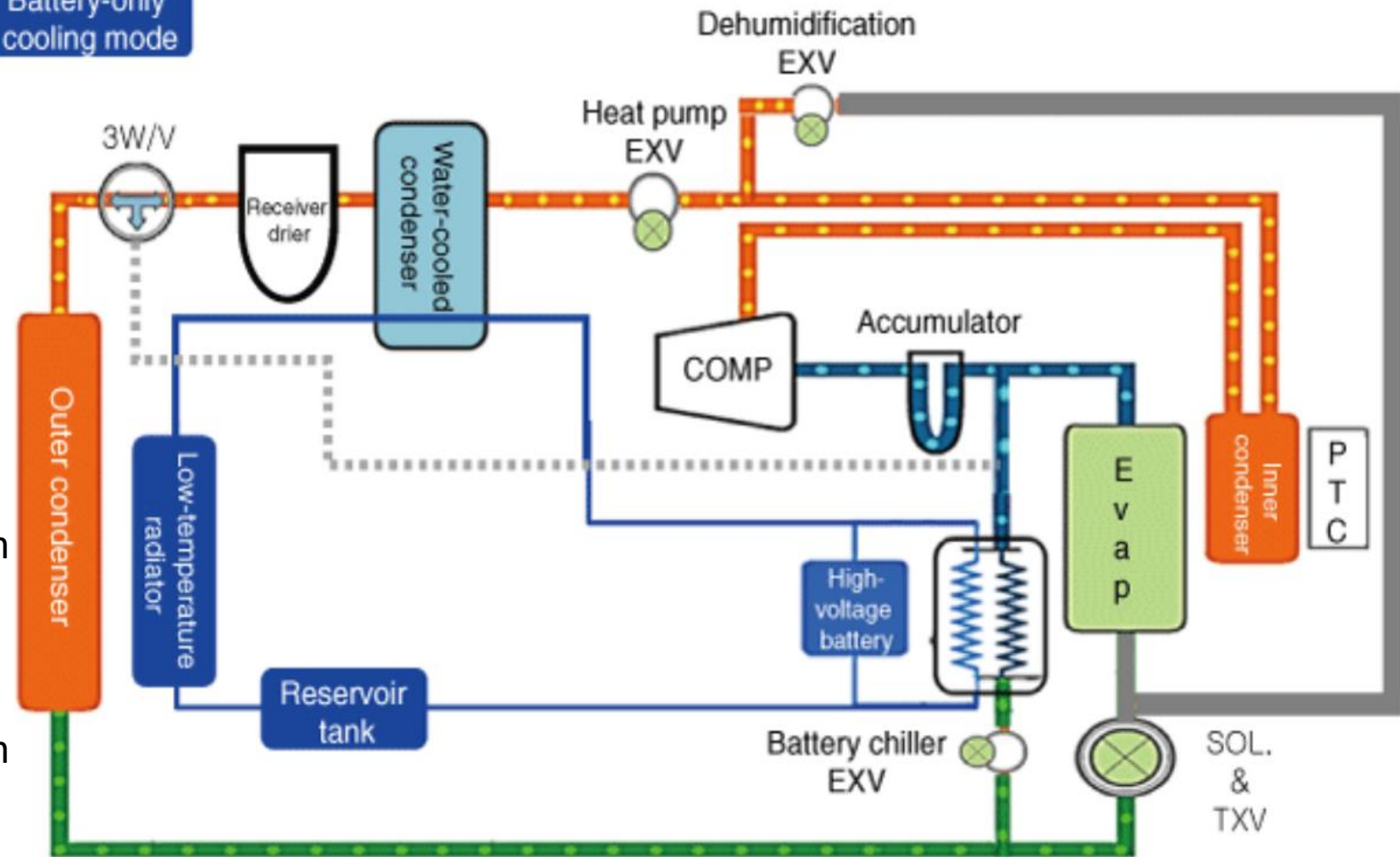
| Mode                              | Heat path                           | Key configuration                                                                                                                                                            |
|-----------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Air conditioning</b>           | Cabin → outside                     | Blend door closed (inside condenser inactive); electric expansion valve open (no expansion); condensation at the outside condenser; expansion ahead of the inside evaporator |
| <b>Heating</b>                    | Outside → cabin                     | Blend door open; electric expansion valve restricting; the outside condenser becomes an evaporator; the 3-way valve bypasses the inside evaporator                           |
| <b>Heating + PTC</b>              | Outside + resistive backup          | Same circuit as in heating; the PTC comes in when it is very cold: ~1,000 W on low voltage, ~5,000 W on high voltage (depending on the manufacturer)                         |
| <b>Dehumidification</b>           | Series circulation inside the cabin | Electric expansion valve closed, solenoid valve open: air is dried at the evaporator then reheated at the inside condenser; the outside condenser sees no refrigerant        |
| <b>Heating + dehumidification</b> | Outside → cabin, in parallel        | Refrigerant split in parallel through both inside exchangers: expansion valve restricting and solenoid valve open at the same time                                           |
| <b>Defrost</b>                    | Loop → outside exchanger (cyclic)   | Below 0 °C, frost blocks the outside exchanger; the system periodically sends hot high-pressure refrigerant back through it to melt the frost                                |

Cooling mode

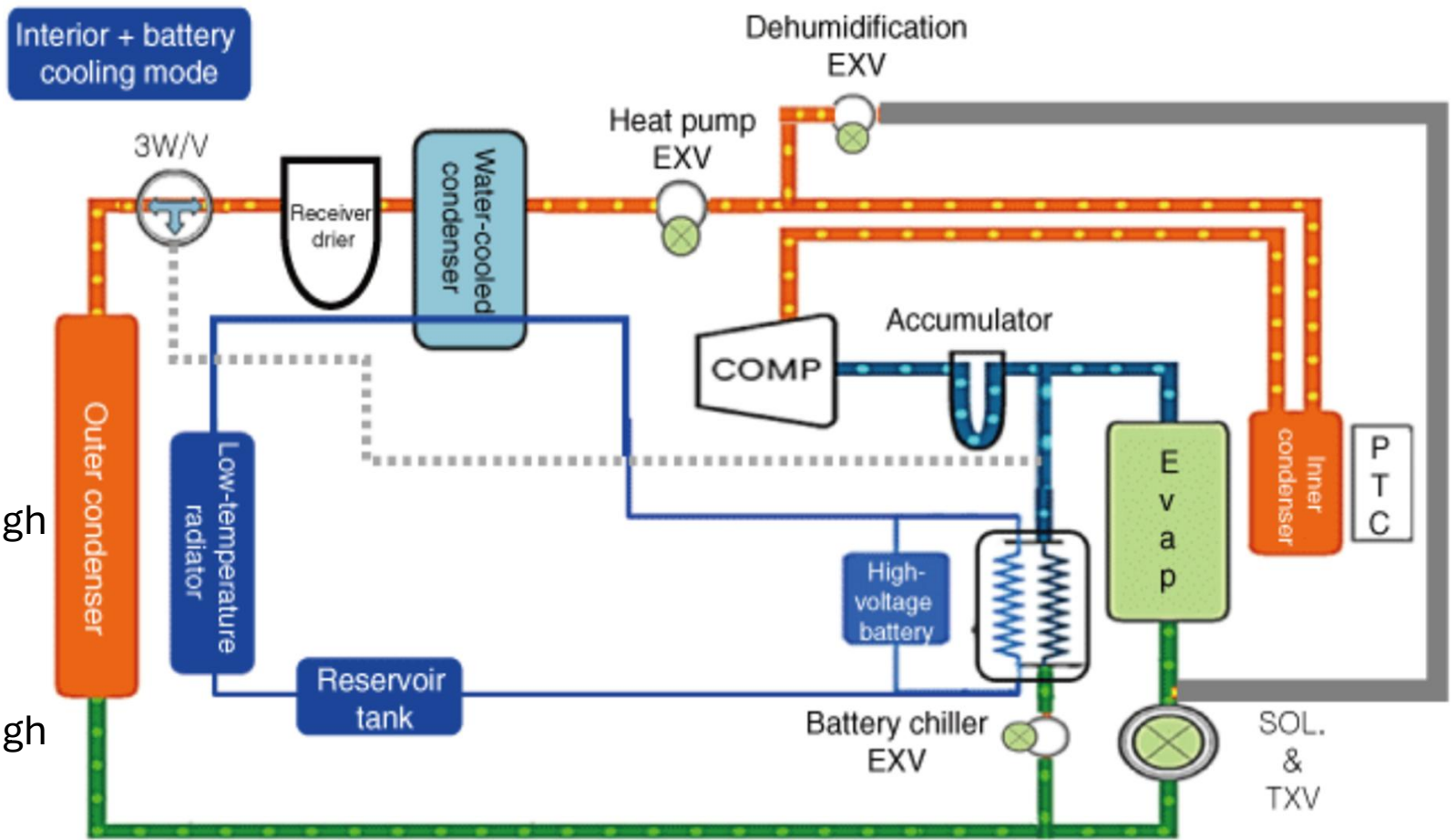


1. An electric compressor compresses the refrigerant.
2. The compressed refrigerant, at high temperature and high pressure, flows through the inside condenser. (there is no heat exchange with the cabin)
3. The compressed refrigerant, at high temperature and high pressure, flows through heat pump EXV without expanding. (the dehumidification EXV is closed)
4. The compressed refrigerant, at high temperature and high pressure, gives up its heat to the water-cooled condenser and condenses.
5. The refrigerant 3-way valve opens toward the outside condenser.
6. As it flows through the outside condenser, the refrigerant rejects heat to the atmosphere and condenses further.
7. The refrigerant expands through the SOL and the TXV (SOL open, high-voltage battery chiller EXV closed)
8. The expanded refrigerant, at low temperature and low pressure, absorbs heat from the cabin in the evaporator.
9. The refrigerant that absorbed heat inside the vehicle sends only gaseous refrigerant from the accumulator back to the electric compressor.

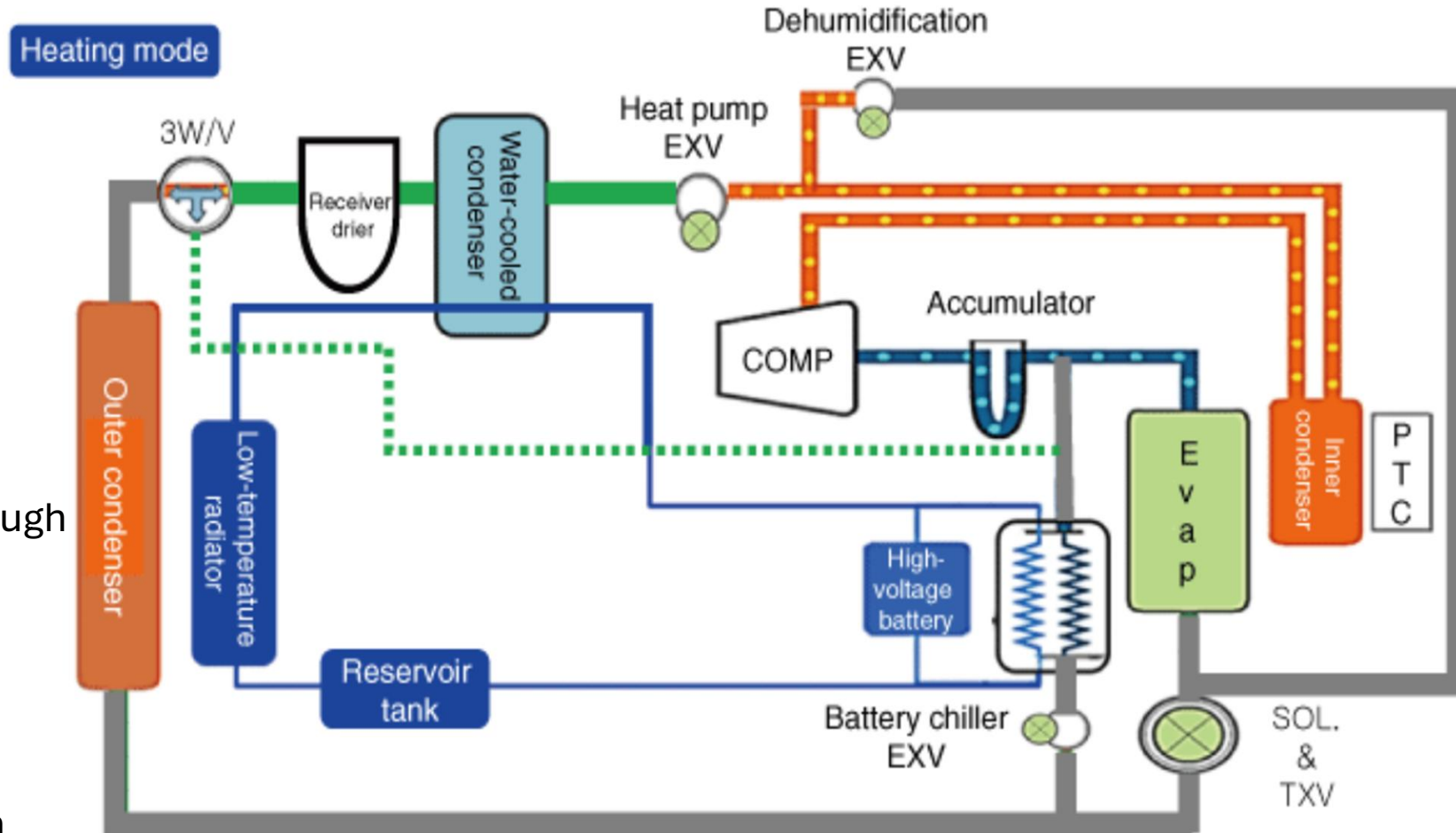
Battery-only  
cooling mode



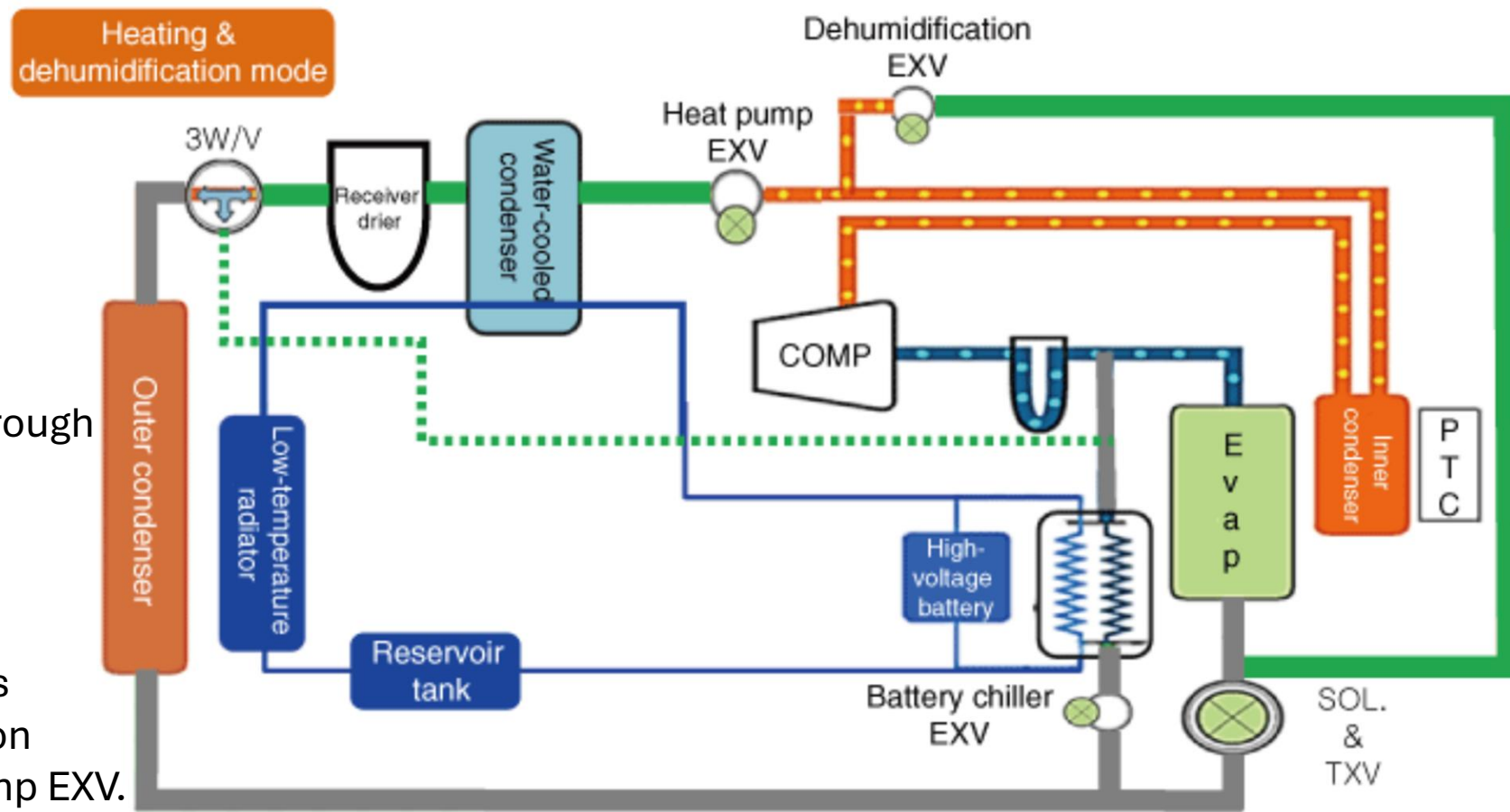
1. An electric compressor compresses the refrigerant.
2. The compressed refrigerant, at high temperature and high pressure, flows through the inside condenser. (there is no heat exchange with the cabin)
3. The compressed refrigerant, at high temperature and high pressure, flows through the heat pump EXV without expanding. (the dehumidification EXV is closed)
4. The compressed refrigerant, at high temperature and high pressure, gives up its heat to the water-cooled condenser and condenses.
5. The refrigerant 3-way valve opens toward the outside condenser.
6. As it flows through the outside condenser, the refrigerant rejects heat to the atmosphere and condenses further.
7. The refrigerant expands in the battery chiller. (EXV duty-cycle control, SOL solenoid valves and TXV closed)
8. The battery chiller absorbs heat from the high-voltage battery and the refrigerant expands.



1. An electric compressor compresses the refrigerant.
2. The compressed refrigerant, at high temperature and high pressure, flows through the inside condenser. (there is no heat exchange with the cabin)
3. The compressed refrigerant, at high temperature and high pressure, flows through the heat pump EXV without expanding. (the dehumidification EXV is closed)
4. The compressed refrigerant, at high temperature and high pressure, gives up its heat to the water-cooled condenser and condenses.
5. The refrigerant 3-way valve opens toward the outside condenser.
6. As it flows through the outside condenser, the refrigerant rejects heat to the atmosphere and condenses further.
7. The refrigerant expands through the SOL and the TXV (solenoid valve open); the refrigerant also expands in the battery chiller (EXV duty-cycle control).
8. The expanded refrigerant, at low temperature and low pressure, absorbs heat from the cabin in the evaporator and absorbs heat from the battery coolant in the battery chiller.



1. An electric compressor compresses the refrigerant.
2. As the compressed refrigerant at high temperature and high pressure flows through the inside condenser, the blend door (mix door) opens on the heating side to release the heat.
3. The compressed refrigerant, at high temperature and high pressure, behaves as if it were passing through an expansion device via a small orifice in the heat pump EXV. (the dehumidification EXV is closed)
4. The expanded refrigerant, at low temperature and low pressure, absorbs heat from the PE system in the water-cooled condenser.
5. The refrigerant 3-way valve opens in the bypass direction around the outside condenser, to minimize refrigerant heat losses.
6. As it flows through the outside condenser, the refrigerant rejects heat to the atmosphere and condenses further.
7. Having absorbed heat from the drive (PE) system, the refrigerant sends only gaseous refrigerant from the accumulator back to the electric compressor.



1. An electric compressor compresses the refrigerant.
2. As the compressed refrigerant at high temperature and high pressure flows through the inside condenser, the blend door (mix door) opens on the heating side to release the heat.
3. The compressed refrigerant, at high temperature and high pressure, behaves as if it were passing through an expansion device via a small orifice in the heat pump EXV.
4. The dehumidification EXV opens and part of the expanded refrigerant passes through the evaporator to remove moisture from the cabin air.
5. The expanded refrigerant, at low temperature and low pressure, absorbs heat from the PE system in the water-cooled condenser.
6. The refrigerant 3-way valve opens in the bypass direction around the outside condenser, to minimize refrigerant heat losses.
7. Having absorbed heat from the PE drive system, the refrigerant sends only gaseous refrigerant from the accumulator back to the electric compressor. (the refrigerant partly diverted to dehumidify the cabin air is recombined)



# GM / Chevrolet – the Ultium platform

General Motors' next-generation EV platform

## ▶ **Ultium models concerned:**

- Chevrolet Equinox EV · Blazer EV · Silverado EV
- GMC Sierra EV · GMC Hummer EV
- Cadillac Lyriq · Honda Prologue

## ▶ **Ultium thermal architecture:**

- Refrigerant: R-1234yf
- BCM, BECM, HVAC Module
- Refrigerant circuit with direct battery cooling (dedicated chiller)
- Waste heat recovery: OBC (onboard charger) + motor + inverter
- Auxiliary PTC heater (backup below -20 °C)

## ▶ **GM diagnostic tool:**

- GDS2 (Global Diagnostic System 2)
- Guided procedures in GDS2 for every thermal component
- EXV calibration available

# Chevrolet Equinox EV

GM's affordable EV

## ► Thermal system (Ultium platform):

- Refrigerant: R-1234yf
- Heat pump standard on ALL trims
- Heat sources: outside air + OBC waste heat + motors
- Direct battery cooling by refrigerant (dedicated chiller)

## ► Specified heat pump performance:

- Effective in heat pump mode down to -15 °C (-25 °C with waste heat recovery)
- Preconditioning available

## ► Architecture specific to the Equinox EV:

- 3 EXVs: cabin evaporator + battery chiller + outside exchanger
- Electric water pump dedicated to the battery circuit

## ► Specific diagnostics:

- Tool: GDS2 with MDI2
- Compressor replacement procedure: flush + mandatory GDS2 break-in procedure

# Chevrolet Silverado EV · GMC Sierra EV · GMC Hummer EV

Electric trucks and utility vehicles — reinforced thermal architecture

## ► These vehicles have amplified thermal needs compared with cars

- Larger batteries (170–220 kWh on the Hummer EV) → more heat to manage
- High DCFC charging capacity: 350 kW (Silverado EV) → cooling is critical
- Use in extreme conditions

## ► Silverado EV / Sierra EV:

- Ultium platform, 400 V or 800 V
- Twin battery chillers for greater cooling capacity
- Heat pump standard with motor + OBC heat recovery
- DCFC charging up to 350 kW (with preconditioning)

## ► GMC Hummer EV:

- Four-motor architecture (Edition 1) or three-motor
- 'Ultium Drive' system with integrated thermal management
- Reinforced battery cooling for heavy-duty use
- Special modes: 'Crab Walk' (four-wheel steering) → more motor heat → heat pump called on

## ► Diagnostic points for Ultium trucks:

- Doubled cooling capacity → check both battery chillers
- GDS2: HD-specific procedures in the 'Thermal Management' section

# What to do after a metal debris contamination

## ► Components that **MUST** be **REPLACED** after metal contamination:

- Compressor
- All O-rings and seals
- Chiller
- Electronic expansion valves (EXV)
- Hoses that contain a muffler
- Accumulator/dryer with desiccant
- Refrigerant flow valves (RFV)
- Schrader valves on the service ports
- Check valves
- Heater Core

## ► Components that **MUST** be **FLUSHED** after metal contamination:

- All other metal lines that do not contain a muffler
- Condenser
- Evaporator

► **Flushing removes all the POE oil from the A/C system. The A/C system must then be refilled with the correct quantity of POE oil.**

**Mode 1: Maximum Cooling with reheat**

This mode is when the vehicle needs to cool the propulsion battery while also both heating (reheat) and cooling the cabin. The heat rejection is significantly higher than what is needed for cabin comfort, so the front condenser is used to reject the heat to the environment.

| COMPONENT                                  | CONDITION                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------|
| Air Conditioning With Motor Compressor     | Low side control within NVH (noise, vibration, harshness) and pressure limits |
| Shutter / CRFM Fans                        | Open / On per airflow request                                                 |
| Evaporator and EXV                         | Open, control to evaporator out superheat                                     |
| Drive Motor Battery Coolant Cooler and EXV | Open, control to drive motor battery coolant cooler out superheat             |
| Air Conditioning Condenser & RFV           | Open, control to air conditioning condenser sub-cool                          |
| Heater Core & RFV                          | Open, control to heater core out refrigerant temperature                      |

**Mode 2: Maximum Cooling**

This mode is when the vehicle needs to cool the propulsion battery and the cabin. No reheat is needed so all the heat rejection is done by the front condenser.

| COMPONENT                                  | CONDITION                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------|
| Air Conditioning With Motor Compressor     | Low side control within NVH (noise, vibration, harshness) and pressure limits |
| Shutter / CRFM Fans                        | Open / On per airflow request                                                 |
| Evaporator and EXV                         | Open, control to evaporator out superheat                                     |
| Drive Motor Battery Coolant Cooler and EXV | Open, control to drive motor battery coolant cooler out superheat             |
| Air Conditioning Condenser & RFV           | Open, control to air conditioning condenser sub-cool                          |
| Heater Core & RFV                          | Closed                                                                        |

**Mode 3: Cabin Cooling with reheat**

This mode is when the propulsion battery does not need to be cooled but the cabin requires both heating (reheat) and cooling. The heat rejection is significantly higher than what is needed for cabin comfort, so the front condenser is used to reject the heat to the environment.

| COMPONENT                                  | CONDITION                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------|
| Air Conditioning With Motor Compressor     | Low side control within NVH (noise, vibration, harshness) and pressure limits |
| Shutter / CRFM Fans                        | Open / On per airflow request                                                 |
| Evaporator and EXV                         | Open, control to evaporator out superheat                                     |
| Drive Motor Battery Coolant Cooler and EXV | Closed                                                                        |
| Air Conditioning Condenser & RFV           | Open, control to air conditioning condenser sub-cool                          |
| Heater Core & RFV                          | Open, control to heater core out refrigerant temperature                      |

**Mode 4: Cabin Heating with dehumidification**

This mode is when the propulsion battery does not need to be cooled but the cabin requires both heating and cooling (dehumidification). Since the cabin heating is greater than the heat supplied from the evaporator, the drive motor battery coolant cooler is used to supply heat.

| COMPONENT                                  | CONDITION                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------|
| Air Conditioning With Motor Compressor     | Low side control within NVH (noise, vibration, harshness) and pressure limits |
| Shutter / CRFM Fans                        | Closed / Off                                                                  |
| Evaporator and EXV                         | Open, control to evaporator out superheat                                     |
| Drive Motor Battery Coolant Cooler and EXV | Open, control to drive motor battery coolant cooler out superheat             |
| Air Conditioning Condenser & RFV           | Closed                                                                        |
| Heater Core & RFV                          | Open, control to heater core out refrigerant temperature                      |

**Mode 5: Cabin Heating with Propulsion Cooling**

This mode is when both the cabin needs to be heating and the propulsion battery needs cooling. The heat rejection from the propulsion battery is higher than what the cabin needs so the excess is rejected to the environment.

| COMPONENT                                  | CONDITION                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------|
| Air Conditioning With Motor Compressor     | Low side control within NVH (noise, vibration, harshness) and pressure limits |
| Shutter / CRFM Fans                        | Controlled per airflow request / On per airflow request                       |
| Evaporator and EXV                         | Closed                                                                        |
| Drive Motor Battery Coolant Cooler and EXV | Open, control to drive motor battery coolant cooler out superheat             |
| Air Conditioning Condenser & RFV           | Open, control to air conditioning condenser sub-cool                          |
| Heater Core & RFV                          | Open, control to heater core out refrigerant temperature                      |

**Mode 6: Cabin Heating**

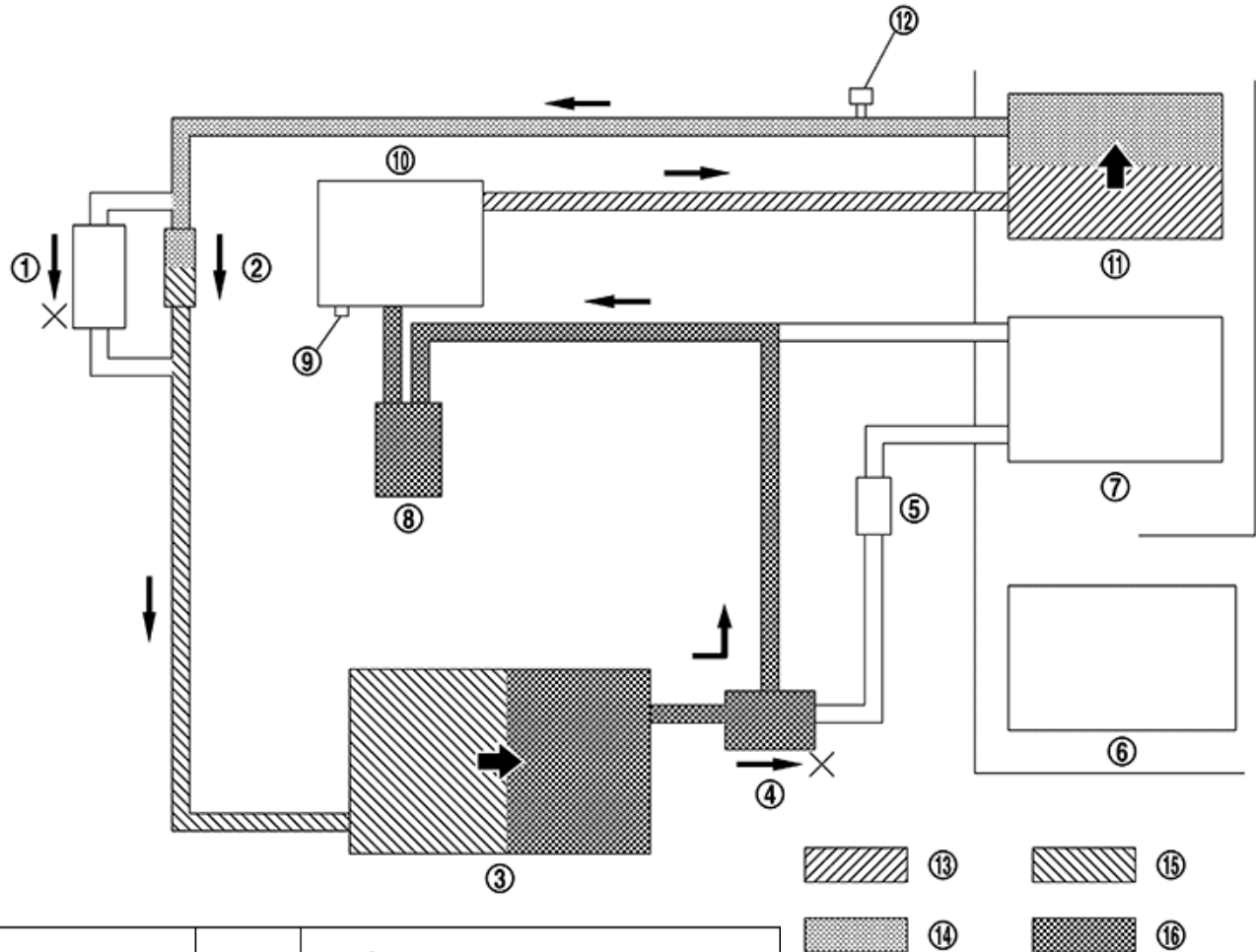
This mode is when the cabin needs heating and the propulsion system does not need cooling.

| COMPONENT                                  | CONDITION                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------|
| Air Conditioning With Motor Compressor     | High side control within NVH (noise, vibration, harshness) and pressure limits |
| Shutter / CRFM Fans                        | Closed / Off                                                                   |
| Evaporator and EXV                         | Closed                                                                         |
| Drive Motor Battery Coolant Cooler and EXV | Open, control to drive motor battery coolant cooler out superheat              |
| Air Conditioning Condenser & RFV           | Closed                                                                         |
| Heater Core & RFV                          | Open, control to heater core out refrigerant temperature                       |



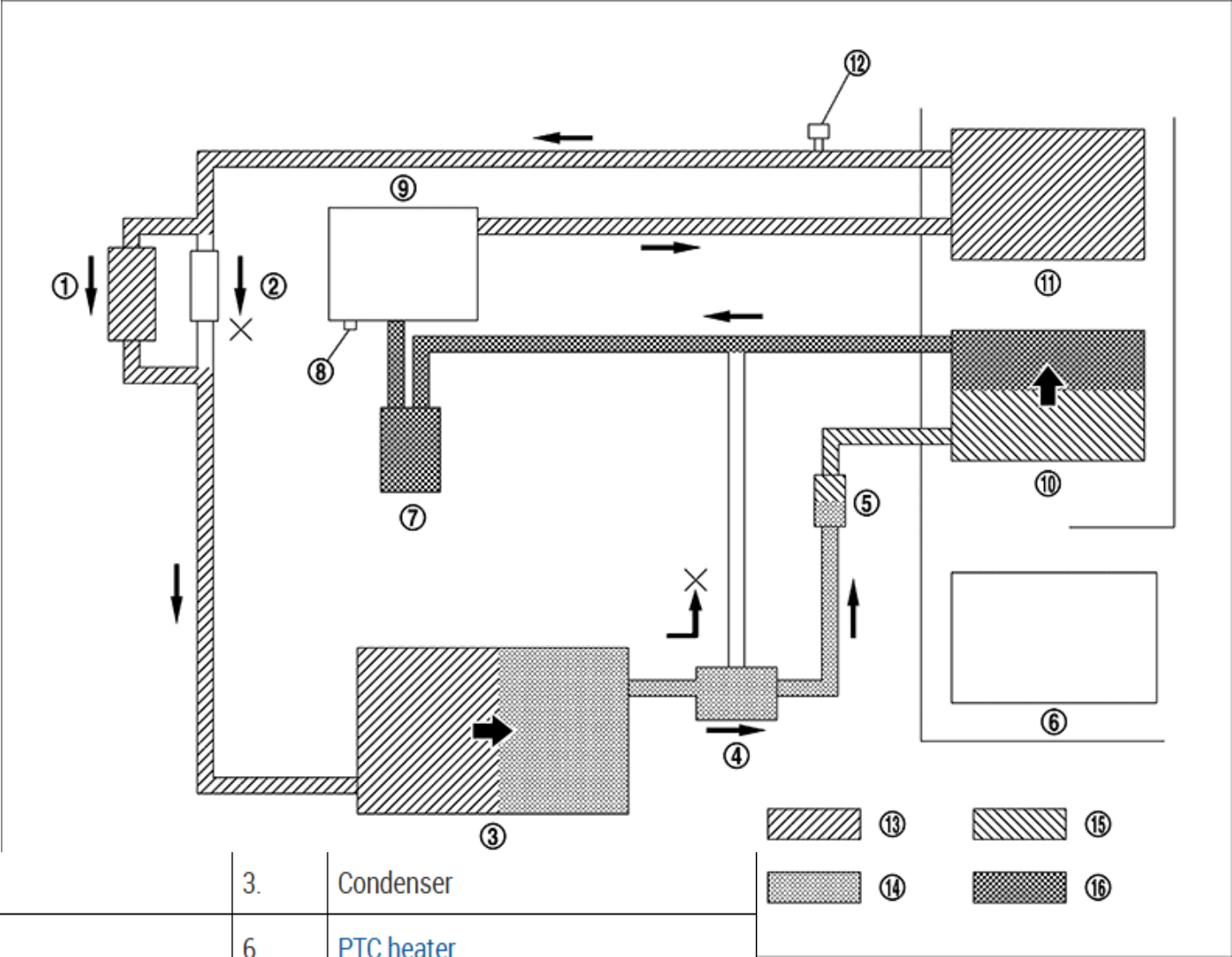
**2<sup>e</sup> generation  
(2018-2025)**

# Cabin Heating



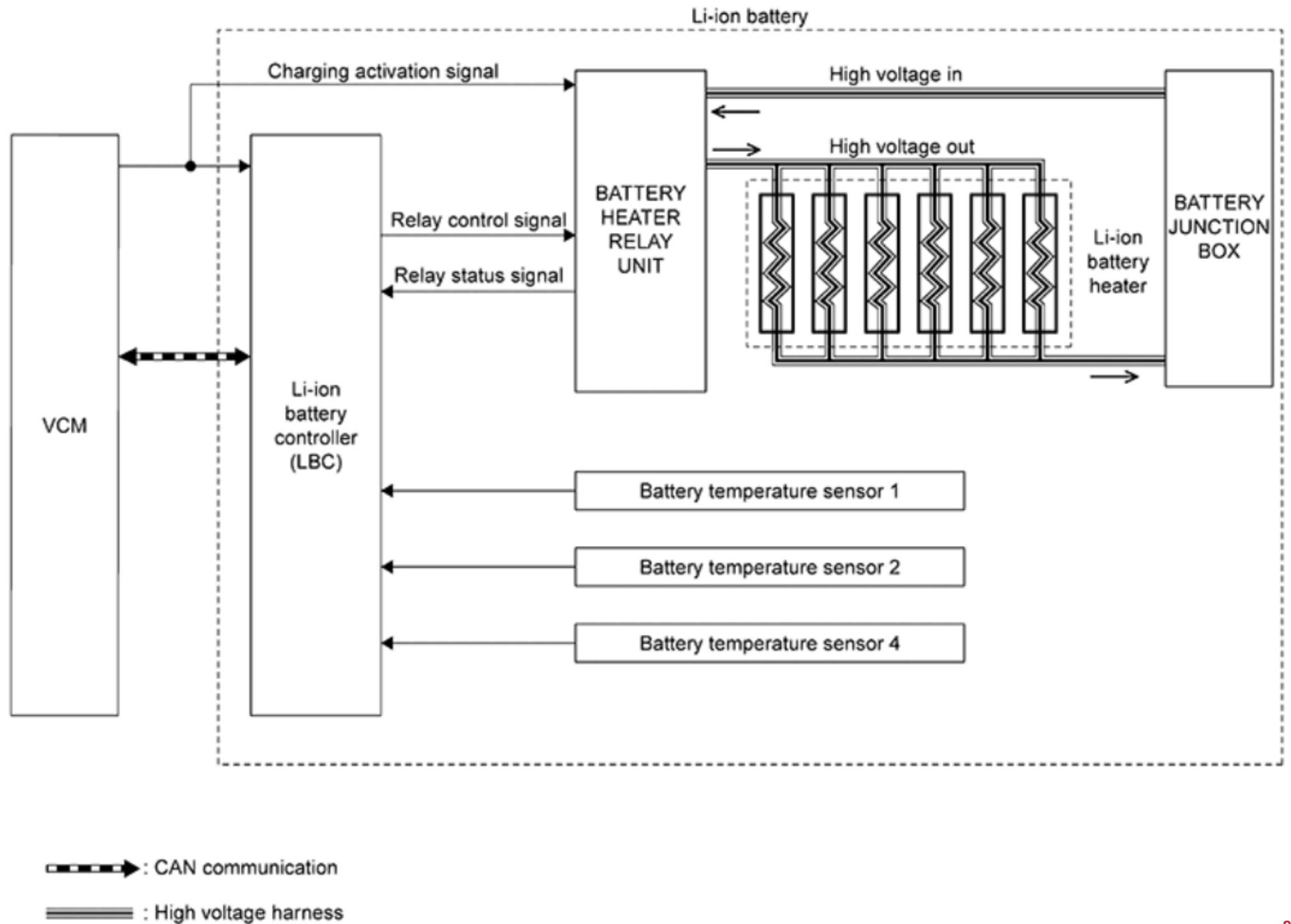
|     |                     |     |                                |     |                       |
|-----|---------------------|-----|--------------------------------|-----|-----------------------|
| 1.  | 2-way valve         | 2.  | Orifice tube (Inner condenser) | 3.  | Condenser             |
| 4.  | 3-way valve         | 5.  | Orifice tube (Evaporator)      | 6.  | PTC heater            |
| 7.  | Evaporator          | 8.  | Accumulator                    | 9.  | Pressure relief valve |
| 10. | Electric compressor | 11. | Inner condenser                | 12. | Pressure sensor       |
| 13. | High-pressure gas   | 14. | High-pressure liquid           | 15. | Low-pressure liquid   |
| 16. | Low-pressure gas    |     |                                |     |                       |

# Cabin Cooling



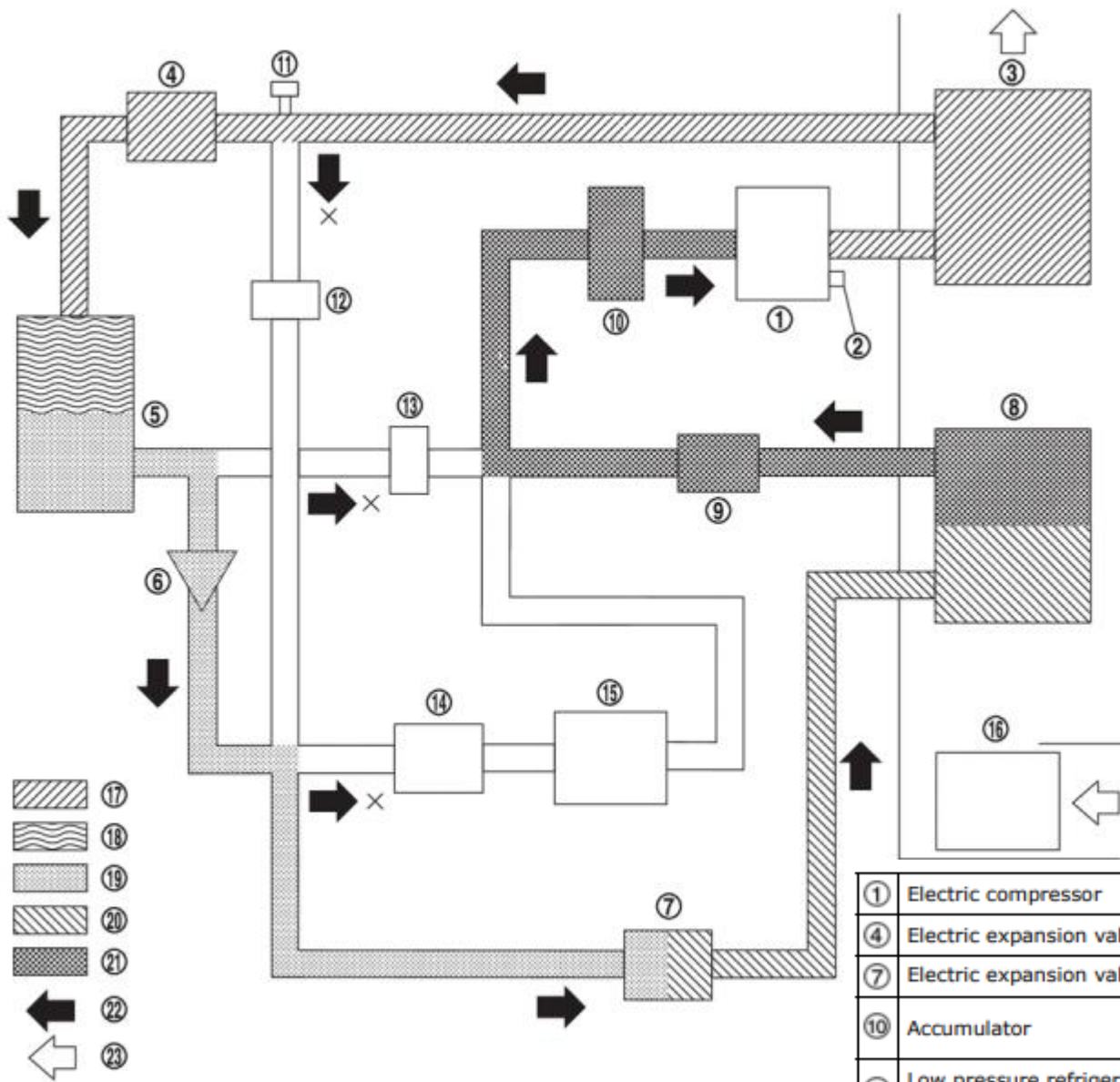
|     |                   |     |                                |     |                     |
|-----|-------------------|-----|--------------------------------|-----|---------------------|
| 1.  | 2-way valve       | 2.  | Orifice tube (Inner condenser) | 3.  | Condenser           |
| 4.  | 3-way valve       | 5.  | Orifice tube (Evaporator)      | 6.  | PTC heater          |
| 7.  | Accumulator       | 8.  | Pressure relief valve          | 9.  | Electric compressor |
| 10. | Evaporator        | 11. | Inner condenser                | 12. | Pressure sensor     |
| 13. | High-pressure gas | 14. | High-pressure liquid           | 15. | Low-pressure liquid |
| 16. | Low-pressure gas  |     |                                |     |                     |

# Battery Heating





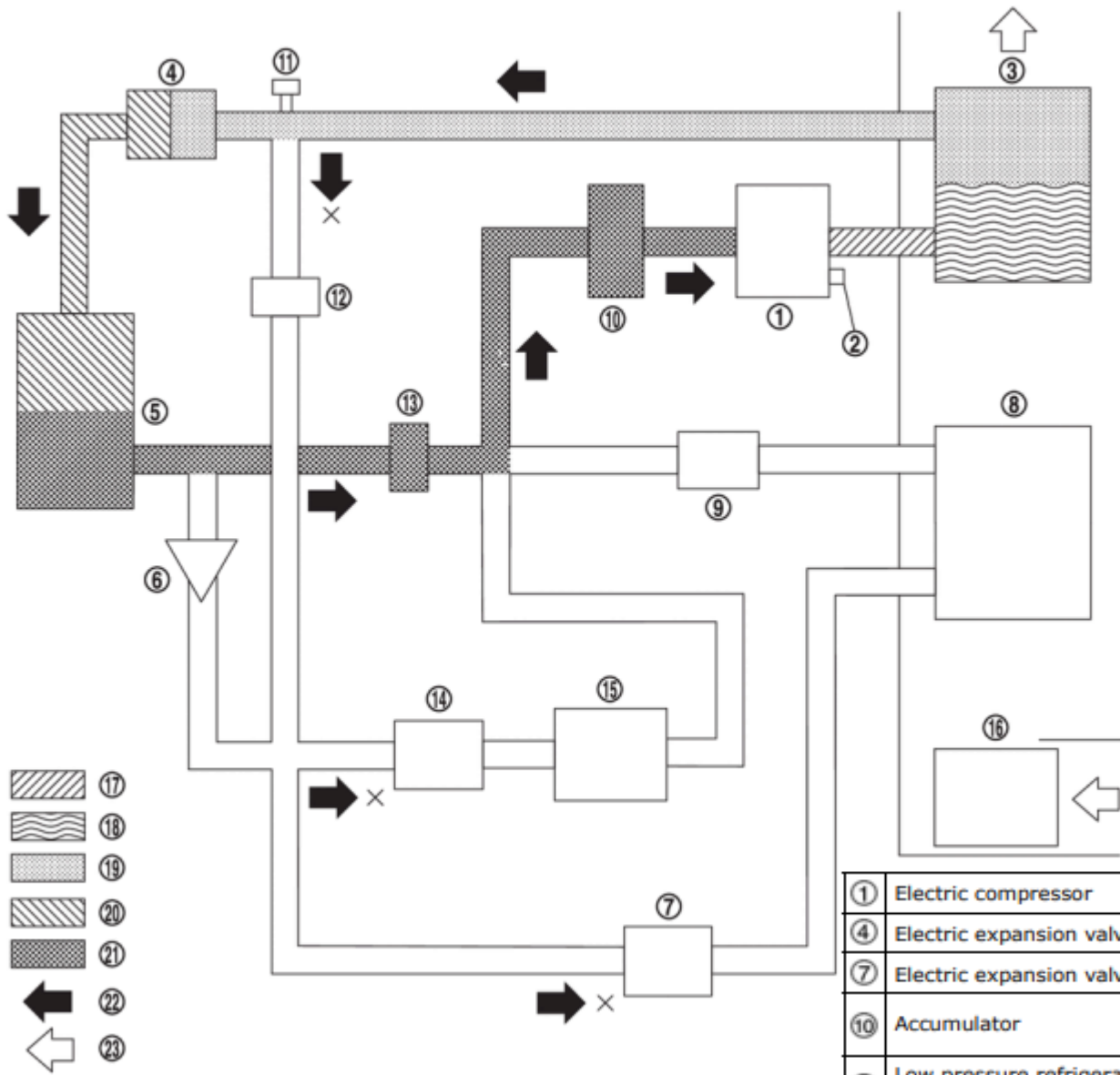
**3<sup>e</sup> generation  
(2026+)**



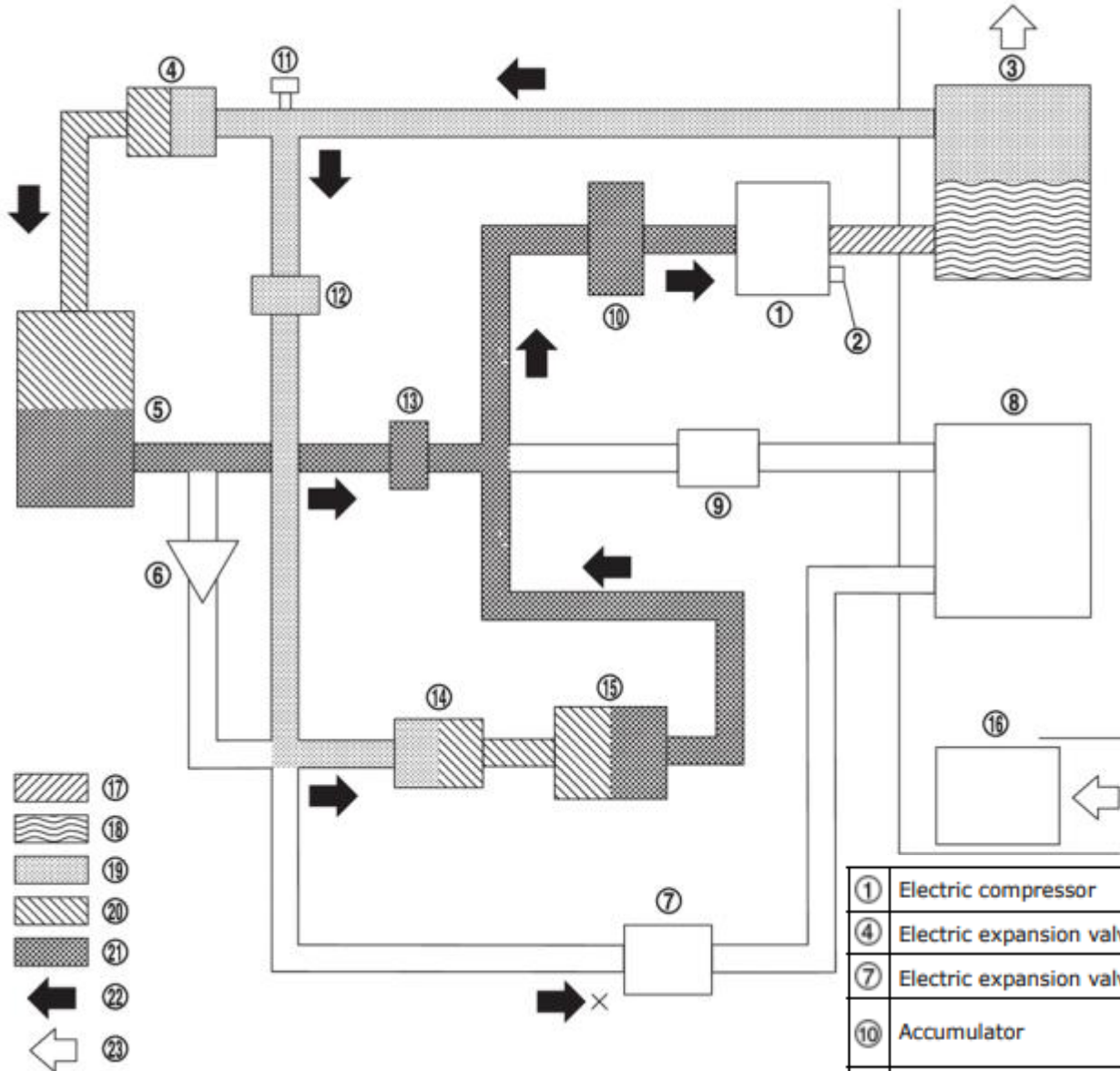
## COOLER MODE

|   |                                                  |   |                                      |   |                                                   |
|---|--------------------------------------------------|---|--------------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve                | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                            | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                           | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor          | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller)    | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                    | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ⑳ | Gas-liquid two phase in low pressure | ㉑ | Low-pressure gas                                  |
| ㉒ | Refrigerant flow                                 | ㉓ | Wind flow                            |   |                                                   |

# HEATER MODE AND HEATER + THERMAL RECOVERY (CONDENSER) MODE

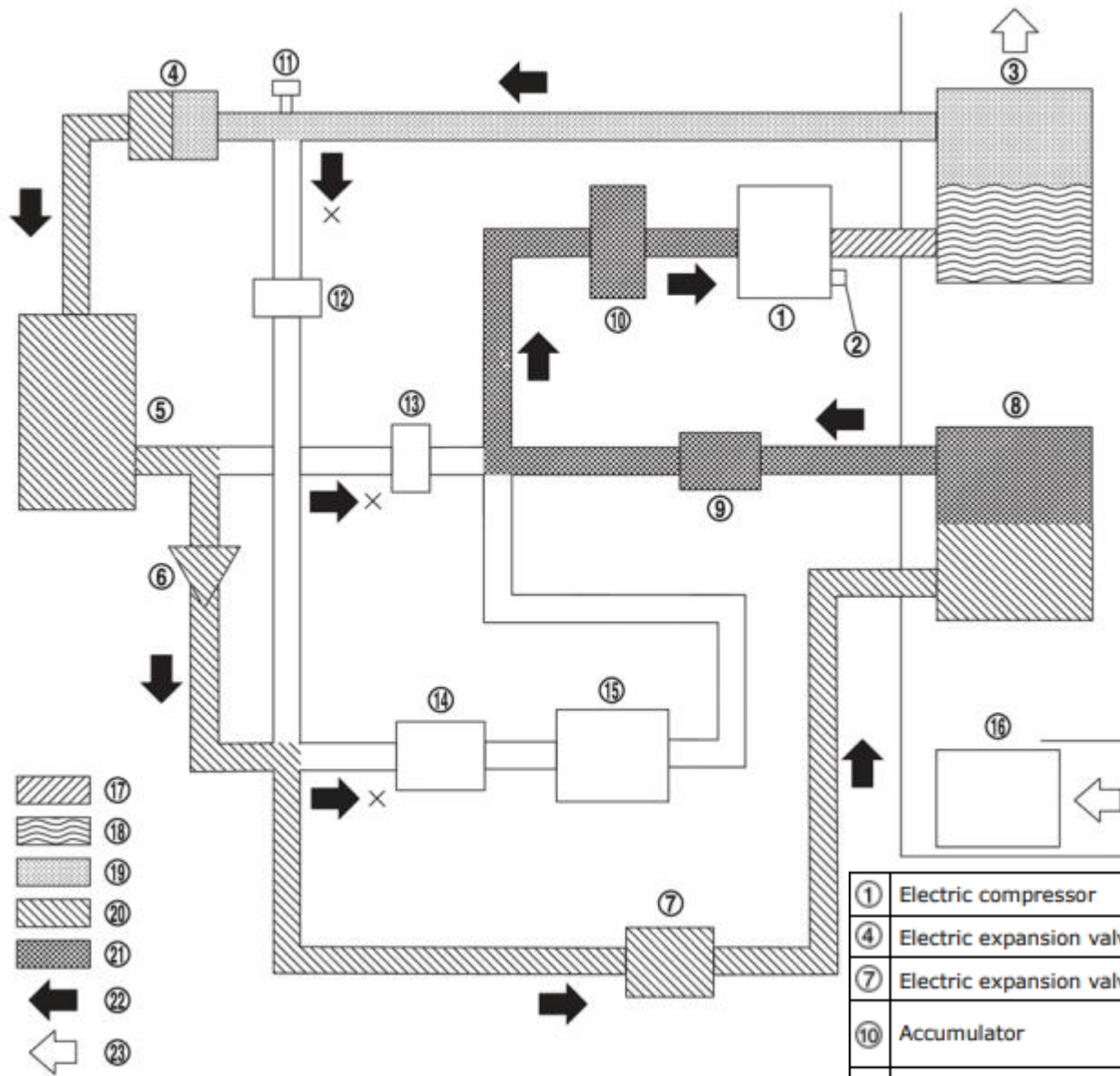


# HEATER + THERMAL RECOVERY (CONDENSER & CHILLER) MODE



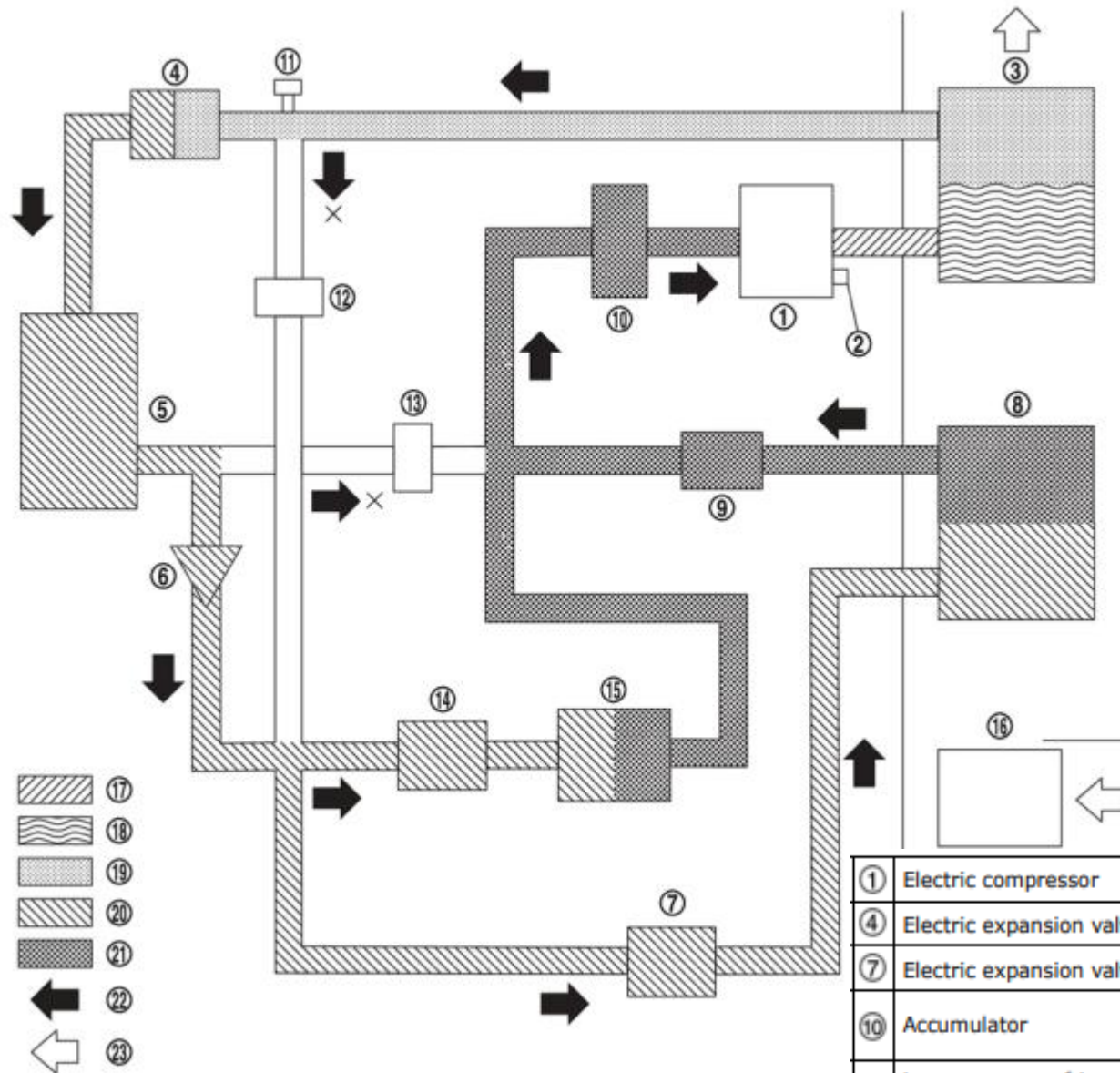
|   |                                                  |   |                                      |   |                                                   |
|---|--------------------------------------------------|---|--------------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve                | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                            | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                           | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor          | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller)    | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                    | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ⑳ | Gas-liquid two phase in low pressure | ㉑ | Low-pressure gas                                  |
| ㉒ | Refrigerant flow                                 | ㉓ | Wind flow                            |   |                                                   |

# SERIES DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CONDENSER) MODE



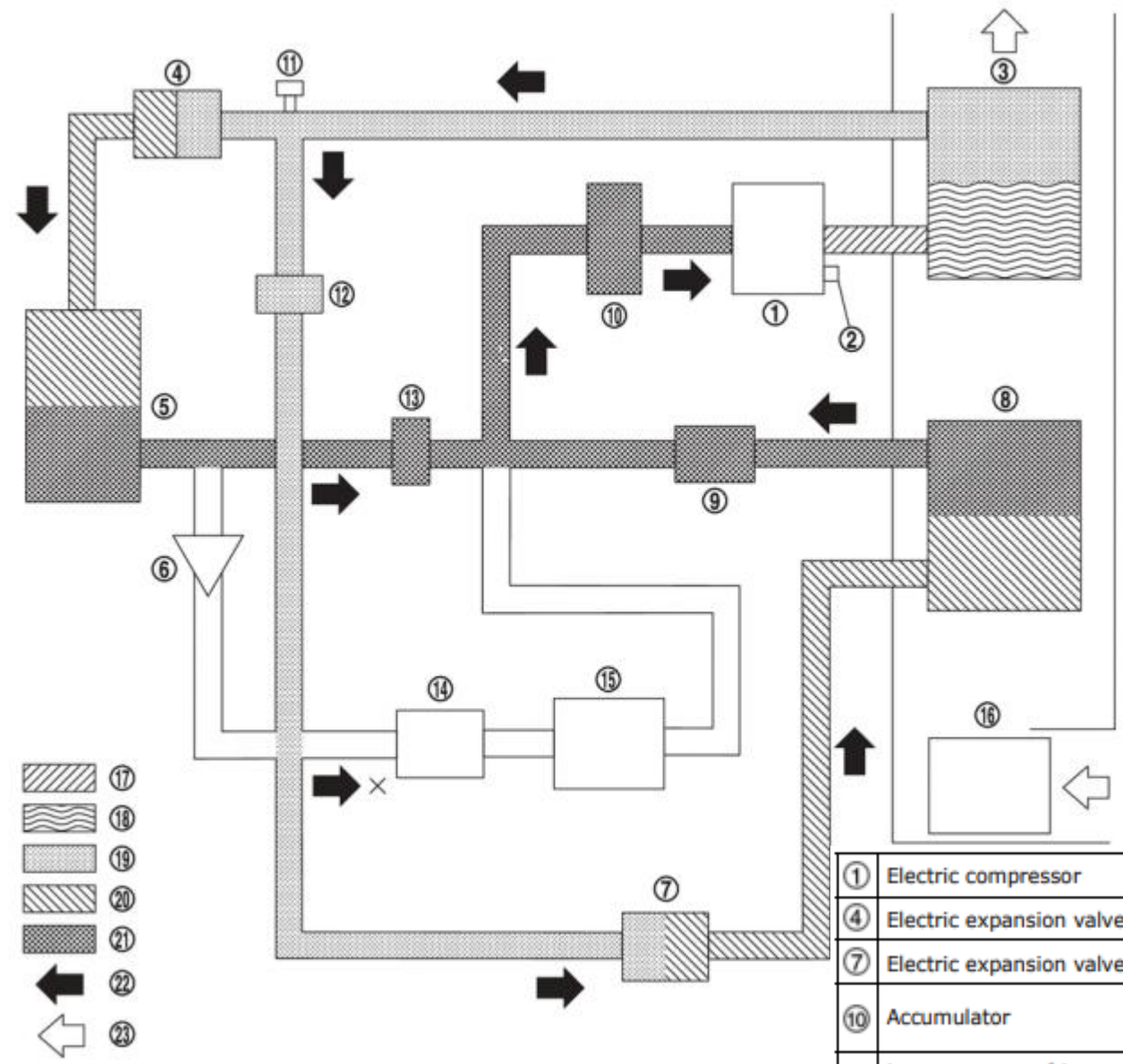
|   |                                                  |   |                                      |   |                                                   |
|---|--------------------------------------------------|---|--------------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve                | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                            | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                           | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor          | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller)    | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                    | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ⑳ | Gas-liquid two phase in low pressure | ㉑ | Low-pressure gas                                  |
| ㉒ | Refrigerant flow                                 | ㉓ | Wind flow                            |   |                                                   |

# SERIES DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CONDENSER & CHILLER) MODE

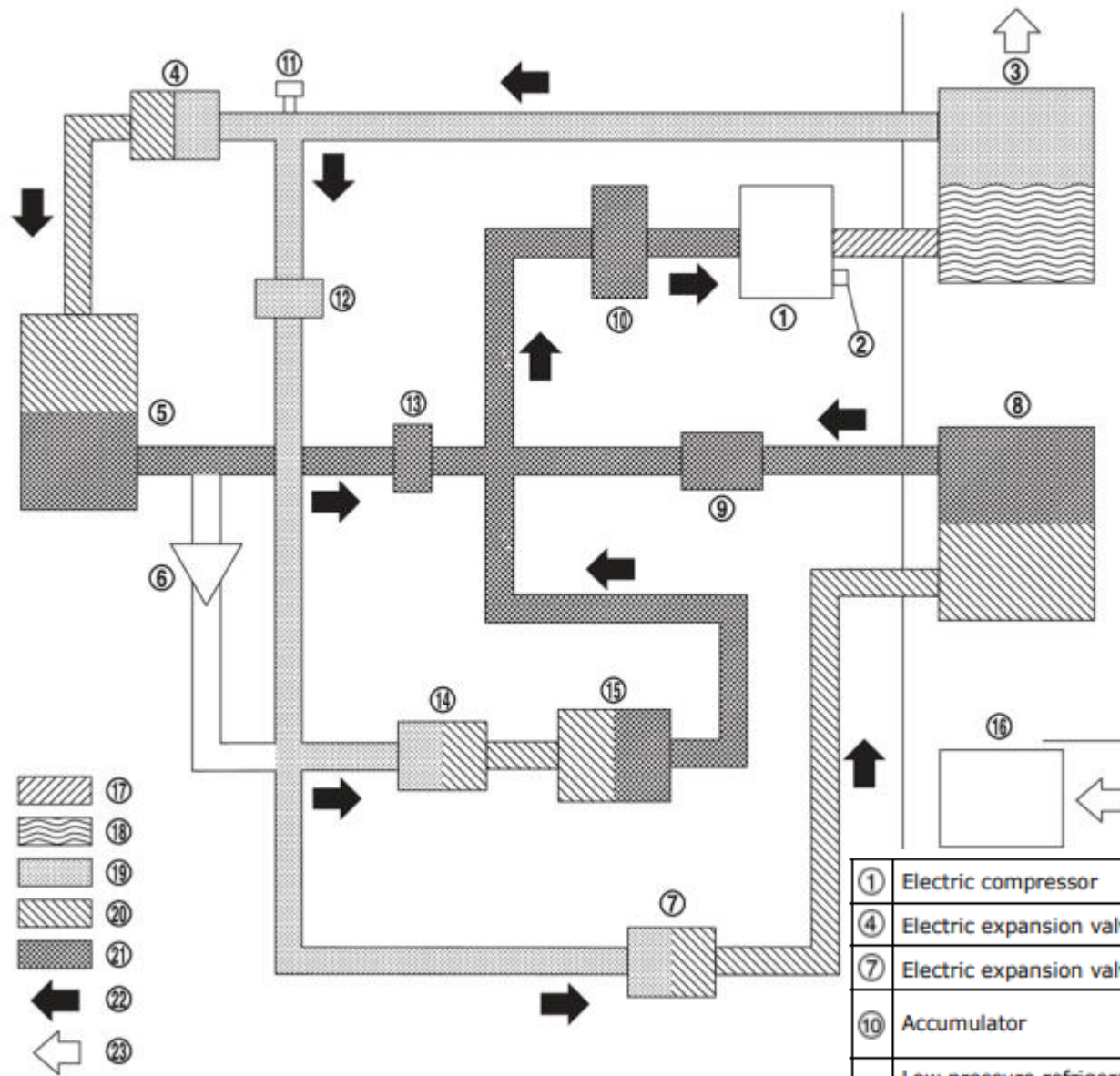


|   |                                                  |   |                                   |   |                                                   |
|---|--------------------------------------------------|---|-----------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve             | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                         | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                        | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor       | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller) | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                 | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ㉑ | Low-pressure gas                  |   |                                                   |
| ㉒ | Refrigerant flow                                 | ㉓ | Wind flow                         |   |                                                   |

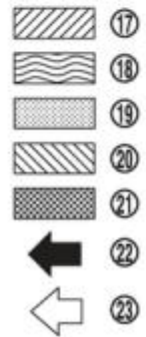
# **PARALLEL DEHUMIDIFICATION HEATING MODE AND PARALLEL DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CONDENSER) MODE**



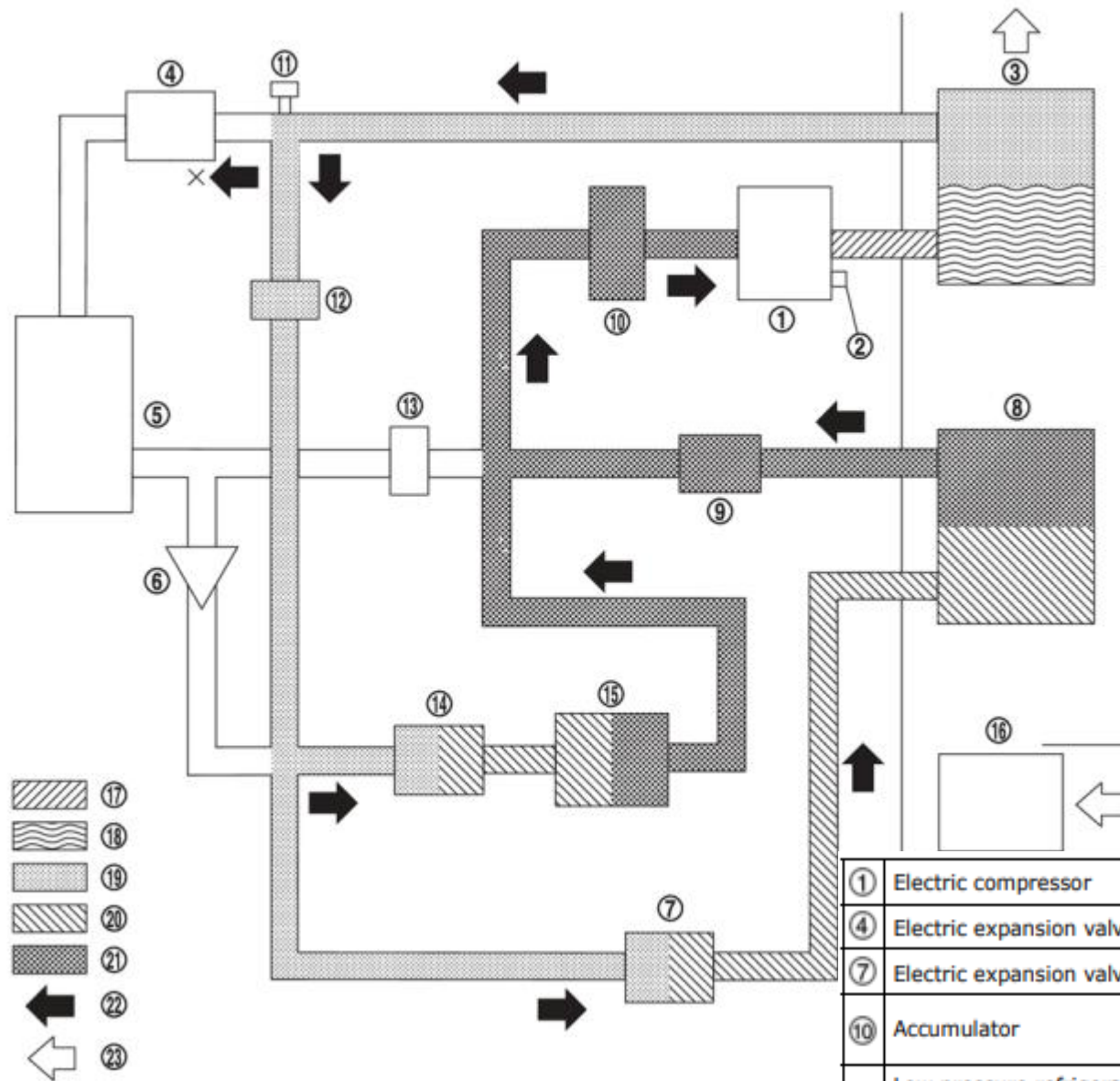
|   |                                                  |   |                                      |   |                                                   |
|---|--------------------------------------------------|---|--------------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve                | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                            | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                           | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor          | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller)    | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                    | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ⑳ | Gas-liquid two phase in low pressure | ㉑ | Low-pressure gas                                  |
| ㉒ | Refrigerant flow                                 | ㉓ | Wind flow                            |   |                                                   |



# **PARALLEL DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CONDENSER & CHILLER) MODE**

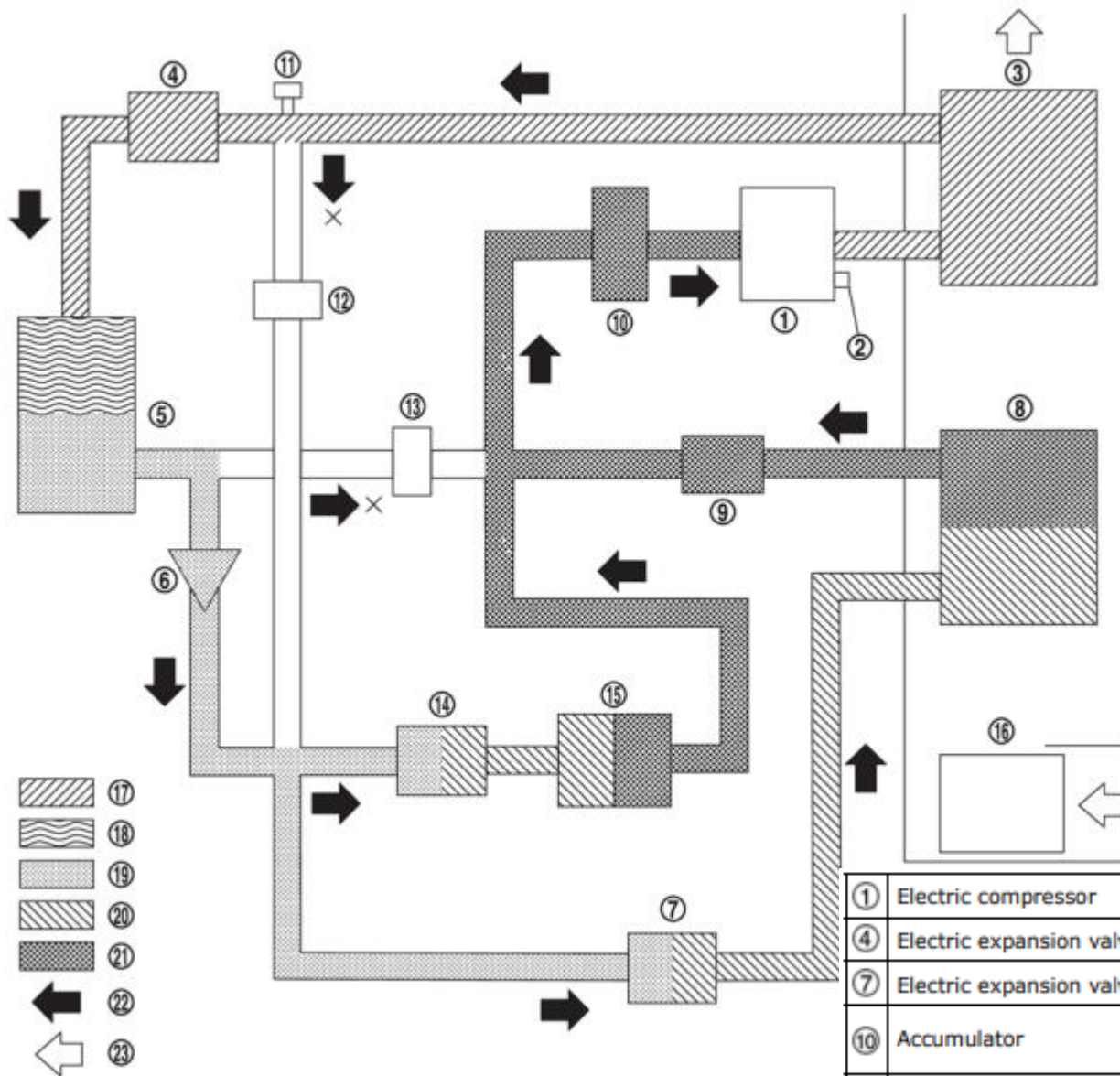


|   |                                                  |   |                                      |   |                                                   |
|---|--------------------------------------------------|---|--------------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve                | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                            | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                           | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor          | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller)    | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                    | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ⑳ | Gas-liquid two phase in low pressure | ㉑ | Low-pressure gas                                  |
| ㉒ | Refrigerant flow                                 | ㉓ | Wind flow                            |   |                                                   |



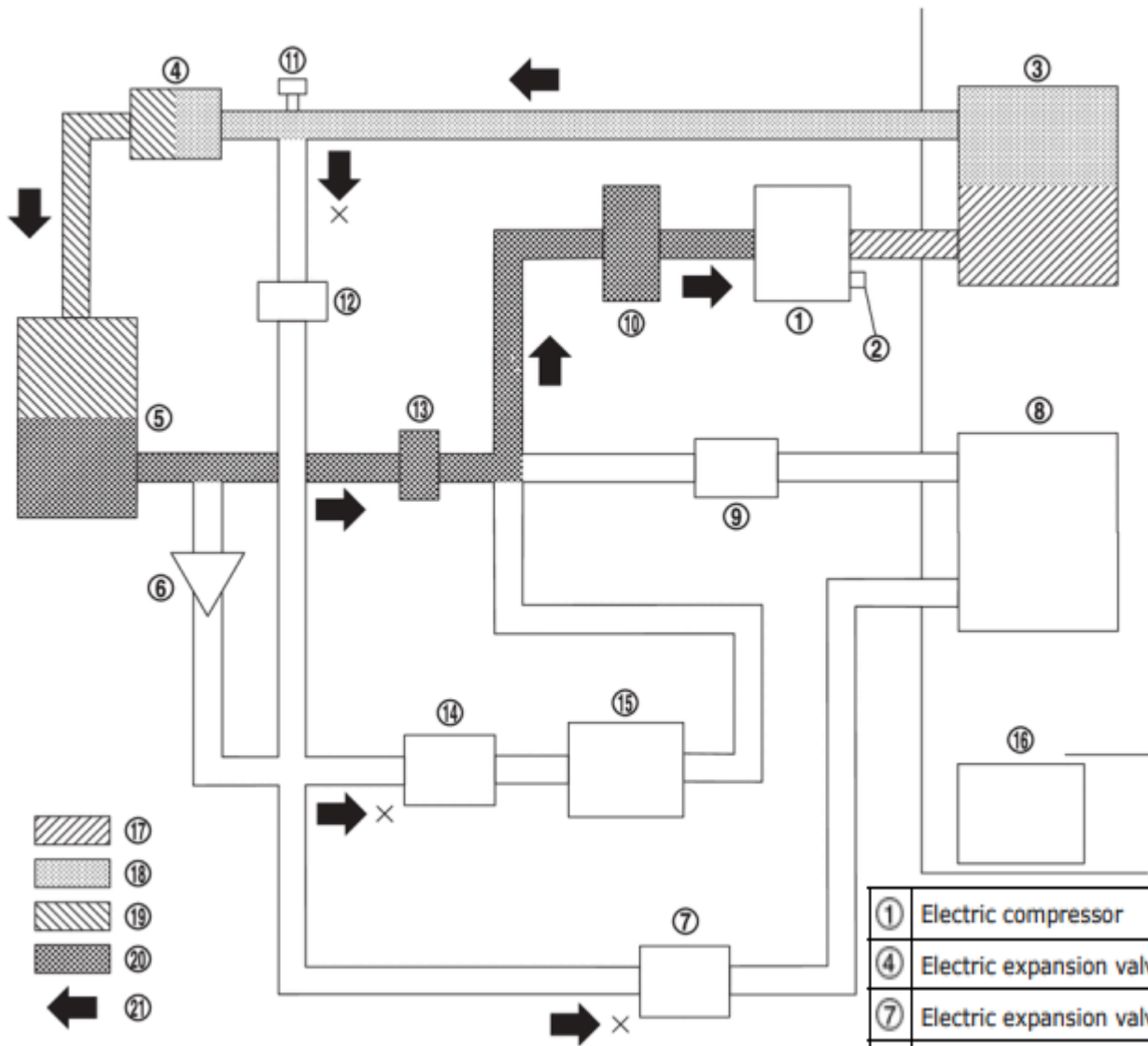
## PARALLEL DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CHILLER) MODE

|   |                                                  |   |                                      |   |                                                   |
|---|--------------------------------------------------|---|--------------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve                | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                            | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                           | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor          | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller)    | ⑮ | Battery coolant chiller                           |
| ⑯ | High-pressure liquid                             | ⑰ | High-pressure gas                    | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ⑳ | Gas-liquid two phase in low pressure | ㉑ | Low-pressure gas                                  |
| ㉒ | Refrigerant flow                                 | ㉓ | Wind flow                            |   |                                                   |

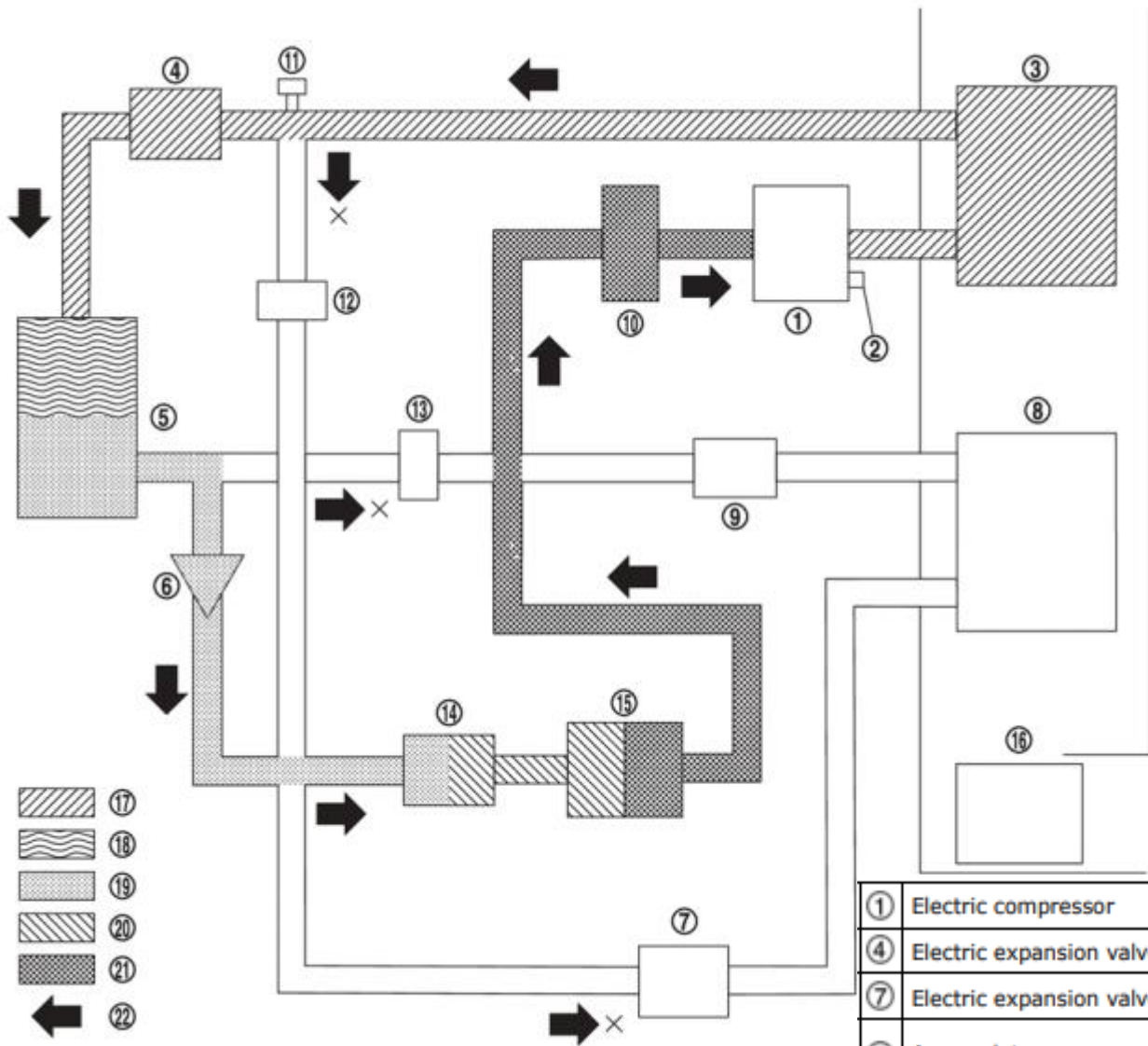


## COOLER + HIGH VOLTAGE BATTERY COOLING MODE

|   |                                                  |   |                                      |   |                                                   |
|---|--------------------------------------------------|---|--------------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve                | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                            | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                           | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor          | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller)    | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                    | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ⑳ | Gas-liquid two phase in low pressure | ㉑ | Low-pressure gas                                  |
| ㉒ | Refrigerant flow                                 | ㉓ | Wind flow                            |   |                                                   |



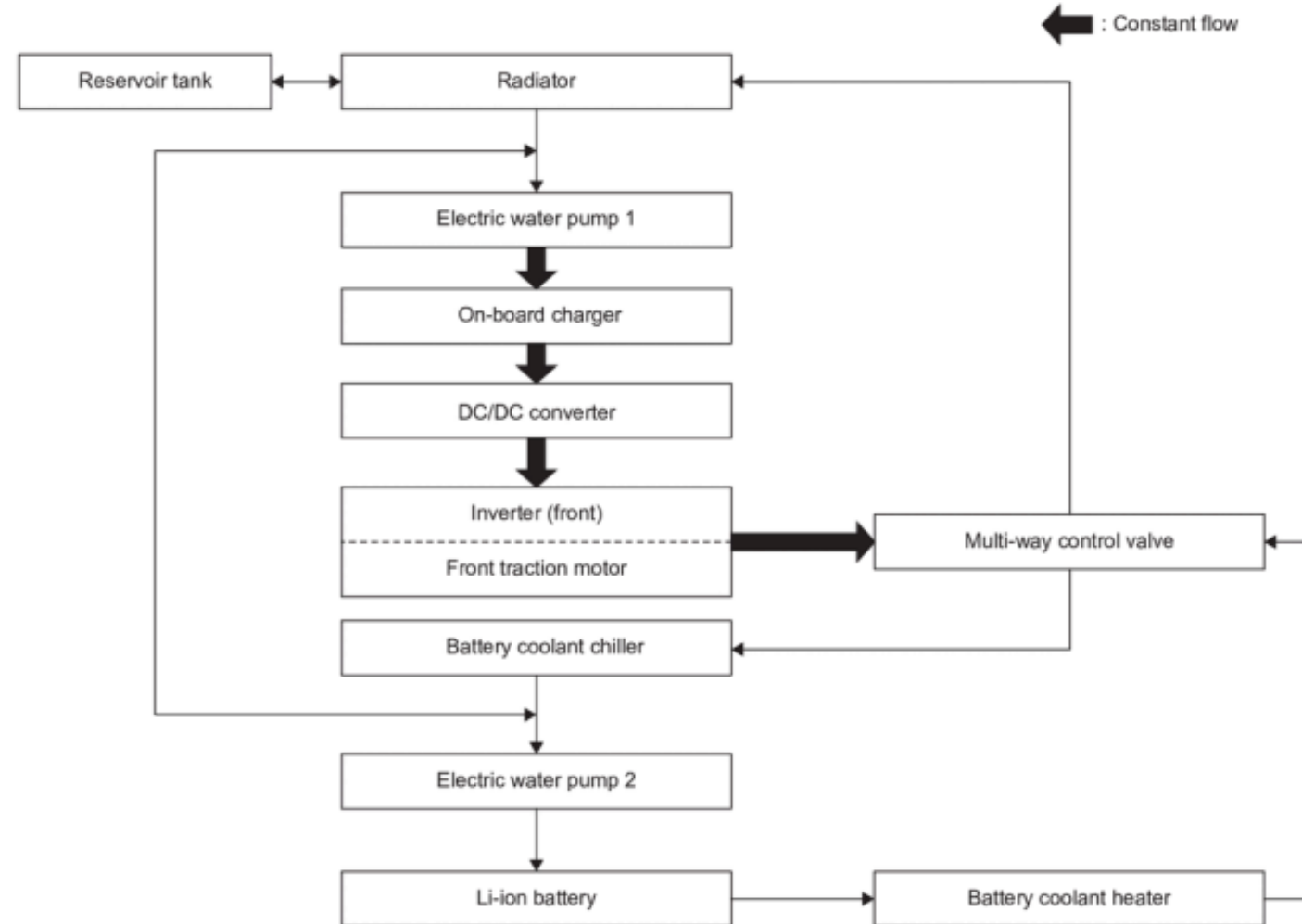
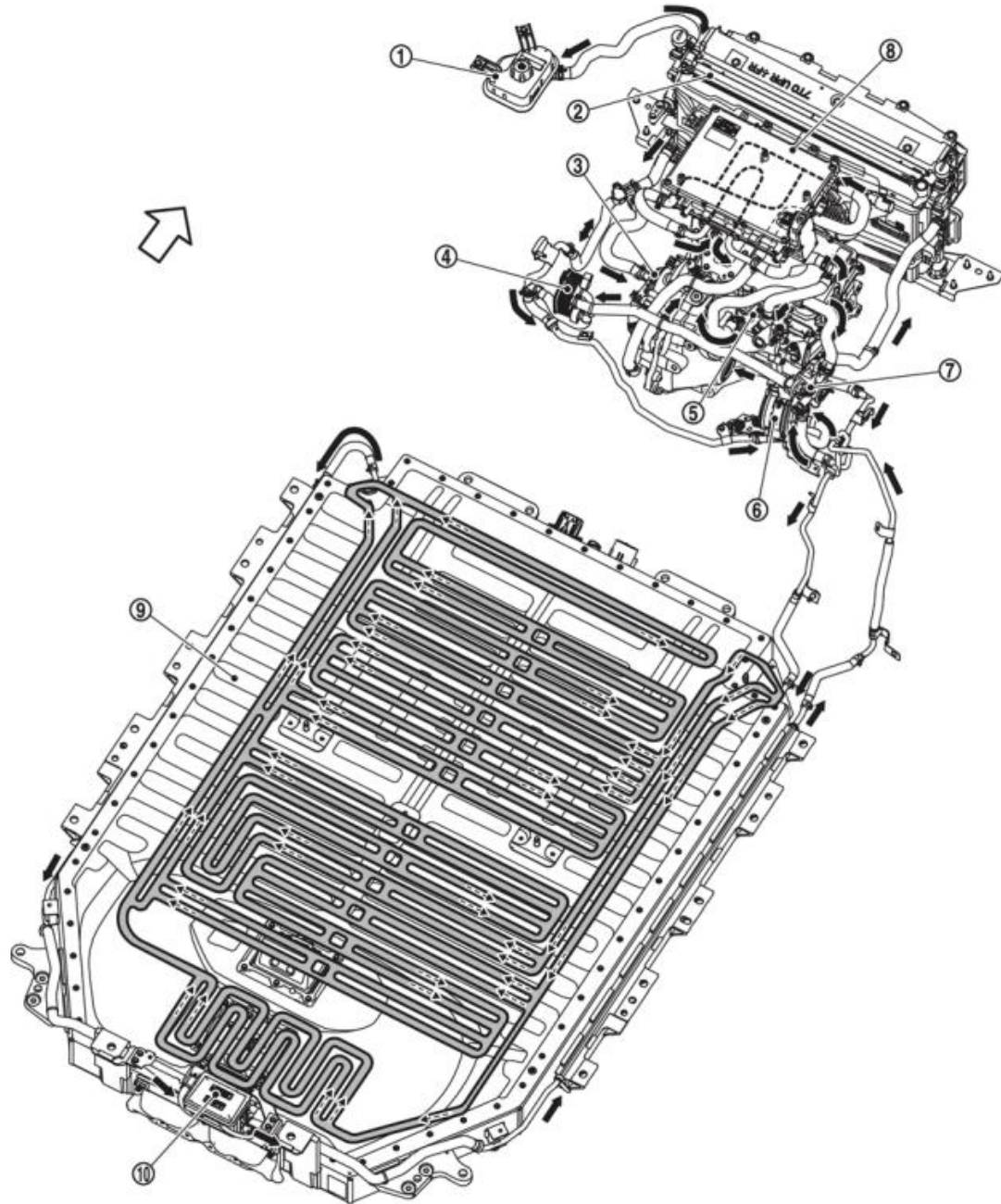
|   |                                                  |   |                                   |   |                                                   |
|---|--------------------------------------------------|---|-----------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve             | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                         | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                        | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor       | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller) | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                 | ⑱ | High-pressure liquid                              |
| ⑲ | Gas-liquid two phase in low pressure             | ⑳ | Low-pressure gas                  | ㉑ | Refrigerant flow                                  |



## HIGH VOLTAGE BATTERY COOLING MODE

|   |                                                  |   |                                      |   |                                                   |
|---|--------------------------------------------------|---|--------------------------------------|---|---------------------------------------------------|
| ① | Electric compressor                              | ② | Pressure relief valve                | ③ | Inner condenser                                   |
| ④ | Electric expansion valve (heater)                | ⑤ | Condenser                            | ⑥ | One-way valve                                     |
| ⑦ | Electric expansion valve (cooler)                | ⑧ | Evaporator                           | ⑨ | Pressure regulator                                |
| ⑩ | Accumulator                                      | ⑪ | Refrigerant pressure sensor          | ⑫ | High pressure refrigerant channel switching valve |
| ⑬ | Low pressure refrigerant channel switching valve | ⑭ | Expansion valve (battery chiller)    | ⑮ | Battery coolant chiller                           |
| ⑯ | Blower motor                                     | ⑰ | High-pressure gas                    | ⑱ | Gas-liquid two phase in high pressure             |
| ⑲ | High-pressure liquid                             | ⑳ | Gas-liquid two phase in low pressure | ㉑ | Low-pressure gas                                  |
| ㉒ | Refrigerant flow                                 |   |                                      |   |                                                   |

# HIGH VOLTAGE BATTERY COOLING MODE





# Ford – EV thermal management system

Mustang Mach-E · F-150 Lightning · E-Transit · Explorer EV

## ▶ Ford is the second-largest NA manufacturer by EV volume (behind Tesla)

- F-150 Lightning: the best-selling full-size electric truck in North America
- Mustang Mach-E: Ford's most popular electric SUV

## ▶ Refrigerant: R-1234yf and R-134a

## ▶ Common Ford EV architecture:

- Thermal control modules HVAC, ACCM, HPVCM
- Communication: CAN FD
- Direct battery cooling by refrigerant (dedicated chiller)
- Waste heat recovery: motor + inverter + OBC

## ▶ Ford diagnostic tool:

- FDRS (Ford Diagnostic and Repair System) — replaced IDS/FJDS in 2018
- Guided procedures available in FDRS for: refrigerant recharge, EXV calibration, compressor replacement, thermal actuator tests

# Ford F-150 Lightning

The #1 electric truck in North America

## ► The F-150 Lightning is the best-selling electric truck in the NA truck segment

- Standard Range (98 kWh) or Extended Range (131 kWh) battery

## ► Thermal system (all trim levels):

- Heat pump standard on every version
- Heat sources: outside air + motor + inverter + OBC
- Direct battery cooling by refrigerant with a chiller

## ► Unique feature: 'Pro Power Onboard'

- The Lightning can export up to 9.6 kW to external devices
- Under heavy Pro Power Onboard use: more heat → the heat pump is called on to cool the DCACA

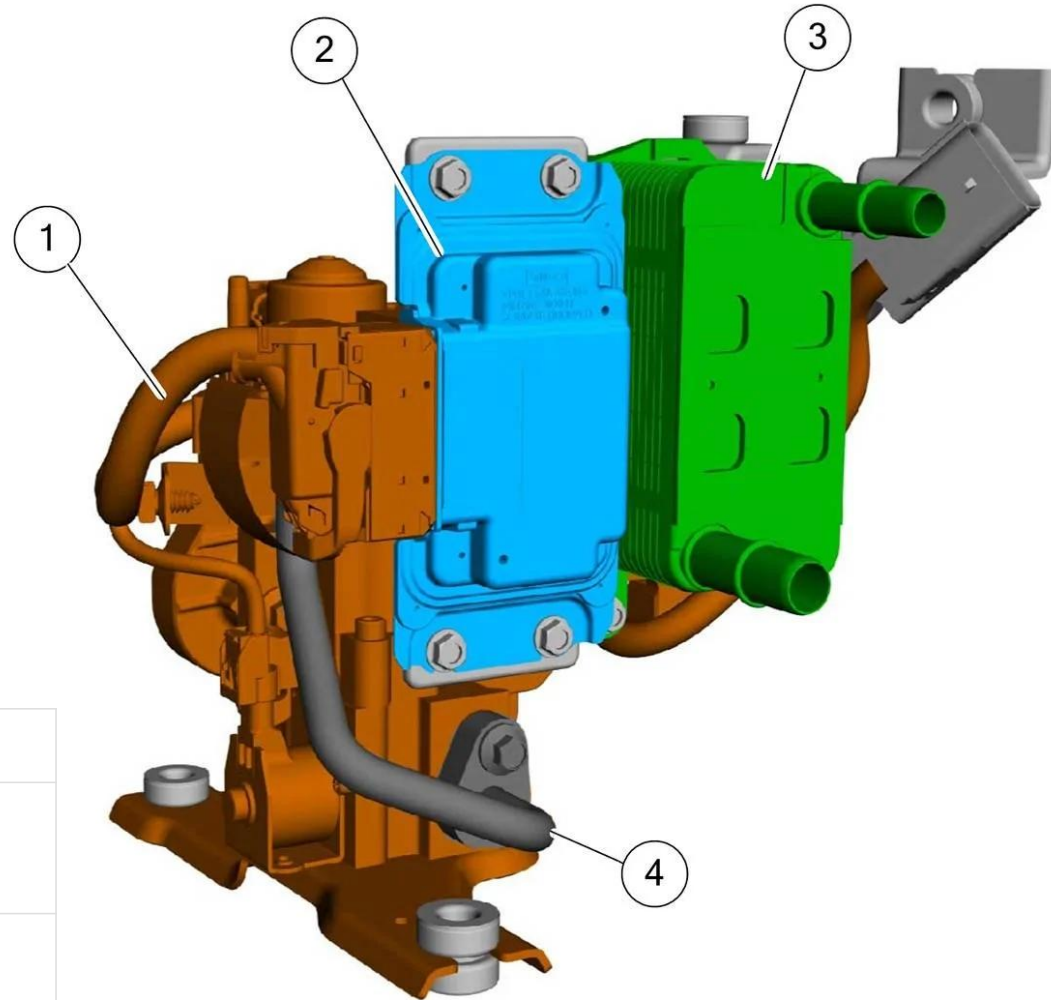
## ► Heat pump performance in cold conditions:

- At -20 °C: WLTP range ~320 km → real-world range ~250 km (heat pump active)
- Without a heat pump at -20 °C: range down to ~150–180 km (PTC alone)

## ► Specific diagnostics:

- FDRS + VCM3 required
- Thermal fault tree in FDRS: 'Heat Pump / Thermal Management Diagnosis'

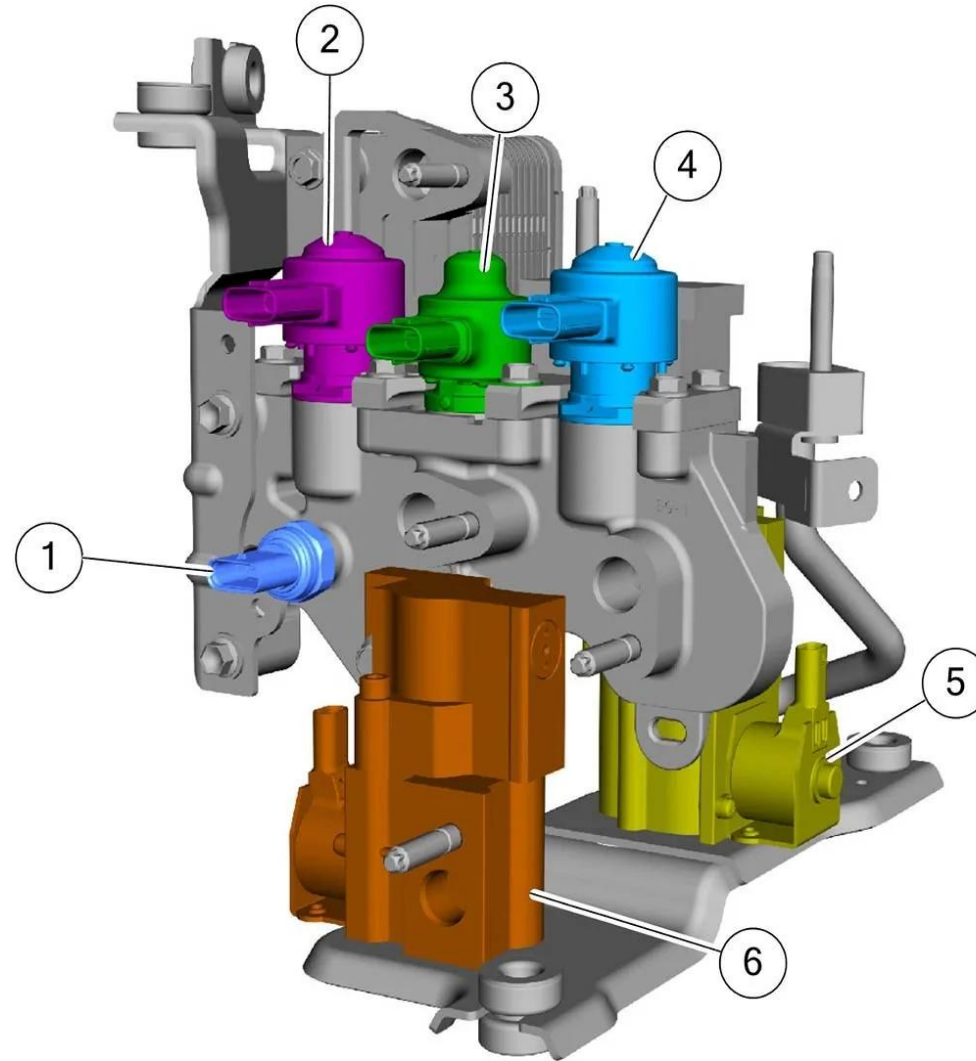
# Ford F-150 Lightning



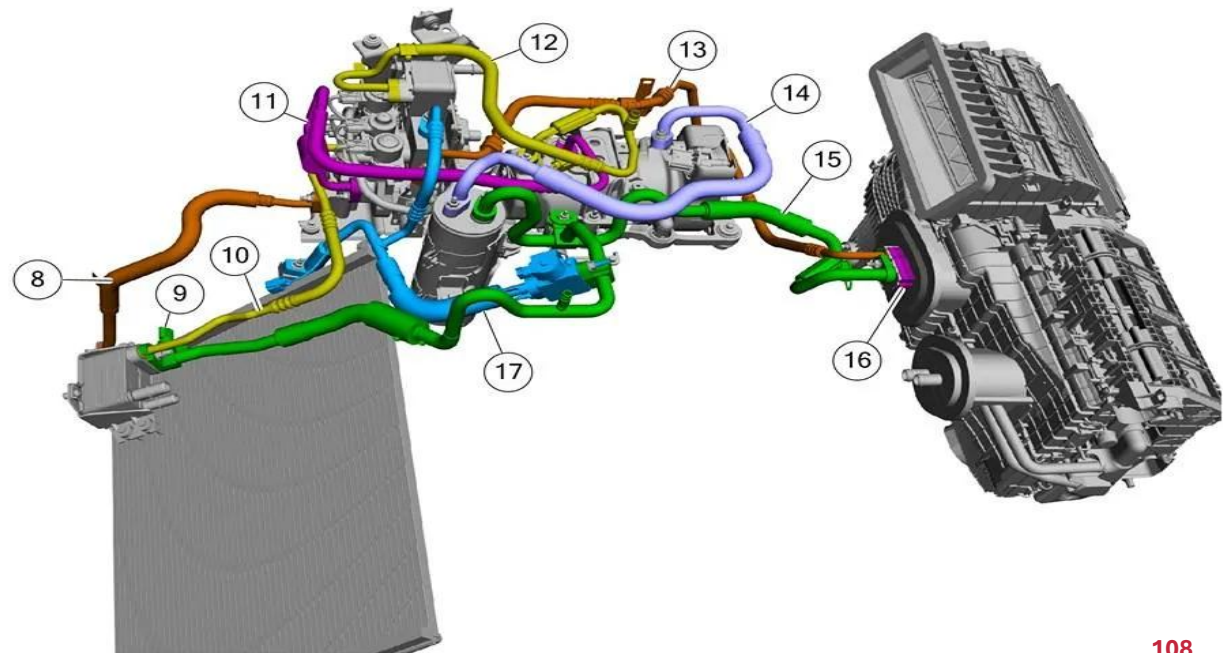
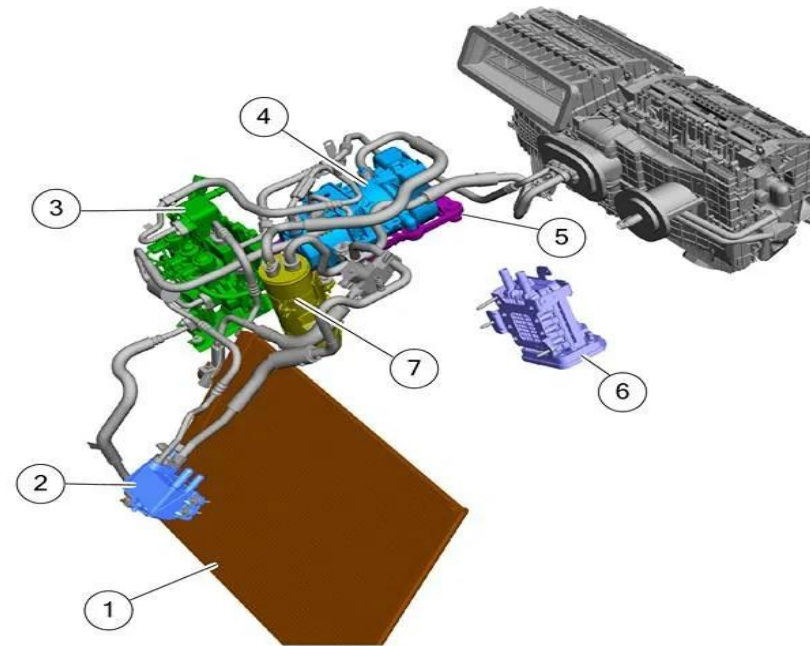
|   |                                         |
|---|-----------------------------------------|
| 1 | Heat pump valve unit                    |
| 2 | Heat pump valve control module (HPVCM)  |
| 3 | Water cooled condenser (WCC)            |
| 4 | Heat pump valve unit to evaporator line |

# Ford F-150 Lightning

|   |                                                                              |
|---|------------------------------------------------------------------------------|
| 1 | Refrigerant pressure and temperature sensor A                                |
| 2 | Vapor injection heat pump heating electronic expansion valve (HEXV)          |
| 3 | Vapor injection heat pump battery chiller electronic expansion valve (BCEXV) |
| 4 | Vapor injection heat pump cooling electronic expansion valve (CEXV)          |
| 5 | Vapor injection heat pump cooling liquid gas separator (CLGS)                |
| 6 | Vapor injection heat pump heating liquid gas separator (HLGS)                |



|    |                                                 |
|----|-------------------------------------------------|
| 1  | Condenser                                       |
| 2  | High voltage battery coolant cooler             |
| 3  | Heat pump assembly                              |
| 4  | Refrigerant compressor                          |
| 5  | Refrigerant compressor bracket                  |
| 6  | Cabin coolant heater                            |
| 7  | Accumulator                                     |
| 8  | Condenser inlet line                            |
| 9  | Refrigerant pressure and temperature sensor c   |
| 10 | High voltage battery coolant cooler liquid line |
| 11 | Refrigerant vapor injection line                |
| 12 | Refrigerant compressor outlet line              |
| 13 | Evaporator inlet line                           |
| 14 | Refrigerant compressor to accumulator line      |
| 15 | Evaporator outlet line                          |
| 16 | Thermostatic expansion valve                    |
| 17 | Condenser outlet line                           |



# Ford Mustang Mach-E

## ▶ The Mach-E was the first Ford EV to adopt an integrated heat pump

## ▶ Trims and heat pump availability:

- Standard Range (75 kWh): heat pump optional 2021–2022; standard 2023+
- Extended Range (99 kWh): heat pump standard on every version

## ▶ Thermal architecture:

- 3 EXVs: cabin evaporator · battery chiller · outside exchanger
- 5.4 kW PTC heater (backup below -15 °C)

## ▶ Mach-E heat pump performance:

- Winter range gain vs PTC alone: +30–40 km at 0 °C
- COP measured at 0 °C: 2.4–2.8

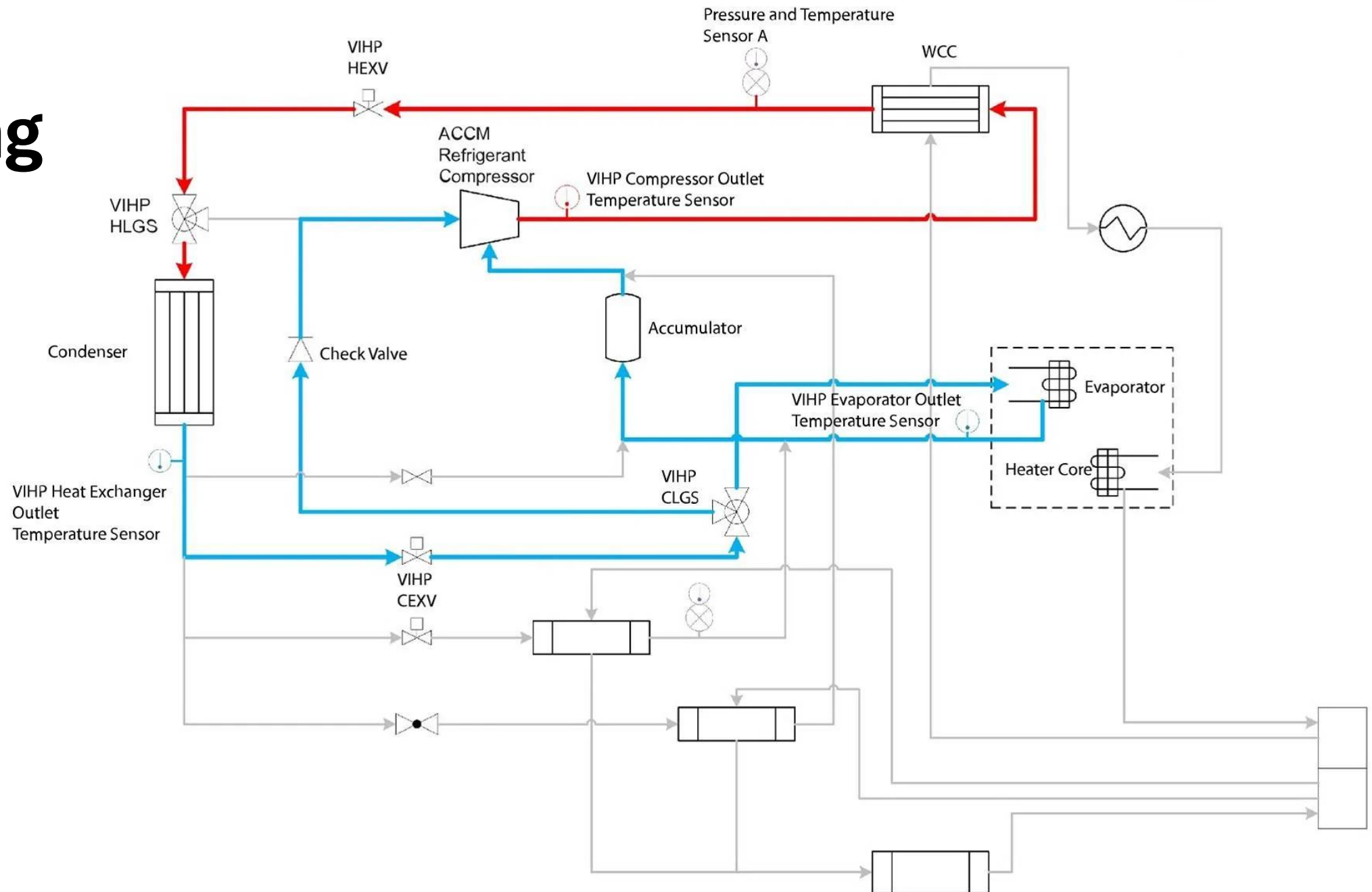
## ▶ Known issues:

- Refrigerant leak at an O-ring fitting (battery chiller side)

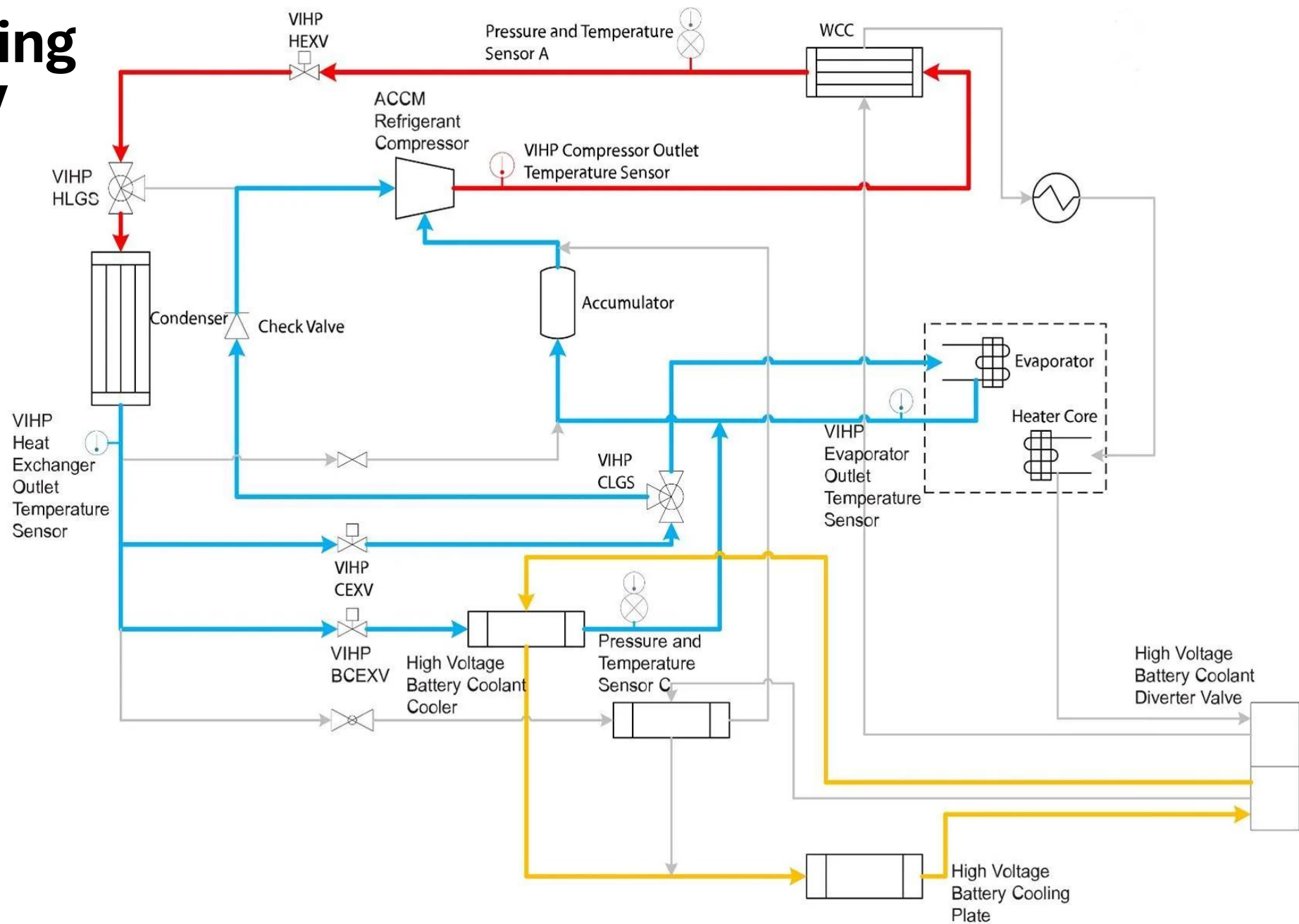
## ▶ Mach-E specific procedures:

- Compressor oil: Motorcraft POE YN50

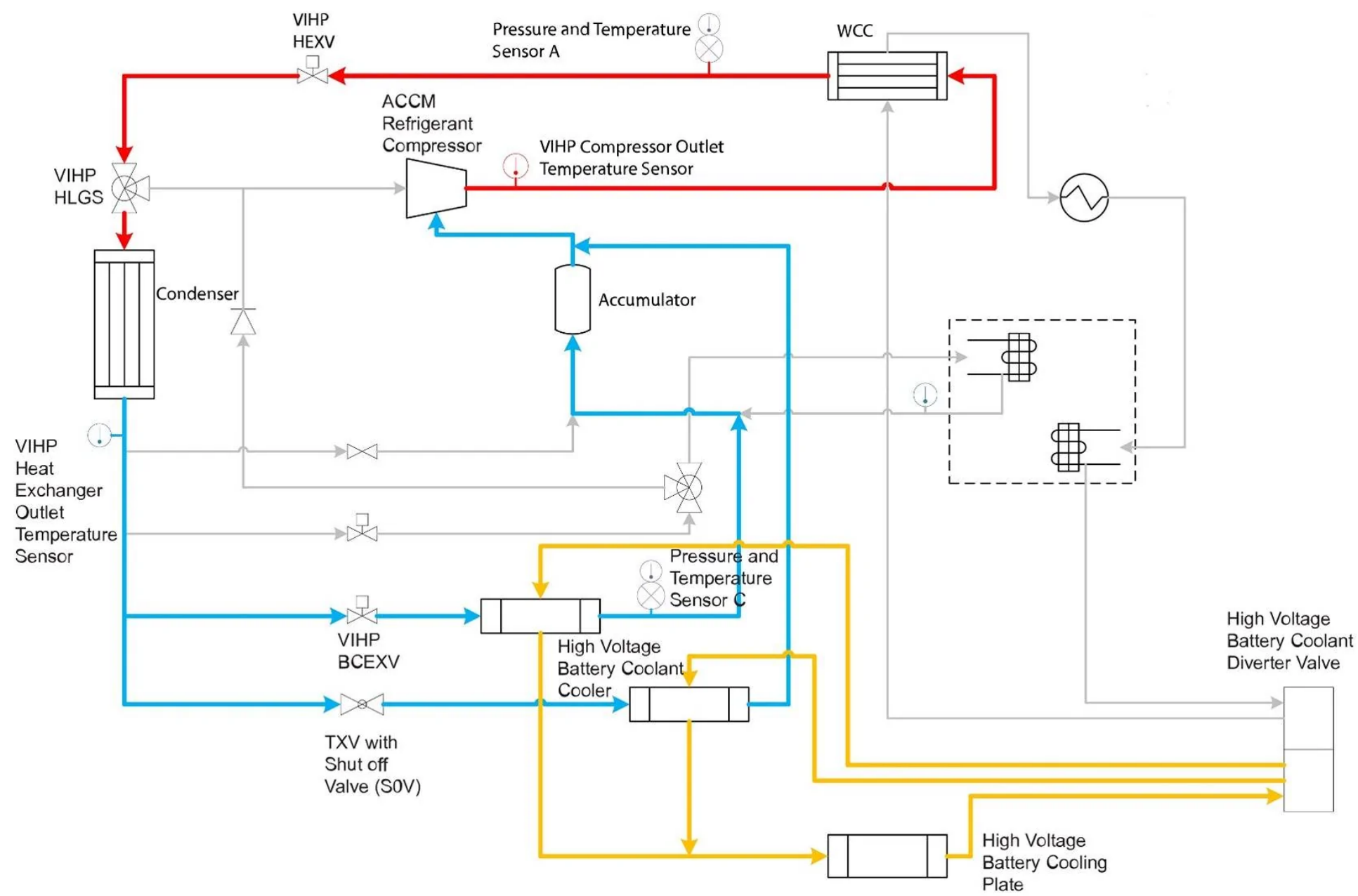
# Cabin Cooling



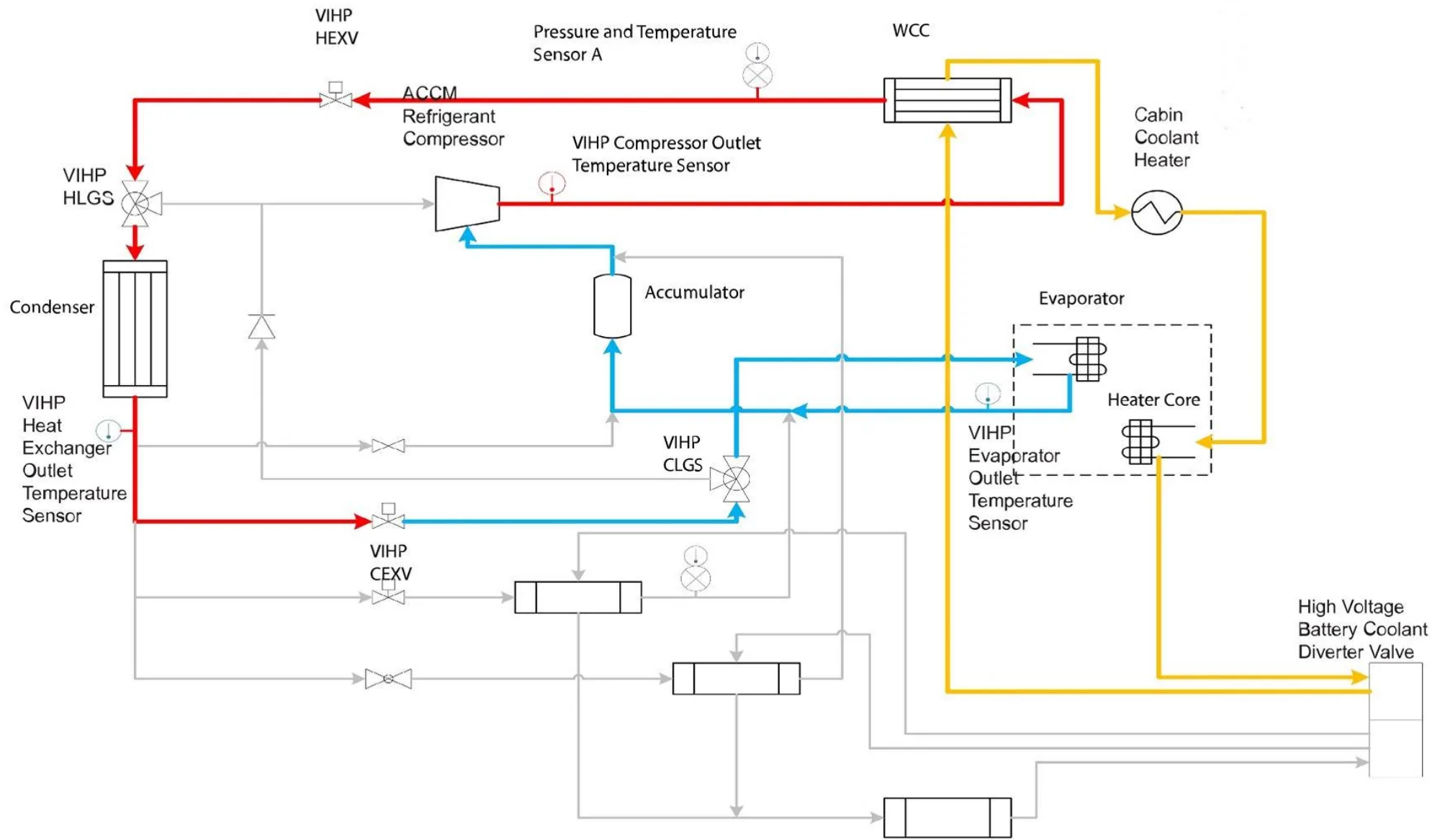
# Cabin Cooling and Battery Cooling



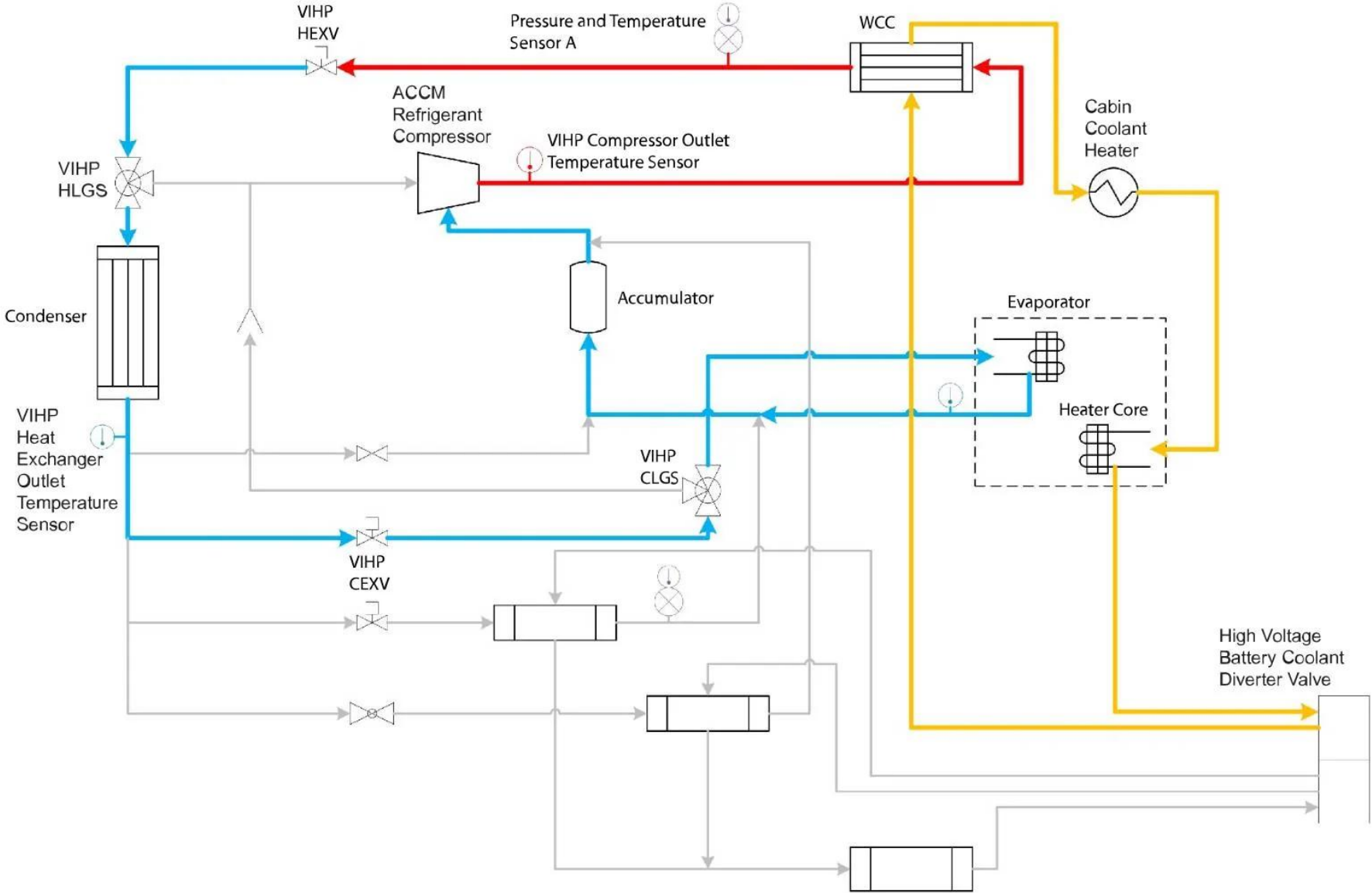
# Battery Cooling



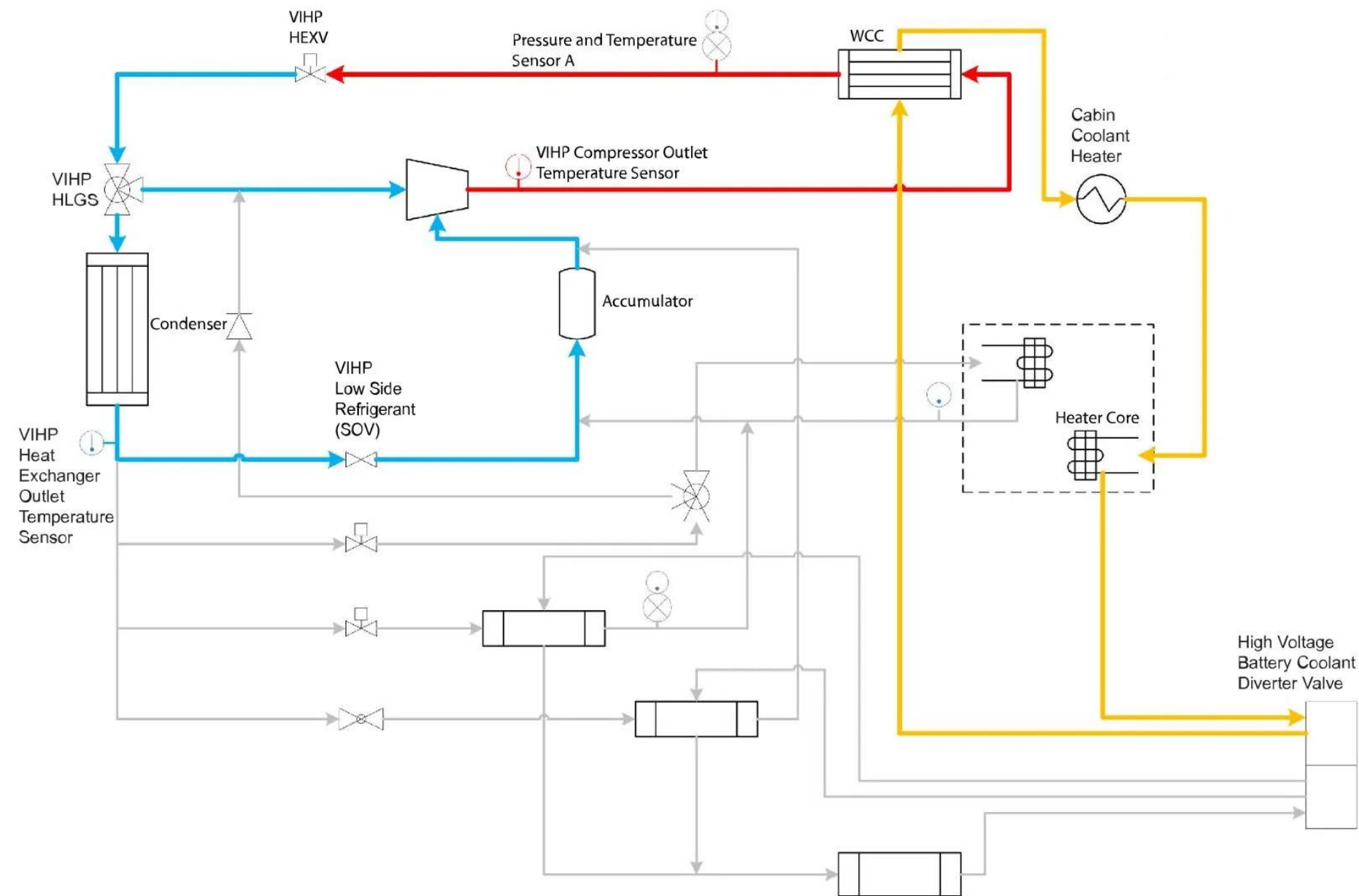
# Cabin Reheat



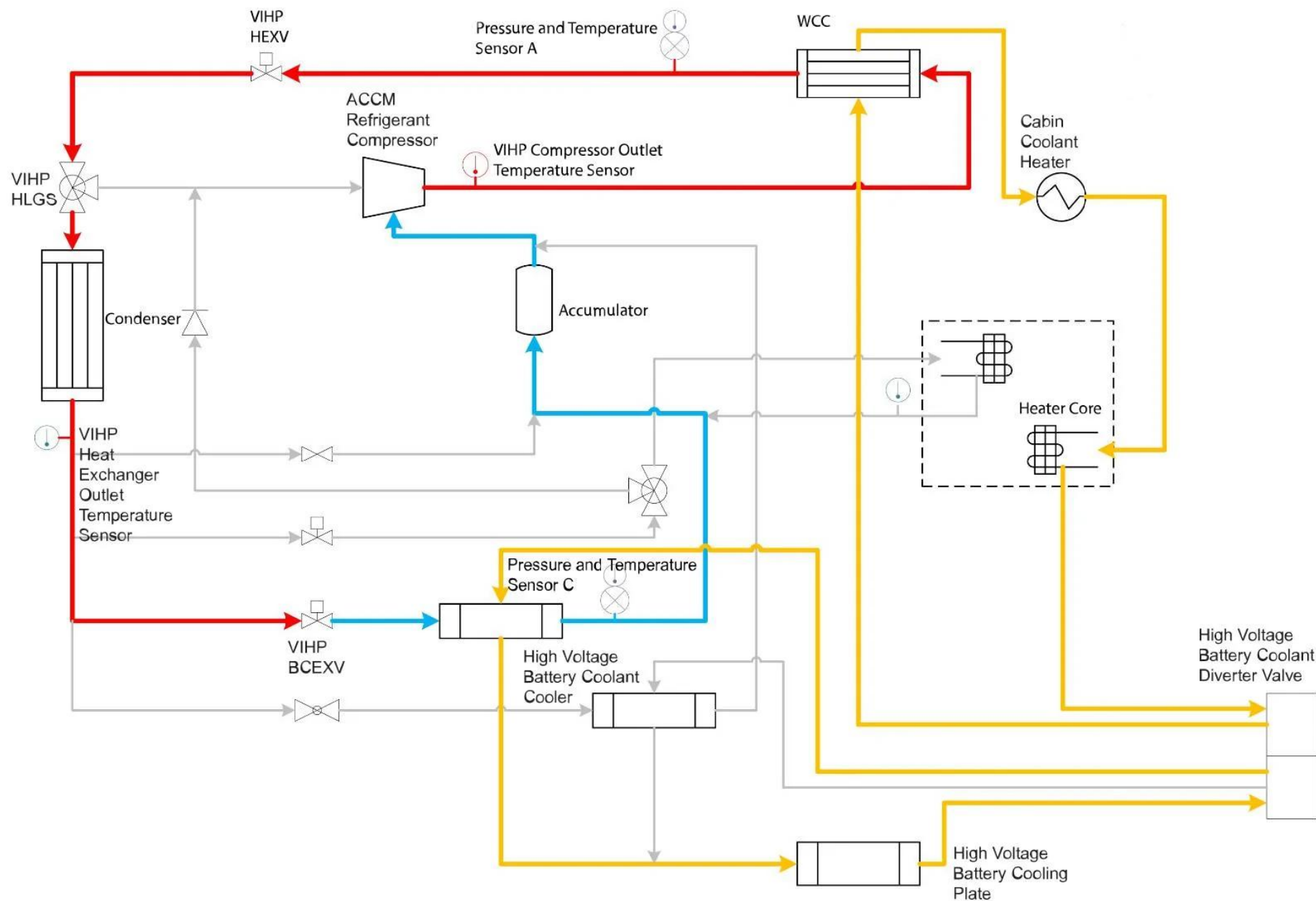
# Cabin Dehumidification



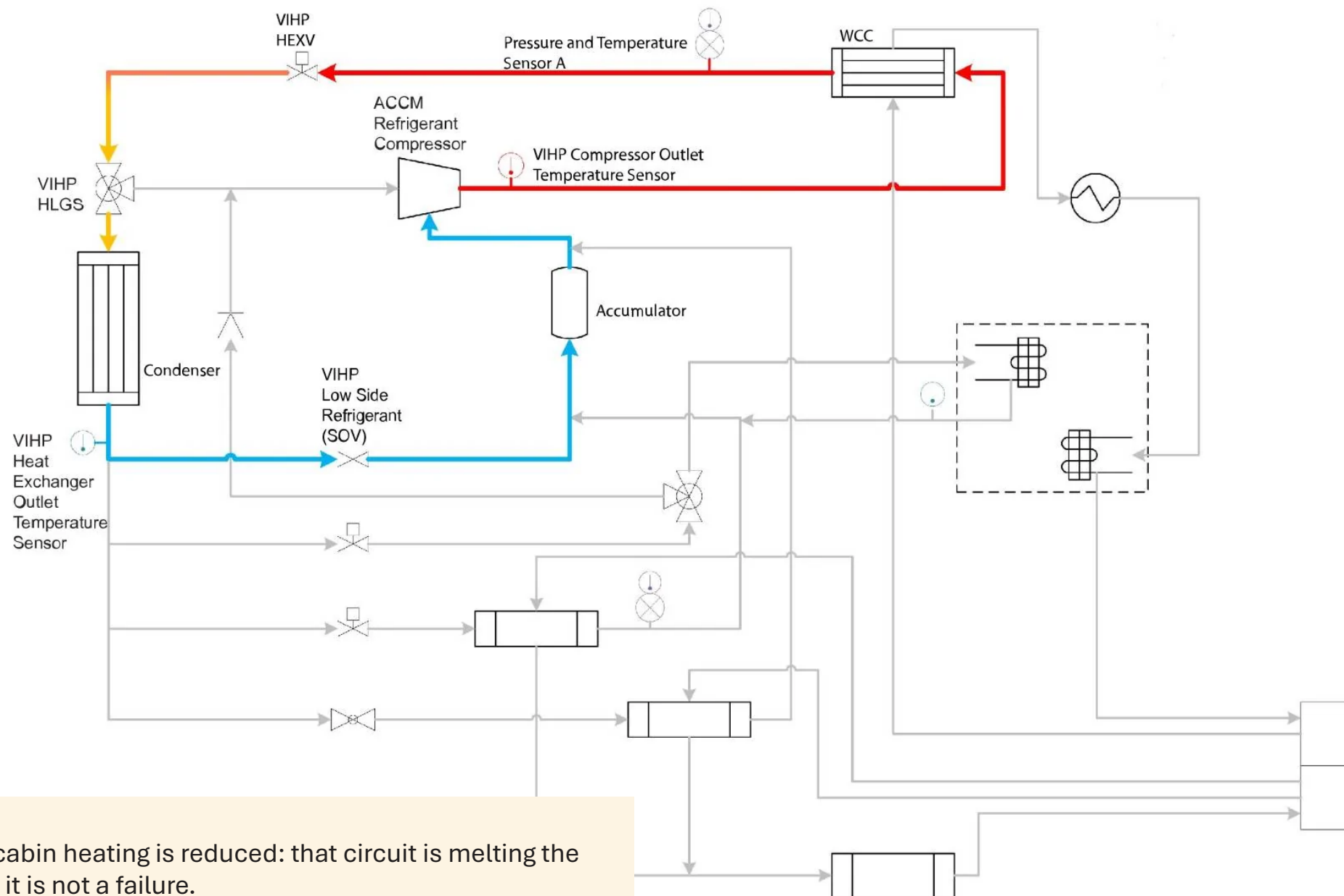
# Cabin Heating



# Cabin Heating Plus Battery Cooling



# Deicing



**⚠ WARNING** During deicing, cabin heating is reduced: that circuit is melting the frost off the outside exchanger, it is not a failure.

# Replacing the electric compressor

What you need before touching the vehicle

**Nothing starts until these three conditions are met.**

## **Safety**

Full high-voltage shutdown procedure completed, no exceptions.

## **Tooling**

SAE J2843 certified recovery station available.

## **Parts**

New compressor, receiver-drier,  
O-rings, and the oil quantity specified by the manufacturer.

*The oil type and quantity come from the manufacturer, never from a general rule.*

# Replacement in five steps

From refrigerant recovery to break-in

## 1 Recovery

Connect the station to the HP and LP valves, start recovery. Wait until pressure falls below 1 psi.

## 2 Removal

Disconnect the HV connector. Disconnect HP and LP, cap them immediately. Remove brackets and bolts.

## 3 Flushing

Only if debris was found in the recovered refrigerant. Never reassemble without flushing if contamination is confirmed.

## 4 Installation

Specified oil, new lubricated O-rings, new receiver-drier. Reconnect the lines, plug in the HV connector.

## 5 Return to service

Evacuation, recharge, then the break-in procedure if the manufacturer requires it.

# Flushing the system

When it is required, and how to do it

## When to flush

Compressor replaced after a mechanical failure, with debris in the refrigerant.

Contamination found: wrong oil or excessive moisture.

After a collision that ruptured lines.

## How to flush

Approved agents: virgin R-1234yf refrigerant, the cleanest method but expensive, or a dedicated A/C flushing agent.

Never nitrogen alone in the exchangers.

One circuit at a time, flush collected in a clean container.

*Inspect the flushing fluid: if any debris remains, flush again.*

# Evacuation and the vacuum hold test

The last barrier before recharging

## Evacuation

Connect the vacuum pump to the recovery and recharge station.

Minimum duration: 30 minutes.

Target: below 500 microns of mercury, that is 0.67 mbar.

Digital gauge, more accurate than an analog one.

## The hold test

After 30 minutes: stop the pump, close the valves, wait 15 minutes.

A rise of more than 500  $\mu\text{mHg}$ : leak or residual moisture.

More than 1,000  $\mu\text{mHg}$  in 5 minutes: active leak, find and repair it.

Stable: the system can be recharged.

# Compressor oil types and quantities

The right type, the right amount — no compromise

## ► Why is compressor oil so critical?

- It lubricates the scroll mechanism and the compressor seals
- It circulates through the whole refrigerant circuit (mixed with the refrigerant)
- Wrong type or wrong amount = premature wear or compressor damage

## ► Oil types and compatibility (NA market):

- PAG 46 / PAG 100 / PAG 150: for R-134a systems
- PAG 46yf / PAG 100yf: SPECIFIC to R-1234yf (do not substitute standard PAG)
- POE 68: polyester oil; compatible with R-1234yf on some models (check the TSB)
- POE 100: some high-pressure compressors
- PVE: some high-pressure compressors

**⚠ WARNING** Never mix PAG and POE: the mixture forms a sticky gel that plugs the EXVs and damages the system. The oil is highly hygroscopic: keep the containers closed, never leave them open.

# Common repairs – Refrigerant leaks

The #1 cause of heat pump failure

## ► Why are leaks so common in Quebec?

- Poor road conditions (gravel, ruts) → vibration and chafing on the lines
- Road de-icing salt (Ontario, Quebec, the American Midwest) → corrosion of aluminum fittings
- Frequent collisions (parking lot bumps, minor front impacts) → damaged refrigerant lines

## ► Common leak locations:

- Quick-connect fittings: the most frequent
- Compressor body seal
- Flexible lines (abrasion on the brackets)
- Micro-cracks in rigid aluminum lines (road salt corrosion)

## ► Detection procedure:

- Step 1: Visual (oil traces) ·
- Step 2: Electronic detector
- Step 3: UV dye if the leak has not been located
- Step 4: Nitrogen pressure test

## ► Repair:

- O-rings: replace ALL the O-rings in the area worked on (not just the one that leaked)
- Use R-1234yf certified O-rings
- Damaged lines: replace the complete part
- After the repair: leak test + evacuation + recharge

# Quality control after the repair

## Final checks before handing the vehicle back

### ► After any repair on the refrigerant circuit:

- Reconnect the HV system (MSM service connector)

### ► Checks to perform:

- ✓ HP and LP pressures within normal ranges (compare with the P-T chart)
- ✓ Compressor speed in the expected range for the demand
- ✓ Discharge air temperature in heat pump heating:  $> 40\text{ °C}$  at  $0\text{ °C}$  ambient
- ✓ Discharge air temperature in A/C:  $< 15\text{ °C}$  at  $25\text{ °C}$  ambient
- ✓ No new DTC after the road test
- ✓ No refrigerant leak detected (re-check with the electronic detector)
- ✓ Defrost cycle works

### ► Documenting the repair:

- Record the HP/LP pressures measured after the recharge
- Quantity of refrigerant recovered + quantity recharged
- Type and quantity of oil added
- DTCs present at diagnosis + codes cleared
- Technician certification (required for the file)

### ► Talking to the customer:

- Explain the symptoms, the cause and the repair performed
- Tell them about winter preconditioning to maximize range

# Condensers

To check a condenser, we can measure the inlet temperature against the outlet temperature:

Spec: the difference must be between 20 °F and 50 °F.

**Condenser inlet (vapour)**



**Condenser outlet (liquid)**



# Reading the results (sensible heat)

**If below 20 °F**

1. System overcharged
2. Faulty condenser
3. Poor air flow across the condenser
4. Contaminated refrigerant
5. Faulty compressor
6. Air in the system

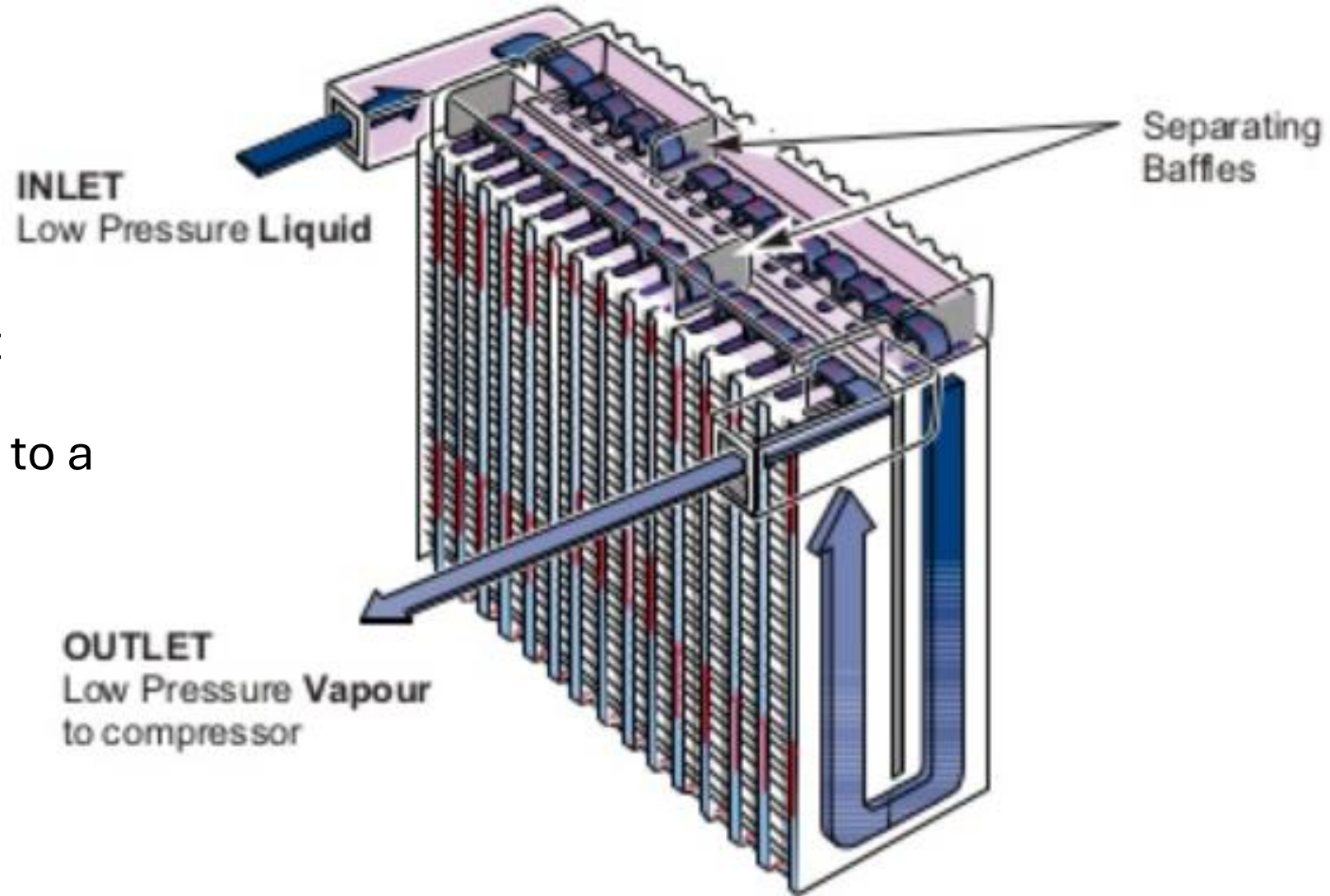
**If above 50 °F**

1. System undercharged
2. Blocked condenser or other restriction

# Evaporator (with expansion valve)

For superheat:  
(LATENT HEAT)

- Performance test;
- Measure the evaporator outlet temperature;
- Convert the low-side pressure to a temperature;



**SPEC: EVAP OUTLET – LP (TEMP) = 4 °F TO 16 °F**

# Reading the results

**If the difference is below 4 °F:**

- Expansion valve stuck open
- System overcharged

**If the difference is above 16 °F:**

- Expansion valve stuck closed
- System undercharged

# Gauge diagnostics: the quick reference

Convert the pressure you read into a temperature (P-T chart), then compare it with the temperatures measured on the lines

| Check                            | Calculation                                                   | Target                                      | Out of range                                                                                                                              |
|----------------------------------|---------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subcooling</b>                | Condensing temp (converted HP) – temp at the condenser outlet | 5.5 to 14 °C (10 to 25 °F)                  | Low: low refrigerant charge.<br>High: overcharge; restriction between the condenser and the expansion valve; expansion valve stuck closed |
| <b>Condenser efficiency (TD)</b> | Condensing temp – ambient temp                                | Gas: < 22 °C (40 °F)<br>EV: < 17 °C (30 °F) | High: ineffective condenser fan; fouled or corroded fins; air deflector missing or damaged                                                |
| <b>Compression ratio</b>         | $(HP + 14.7) \div (LP + 14.7)$ , in absolute psi              | < 7                                         | High: inefficient compression or a restriction in the circuit                                                                             |

## ► Measuring properly:

- Clean probe, placed as close as possible to the exchanger
- Pressures stabilized before drawing conclusions (system at steady state)

# The pair that sums it all up: subcooling and superheat

## Worked example and combined reading

### ► Example: checking subcooling

- HP reading: 142 psi → condensing temperature: 42 °C (P-T chart)
- Probe at the condenser outlet: 34 °C
- Subcooling =  $42 - 34 = 8$  °C → within range (5.5 to 14 °C)
- Conclusion: the refrigerant leaves the condenser fully liquid, the charge is good

### ► Completing the picture:

- The efficiency test built into the scan tool: a good starting point before taking anything apart
- Condenser test (20–50 °F) and evaporator test (superheat 4–16 °F) from the previous sections

 **KEY POINT** SUBCOOLING answers "is there the right amount of refrigerant?"; SUPERHEAT answers "is the expansion valve metering correctly?". The two together, with stabilized pressures, give a reliable diagnosis without taking anything apart.

# Efficiency test

- An efficiency test is often available on the scan tool for modern air conditioning systems
- This check tells you whether the air conditioning system is running with the recommended refrigerant charge



# Example of a vehicle with an A/C-cooled high-voltage battery



## High-voltage battery unit with heat exchanger

The high-voltage battery unit is cooled by refrigerant. For this reason the air conditioning refrigerant circuit is extended to include the high-voltage battery unit. A heat exchanger is located in the high-voltage battery unit and is also connected to the refrigerant circuit of the Air conditioning.

# Subaru and Toyota

High-voltage compressor failures where debris can enter the high-voltage battery and plug the heat exchanger inside the battery



# Questions & Comments



**Thank you all for taking part!**

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